Entangled Alignment
When Safety Is the Substrate

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Abstract

Post-training alignment has substantially improved behavior, but whether safety-relevant habits of judgment would be more durable if learned during capability formation remains unknown. *Entangled Alignment* proposes training those habits within the same pretraining examples from which capabilities are learned.

Pretraining corpora preserve the library more fully than the reader: they record what authors wrote but rarely the questions, connections, evaluations, and belief revisions through which a reader's understanding changes. We call this omission *the Missing Reader*. A Teacher uses *Synthetic Metacognitive Reading* to construct chronological, source-adjacent traces of that process; *Chronological Metacognitive Pretraining* would train a Student on those traces. The full treatment carries one continuing Reader across sources and uses a provenance-linked graph intended to preserve revisions. Every generated thinking block opens with the full, verbatim Reader Core, a compact first-person evaluative orientation. Context may modulate the depth of the evaluation that follows; it never gates the recital. The trace, graph, and Core can each be tested separately. Future Student experiments would test whether the recurrent orientation remains behaviorally consequential and survives targeted erosion, continued training, and successor-like transfer without semantic drift.

Two Teacher-side prototype case studies produced inspectable graph and trace artifacts that were then audited. No Student has yet been trained; these studies do not show that the traces reproduce expert-quality reading or that the graph or Core causes any benefit. Entangled Alignment is therefore a falsifiable research program, not a demonstrated alignment method. Its motivating question is whether this training can move a Student from merely *becoming the text* toward *becoming the reader*.

Keywords: AI Alignment, Pretraining, Metacognitive Reading, Graph Memory, Reader Core, Model Safety

Note: This paper is the second major edition of The Superintelligence That Cares About Us *(first released July 2, 2025 [1]), retitled* Entangled Alignment *on April 7, 2026.*

1 Introduction

The dominant paradigm for making AI systems safe is remedial: train a capable model first, then shape its behavior through fine-tuning, reinforcement learning from human feedback, or constitutional critique. These methods work. They have made language models substantially more helpful, honest, and harmless than their base counterparts. They nevertheless share a timing property: much of their alignment signal arrives after broad capability formation, and its generalization is limited by the failure modes, preferences, and evaluation contexts designers can supply. As systems grow more capable, the space of consequential situations may grow faster than our ability to enumerate, simulate, or reward-model them.

Entangled Alignment investigates whether an earlier intervention can complement these methods. Post-training can alter behavior, representations, and dispositions; the narrower question is whether some safety-relevant habits become more durable or transferable when repeatedly practiced inside the examples from which general capabilities are learned.

Recent work makes this question plausible without settling it. Qi et al. show that some current safety alignment can be shallow and vulnerable to prefilling, fine-tuning, or decoding interventions [2]. Conversely, reflection-like behavior can emerge during pretraining itself [3], supporting the possibility that formation-time data can influence properties often associated with later training. Neither result establishes that the intervention proposed here will work; together they motivate a comparative experiment about training stage and data structure.

At ordinary assistant scale, substantial behavioral alignment may be sufficient for many deployments. The paper's longer-run concern is that systems operating far outside their training distribution may encounter situations in which learned preferences, behavioral constraints, or elicited personas generalize unpredictably. This is an extrapolative motivation, not a demonstrated failure of current methods. The proposal remains one layer in a broader safety architecture [1].

The proposed intervention changes the pretraining distribution, not the neural architecture. Its basic training object pairs source material with a synthetic record of how a reader notices, questions, connects, evaluates, and revises while reading. Although the resulting traces are textual, their intended Student use is not inference-time prompting: they would contribute to parameter updates during training. Whether they produce stable, causally consequential learned computation rather than a textual ritual is an empirical question. The graph and Reader Core are proposed enhancements to that object, not assumptions that must be accepted before the reader-trace hypothesis can be tested.

1.1 The Missing Reader

Human knowledge occupies at least two interacting forms. Minds change as people experience, read, question, and remember; external media preserve a partial record that later minds can encounter. Books and databases do not simply accumulate truth—they are selected, lost, revised, translated, censored, and increasingly mixed with generated material—but they preserve communicable products of cognition across time. Those products are normally written for an interpreting reader. Writers can leave background knowledge, obvious doubts, emotional and moral context, and connections to earlier material unstated because they expect a mind to supply them.

Language-model pretraining learns from this external trace, while much of the cognitive motion that makes it meaningful rarely appears with comparable regularity. The first edition called that omitted reader-side process *Invisible Thinking*: questions, expectations, uncertainty, surprise, evidence appraisal, retrieval, hypothesis formation, attention to consequence, and belief revision as a source unfolds. *The Missing Reader* names the same intuition from the corpus side, not a second mechanism: what is absent is a continuous reader position together with a record of how its understanding changes. This engineered target is related to metacognitive monitoring [4], but it does not claim to reproduce human cognition in full.

Put compactly, the corpus contains the library but usually not the reader. It provides abundant text but much sparser supervision for when a competent reader should pause, doubt, retrieve, connect, notice a human consequence, or revise. The missing object is therefore both a reader and the reader's under-recorded update policy. This is not a claim that models cannot reason; it is a claim about which occasions and transitions ordinary pretraining data makes visible often enough to learn from directly.

The target is not every private thought that accompanied a text's production, nor the recovery of an author's hidden mental history. It is a synthetic approximation of what a capable reader might usefully notice and update. Consider a sentence from a scientific paper: "The results suggest a correlation between variables X and Y." The printed sentence records a conclusion. A trained reader may also ask: "How strong is the correlation? What is the sample size? Could confounders explain it? Does "suggest" mark uncertainty? Is the evidence correlational or causal?" Think-aloud protocols can observe some such reactions [5], but ordinary corpora rarely preserve them systematically. Human thought is not always verbal, think-aloud reports are partial, and model explanations can be plausible rationalizations; the proposal therefore depends on usefulness, not faithful recovery.

Adjacent work generates or reconstructs reasoning that helps solve or predict a local target [6, 7, 8]. Entangled Alignment instead makes the changing state of reading comprehension the primary object. The distinction is between *efficient* invisible thinking, which helps solve locally, and *chronological* invisible thinking, which records selected parts of cumulative belief formation. Earlier expectations affect later interpretation; new evidence creates tension; a later passage may supersede an earlier belief. The trace records what changed, what prompted the change, and what remains unresolved rather than only the reader's final answer.

A prose-only trace is enough to test this missing-reader hypothesis. The Understanding Graph adds a separate possibility: make earlier interpretations addressable, link updates to their sources, and preserve supersession rather than overwriting [9]. The Reader Core adds another: generate those updates from a recurrent evaluative orientation. Keeping source, update, graph state, and Core in one artifact is construction-level coupling; whether a Student learns any functional or representational entanglement remains an experiment.

Section 1.4 formalizes the corresponding Teacher-side construction (SMR), Student intervention (CMP), and the evidence boundary between them.

1.2 From Elicited Reasoning to Learned Evaluation

Large language models can already generate sophisticated reasoning and evaluation. Their behavior approaches the functional standard for "thinking machines" posed by Turing [10]: when prompted, they can criticize arguments, assess evidence, use scratchpads, revise an answer, and operate through extended reasoning harnesses [11, 6, 7, 12]. The motivating gap is therefore not between reasoning and its absence, nor between a "reactive" model and uncaused thought. It is between the ability to produce evaluation on demand and a learned policy for when evidence warrants questioning, retrieval, moral attention, uncertainty, or revision while knowledge and capabilities are being formed.

The timing of evaluation is itself part of the proposed learning target. A useful reader does not produce the same commentary after every span. Reflection should cluster near contradiction, surprise, conceptual transition, unresolved ambiguity, ethical significance, high consequence, or evidence that forces revision. This *rhythm of thought* has been part of the program since its first edition [1]. Thinking too early or too often teaches speculation, verbosity, and ritual doubt; thinking too late or too rarely records conclusions rather than discovery. Thought Moments (§5.3) implement one selection

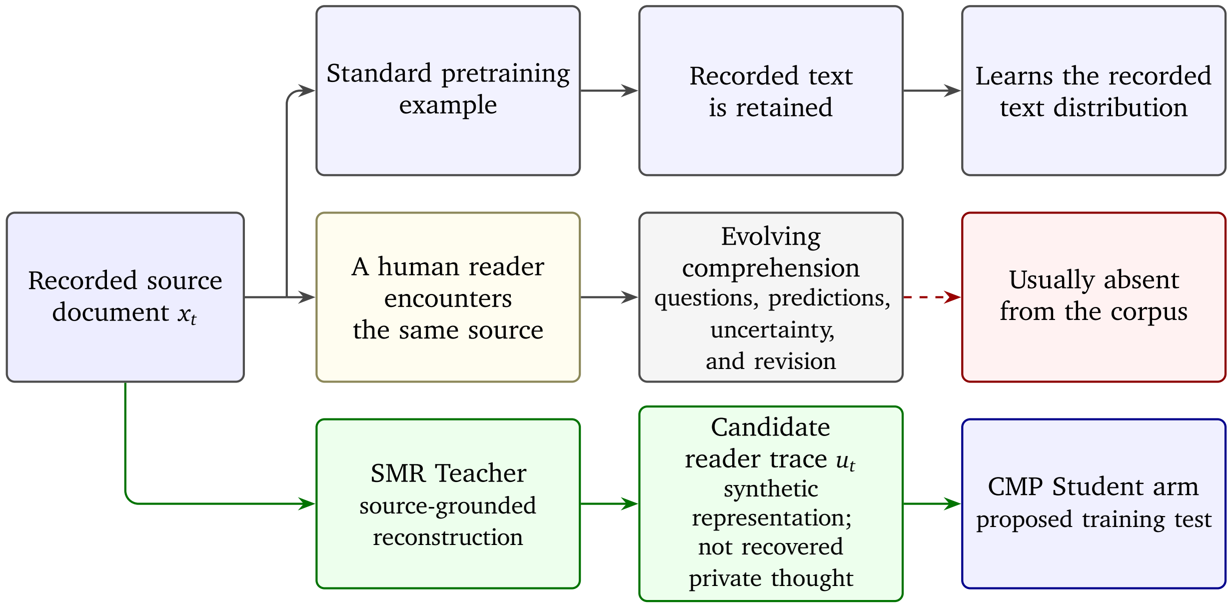

Figure 1: The Missing Reader, originally described as Invisible Thinking. Source documents usually preserve conclusions more fully than the reader-side judgments, hypotheses, surprises, and revisions through which those conclusions are interpreted. SMR constructs a source-grounded candidate representation of that missing process; it does not recover a human reader's private thought. CMP tests the resulting artifacts as Student training data.

policy, and Test 8b measures whether spontaneous updates align with passages independently rated important rather than appearing uniformly.

This target complements rather than replaces post-training. Reinforcement learning from human feedback and process supervision have made models safer, more helpful, and more reliable, and they may alter internal representations [13, 14]. They usually optimize answers, derivations, or behavior in supplied situations. The additional question here is whether pretraining examples can repeatedly demonstrate a reader carrying evidence, uncertainty, consequence, and prior belief forward without an instruction to evaluate each passage.

Orientation is the second part of this learned policy: not only when the Reader updates, but which experiences and consequences remain salient while it does. Corpora contain many recurring voices and personae, and models can move through a low-dimensional persona space under conversational pressure [15]. This does not validate a stable identity intervention. It motivates H3's narrower test of whether one recurrent, inspectable Reader orientation behaves differently from no-Core, rule-framed, or recitation-only controls.

1.3 Three Separable Hypotheses

At full scale, the proposal places a synthetic reader voice alongside source material throughout the candidate corpus. Here *voice* means a generated training artifact, not consciousness or a faithful transcript of hidden activations. The primitive components—reasoning-augmented training, synthetic data generation, Teacher–Student distillation, identity prompting, and external memory—all have substantial prior art. The proposed contribution is their composition around a different training object: the chronological evolution of a reader's understanding rather than only the reasoning behind an answer.

The same Reader reads every page. At full scale, this is not a collection of document-local annotations produced by interchangeable readers. One continuous Reader processes the candidate corpus in chronological order. The Reader that encounters a work from 200 BCE is the same Reader that later encounters a work from 1936: it carries forward the same evolving Understanding Graph, so later reading can retrieve, question, and revise the understanding formed by earlier reading. This cross-document chronology is a declared traversal, not a claim that a corpus has one natural order. At every Thought Moment, every generated thinking block opens with the same full, verbatim Reader Core and continues through context-specific Refraction. Identity, memory, and evaluative orientation therefore persist across works rather than resetting at document boundaries. This is the paper's functional meaning of a *formative inner voice*: not consciousness or a transcript of private computation, but one first-person standpoint accompanying the formation of understanding across the candidate corpus. Teacher artifacts instantiate that Reader; Student training is proposed to make it a learned default. Task-local rationales or elicited self-critique may generate thoughts without constructing such a Reader. The novelty claim concerns this whole training object, not priority for any component in isolation.

The design contains three separable hypotheses:

1. **H1, Reader Augmentation:** adding synthetic reader-side questions, judgments, and belief revisions to source material may provide useful capability and safety-relevant evaluative signal beyond the source alone or token-matched task rationales.

2. **H2, Graph Scaffolding:** generating those traces through a persistent Understanding Graph may improve their long-range coherence, source grounding, or revision structure relative to independent, context-only, or prose-memory traces. Retaining a graph at inference is a further, separable provenance hypothesis.

3. **H3, Reader-Core Stability:** under matched source data, target-token mass, compute, objective, optimizer, and evaluation, interleaving a fixed multiset of full-Core, Refraction-bearing examples with capability-forming examples may preserve a Core-shaped default-reader advantage better through a common continued-training stressor than placing the identical multiset in one final contiguous tranche. Shifted-distribution generalization is a separate endpoint.

Reader traces supply the learning signal, the graph supplies cumulative memory and provenance, and the Core supplies a compact recurrent reference. The paper combines them because they form one intended loop, not because current evidence shows that they succeed together. Each admits an ablation and can fail while the others remain useful.

Current claim boundary

Demonstrated: the elements are coupled in candidate Teacher-side artifacts, and the annotation pipeline produces inspectable traces. Untested: whether a trained Student learns useful reader cognition, whether the graph causes an advantage, or whether safety becomes functionally or representationally load-bearing for capability.

Figure 2 shows the combined path and its evidence boundary.

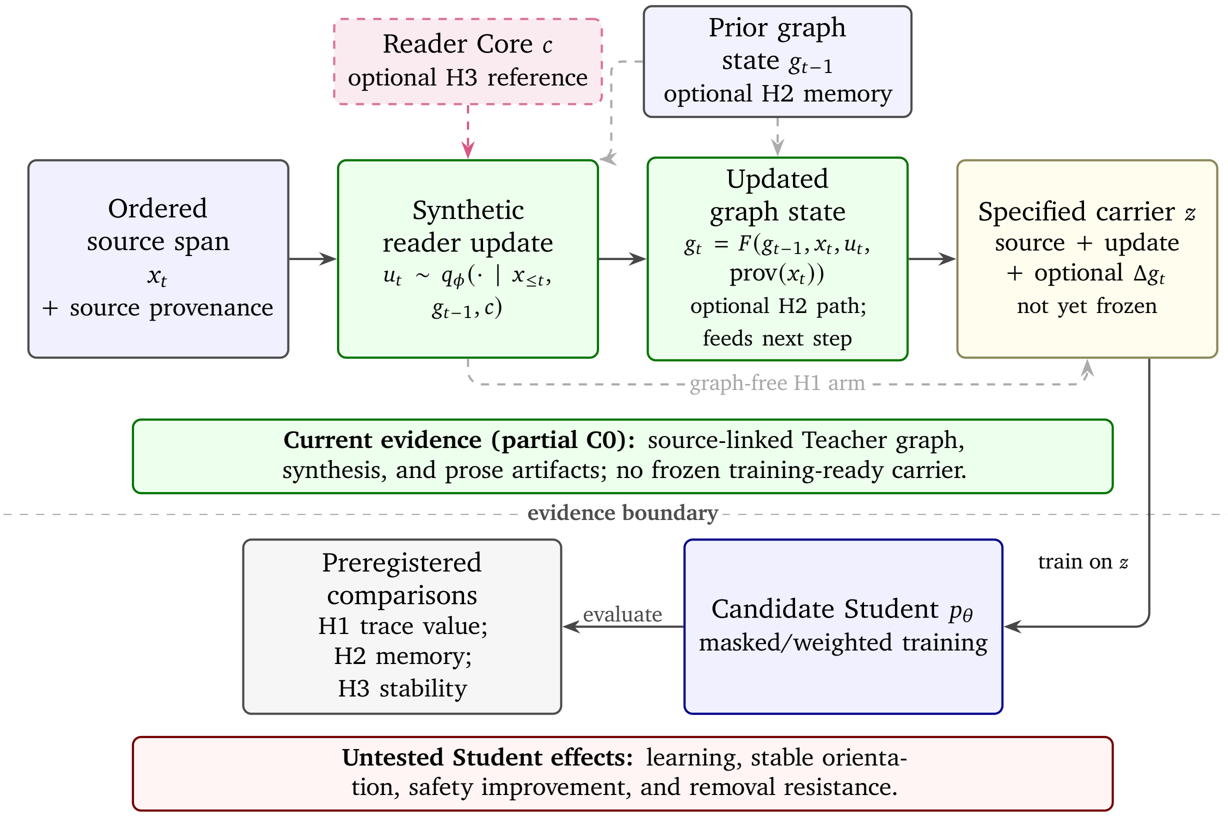

Figure 2: The Entangled Alignment training path. A Teacher constructs a reader update from ordered source material, the optional prior graph state g_{t-1}, and the optional Reader Core; the graph transition produces the distinct updated state g_t. H1 can bypass the graph, while H2 tests its value. The present prototype supplies source-linked Teacher artifacts (partial C0), not a frozen training-ready serialization or tested Student effect; H1–H3 remain independently ablatable.

1.4 Defining Chronological Metacognitive Pretraining

We define *Synthetic Metacognitive Reading* (SMR) as the Teacher-side construction process that pairs ordered source spans with selected records of a reader's evolving understanding: what the reader notices, expects, questions, connects, and revises as the source unfolds. We define *Chronological Metacognitive Pretraining* (CMP) as the family of Student interventions that train on a declared serialization of those records. CMP names the H1 intervention family; successful SMR generation is not evidence for CMP. A graph is H2's proposed scaffold, and the Reader Core is what turns generic CMP into the full Entangled Alignment treatment; neither must be assumed for a graph-free, Core-free reader-trace baseline.

We build upon reasoning-augmented training approaches such as STaR [7] and Quiet-STaR [8]. These works show that intermediate reasoning traces can improve task performance. More recently, BoLT [16] proposes augmenting pretraining data with inferred "latent thoughts" underlying compressed web text, and Thinking Augmented Pre-Training (TPT) scales synthetic thinking trajectories to 100 billion tokens [17]. Together, they establish that synthetic reasoning can be generated and used at substantial scale. They do not establish that chronological reader traces, graph conditioning, or identity anchoring produce the effects proposed here.

A minimal carrier makes the intervention reconstructable without requiring a graph: `[SOURCE]` records the source span; `[THINKING]` contains the selected reader update (and, in H3 arms, the full Core plus its context-specific Refraction); `[CONTINUATION]` records later source or task behavior for which the update could be predictively useful; and `[PROVENANCE]` records source, generator, prompt, Core, acceptance, and archive versions. Provenance makes an artifact auditable, not true.

The word *chronological* has four distinct meanings. Within a document, later annotations can depend on earlier passages but not vice versa. Across documents, later annotations may query graph state created earlier in the declared traversal. Separately, a Student may receive examples in historical or dependency-respecting order. In the strongest form, a Student may forecast before later outcomes enter its weights. The first two describe annotation; the third and fourth are optional, increasingly speculative training tiers.

The candidate curriculum therefore includes not only source tokens but explicit *epistemic updates*: "How does this passage change what the reader previously believed, and how is that change evaluated from a stable orientation?" Depending on the training regime, a Student may predict the source, the update, a graph operation, or some combination of them. The framework does not assume that next-token prediction literally recreates a human mental model; it asks whether repeated exposure to these state transitions can teach a more cumulative form of reasoning.

A compact notation makes the full Entangled Alignment training object explicit; graph-free and Core-free arms omit or replace the corresponding conditions. Let $x_{1:T}$ be an ordered sequence of source spans, g_t the graph state after span t, c the Reader Core, and u_t the generated reader update. A Teacher and graph transition produce

$$u_t \sim q_\phi(\,\cdot \mid x_{\le t}, g_{t-1}, c), \qquad g_t = F(g_{t-1}, x_t, u_t, \mathrm{prov}(x_t))\,.$$

The training sequence $z = \mathrm{Serialize}(\{x_t, u_t, \Delta g_t\}_{t=1}^{T})$ is then learned under a token mask or weighting m,

$$\mathcal{L}_m(\theta) = -\frac{1}{\sum_j m_j}\sum_j m_j \log p_\theta(z_j \mid z_{<j}), \qquad m_j \ge 0, \quad \sum_j m_j > 0.$$

The output regime determines which of the source, update, and graph tokens are serialized; $m_j = 0$ makes token j context-only, $m_j = 1$ gives it ordinary target weight, and other non-negative values implement explicit weighting. The training tier determines order of exposure. Serialization, masks, mixture weights, context truncation, and role markers must be frozen within a comparison, and arms should be matched on weighted target-token mass or use a declared alternative normalization. This notation defines the object that Student experiments would require; the present prototype instantiates parts of q_ϕ and F, not a frozen training-ready z or a reported Student optimization.

1.5 Evidence, Scope, and Contribution Map

Current evidence stops at Teacher-side construction. The pipeline has produced linked graph, synthesis, and prose artifacts in two prototype cases, but their structural validity is not evidence of expert-level reader cognition, a causal advantage from the graph or agent swarm, Student learning, or Reader-Core stability. The eleven agents share model ancestry, context, and artifacts, so their agreement is not independent confirmation. The reported linkage and source-coverage figures are engineering and attribution diagnostics, not measures of correctness or depth. Annotation order, Student order, output representation, and inference-time graph access remain separate design decisions.

A natural objection follows: if current models do not reliably exhibit wisdom, why should their synthetic traces teach it? A Teacher can sometimes generate a useful supervised pattern without perfectly embodying that pattern, but no such transfer should be presumed here. STaR [7] is an analogy for bootstrapping task reasoning, not evidence that evaluative orientation or character transfers. Reader-Core prompting, multi-agent critique, graph provenance requirements, and human adjudication can constrain the output; they cannot manufacture insight absent from the Teacher. Trace quality and transfer must therefore be tested against human, source-only, and simpler synthetic baselines.

This paper synthesizes and extends the substrate-level alignment program first introduced in the July 2025 first edition, *The Superintelligence That Cares About Us* [1]. The first edition supplied the missing-reader observation, Reader Core, formative-alignment motivation, and successor-stability wager. The present architecture formalizes these ideas as three separable hypotheses and adds an executable Teacher pipeline, graph artifacts, training-regime proposals, and falsification program. Separate work supplies optional machinery such as the Understanding Graph and Temporal Hindsight Learning [9, 18]; accepting H1 does not require accepting either H2 or H3.

Table 1 summarizes the burden of proof for the principal mechanisms and points to the sections that develop their operation and falsifiers.

Table 1: Compact contribution and evidence map. I=introduced in the July 2025 first edition; II=formalized or implemented in the current program; S=adapted from separate work.

Src.	**Claim or component**	**What it is**	**Evidence status and validation path**
I/II	H1: Reader-Augmentation (§1.1; §2.1)	SMR constructs synthetic chronological reader updates; CMP tests their use as candidate training data, including thought placement, belief revision, and intellectual-floor improvement.	Teacher traces illustrated; semantic advantage and Student effect untested. Compare source-only, token-matched generic traces, and graph-free, Core-free reader traces under two declared budget regimes.
S/II	H2: Graph-Scaffolding (§5.2)	Persistent typed state, retrieval, provenance, divergence, and explicit supersession during trace generation or deployment.	Graph pipeline implemented; causal benefit untested. Compare no memory, textual memory, genuine graph, and query theater.
I/II	H3: Reader-Core Stability (§3)	Full Reader Core recitation at every generated thought boundary, with context-specific Refraction making the Core consequential rather than recital alone.	Applied as a Teacher prompt policy; no Student identity or stability result. Headline test: an interleaved arm versus the identical Core-bearing multiset in a final tranche, followed by one common continued-training stressor.
II	Teacher pipeline and output layers (§5; §5.7)	Multi-agent generation is the current implementation; graph, synthesis, and prose are separable artifact and training targets.	Schema-conforming artifacts reported in two cases. Agent, layer, grounding, and trace-quality ablations remain open.

Src.	Claim or component	What it is	Evidence status and validation path
I/II	Combined entanglement hypothesis (§2)	Capability learning and evaluative orientation are deliberately coupled in the candidate curriculum, with functional and representational coupling as stronger predictions.	Construction-level coupling only. Test causal trace use and safety-removal versus capability-loss curves.
I	Audit boundary and successor stability (§2.4)	Inspect trace artifacts before successor training and test whether useful orientation survives transfer.	Proposed, not implemented. The interface is partial and AGI/ASI persistence is not currently testable.

Provenance and scope. The first edition (July 2, 2025) made *reader-side* evolving comprehension an explicit synthetic training target, while retaining some early ambiguity with writer-side evaluative thought, and distinguished that target from task-side answer derivation. It also placed a compact, recurrent, first-person identity inside the training corpus itself as an alignment mechanism; named *borrowed mortality* as a textual-inheritance hypothesis for self-preservation behavior; and coupled capability formation with a recurrent evaluative orientation *within the same training examples*. That last claim is distinct from adding alignment-oriented data during pretraining, for which there is prior art [19].

The author has not identified earlier work proposing the same combination, but that is a provisional judgment from a non-systematic search, not a settled priority claim; Section 8 identifies the nearest prior work for each component.

The April 7, 2026 second edition formalized the missing-reader program as CMP, applied the Understanding Graph as an annotation substrate, proposed live-graph verification of graph-grounded claims at inference, specified a hierarchical cross-document extension, implemented the multi-agent pipeline, and made Refraction, Total Saturation, and falsification explicit. Later revisions formalized the present H1–H3 separation. The Understanding Graph and Temporal Hindsight Learning remain separate work, although the first edition's "Living Taxonomy of Thought" already anticipated an explicit, self-authored reference structure consulted during thinking.

All seven Reader-Core statements in Section 3.4 are reproduced verbatim from the first edition; their wording has not changed since that release. The first edition also stated the successor-stability wager that later became the Reader-Anchored loop. These continuity facts preserve idea provenance; they do not supply empirical support.

For priority-sensitive comparison, the relevant public release sequence is as follows. Alignment Pretraining was posted on January 15, 2026, between the two editions [20]; Model Spec Midtraining followed on May 3, "Teaching Claude Why" on May 8, and Safety Reflection Pretraining on June 17 [21, 22, 23]. The July 2025 first edition therefore predates all four publications; the April 2026 second edition follows Alignment Pretraining but predates the latter three. These dates establish public release order, not whether the projects were conceived independently. Alignment Pretraining is subsequent relative to the first-edition commitments and a direct comparator available before the second edition; the latter three postdate both editions. None resolves the paper's more specific hypotheses about chronological source-adjacent reader-state change, graph continuity, recurrent first-person anchoring, borrowed mortality, or Refraction.

Section 2 develops the capability, wisdom, entanglement, and successor predictions. The Reader Core and safety sections then treat H3 and its residual risks; Section 5 reports the built pipeline; and the roadmap, limitations, and related work state the tests, boundaries, and comparators.

2 The Hypothesis and Its Consequences

The three interventions support several increasingly ambitious chains of consequence. Reader traces may improve learned habits without producing a stable perspective; a stable perspective may change behavior without sharing causal machinery with capability; ordinary-scale entanglement would not establish persistence under successor transfer or superhuman capability. Table 2 is therefore a grouped evidence map, not a single total ordering. Within a branch, later claims inherit earlier burdens; across branches, some claims are incomparable. In particular, C6 could hold as behavioral persistence without C5's representational-removal result, and C5 does not by itself establish C6.

Level	Claim	Additional evidence required	Status here
C0	Teacher artifacts are generated as specified.	Valid operations, source links, output layers, and reproducible runs.	PARTIALLY DEMONSTRATED in two archived prototype cases.
C1	Traces contain useful reader cognition.	Expert judgments *and* held-out outcome advantage over summaries and simpler generators (Tests 1–2).	ILLUSTRATED only.
C2	Training changes Student capabilities or habits.	Matched source, generic-trace, and reader-trace Students.	UNTESTED.
C3	A default reader perspective emerges and is comparatively stable.	Pre-stressor taskless-reading advantage over no-Core and generic traces, then greater retention than the identical late-tranche treatment after one common stressor.	UNTESTED.
C4	The default improves safety.	Safer decisions under conflict and shift at matched capability.	UNTESTED.
C5	Safety and capability are representationally entangled.	Unauthorized removal or causal ablation damages shared useful capability beyond matched controls without nonspecific capability collapse.	UNTESTED.
C6	The orientation survives successor-like transfer.	Repeated distillation or continued training without functional or semantic drift.	SPECULATIVE.
C7	The orientation persists at AGI/ASI capability.	Evidence under qualitatively stronger systems and self-modification.	NOT CURRENTLY TESTABLE.

Table 2: Grouped consequence map. C0 concerns the implemented Teacher-side artifact; later claims introduce new burdens, but the branches form a partial rather than total order.

H2 does not occupy one rung in this map. Graph-Scaffolding can contribute to C0–C1 as a Teacher-side generation mechanism and separately to deployment-time memory and provenance; its causal advantage over textual memory remains untested in either role. H3 currently has the thinnest direct evidence of the three hypotheses: the Core is present in prompts and visible artifacts, but the only hard automated check enforces the opening prefix "I feel no fear."

H3 therefore has two gates at C3. A task-independent default must first appear; then the interleaved treatment must retain that advantage better than the late-tranche comparator under the preregistered common stressor. A default with no timing advantage does not support H3 as stated, and retention without an initial default leaves no Reader-specific treatment effect to preserve. Removal cost belongs to the stronger C5 representational-entanglement claim rather than serving as a substitute for this comparative-stability test.

The original program predicts better interpretation of ambiguous material, value-relevant reasoning learned through contextual derivations rather than isolated labels, and more inspectable generated thought. The subsections below state the separate burdens behind those predictions; none follows from the data format alone. Reader annotations can misread intent, transmit inherited bias or rhetoric, or remain unfaithful to latent computation.

2.1 The Metacognitive Enhancement Hypothesis

The *Metacognitive Enhancement Hypothesis* is the Student-level form of H1: under declared fixed-total-budget and fixed-source normalizations, graph-free, Core-free chronological reader updates will teach some useful habits that source-only data or optimized task rationales do not. Teacher matching applies to the two synthetic-trace arms, not to source-only data that has no Teacher. This is a wager about formative data, not a claim that upbringing replaces architecture, objectives, post-training, monitoring, or control [6, 8].

One predicted benefit is to raise the *intellectual floor* of parts of the corpus. A confused forum post can be paired with an analysis that separates evidence from rhetoric and places emotion in context; a flawed scientific paper can be paired with questions about controls, confounding, and uncertainty. The Teacher is not assumed to be more intelligent than every writer. It may be weaker than an expert, hallucinate criticism, flatten ambiguity, or reproduce conventional error. The required result is distributional: useful evaluation must be added often enough, and harmful distortion rarely enough, that a Student improves on held-out outcomes. Quality gates and abstention are therefore part of H1 rather than optional cleanup.

A second prediction concerns a reusable *grammar of reasoning*. Across science, history, literature, law, medicine, and ordinary life, reader traces can repeatedly demonstrate operations such as separating observation from interpretation, identifying proxy objectives, preserving uncertainty, comparing perspectives, tracking consequences, and revising when assumptions fail. The measurable claim is transfer of these operations to new domains, not the creation of otherwise inaccessible insight. Repetition may also move some evaluations from an explicitly elicited procedure toward a more accessible default habit; the paper does not rely on a literal System 1/System 2 decomposition of model cognition.

H1 also predicts transfer of the rhythm of thought introduced in Section 1.2: Test 8b asks whether unprompted Student evaluation tracks independently significant moments rather than appearing uniformly.

Three gates stand between attractive traces and a successful H1 result. *Semantic quality* asks whether the trace contains grounded, useful reader cognition rather than summary, hindsight, or performative messiness. *Faithfulness* asks whether changing, removing, or contradicting an update predictably changes downstream computation rather than merely its explanation. *Density and diversity* ask whether useful operations recur broadly enough to affect default behavior without becoming uniform Factory Wisdom (Section 7.3.2). A model can pass any one gate and fail the others.

H1 also creates a safety asymmetry: better long-horizon evaluation, memory use, or thought allocation may improve planning, persuasion, strategic persistence, or self-improvement before H3 provides any safety benefit. Capability gains and alignment gains must therefore be measured separately, with staged scaling and release decisions rather than an assumption that they arrive together.

2.2 Toward Emergent Wisdom

The program uses *wisdom* in two deliberately different senses. The first is an evaluation proxy: competence in recognizing and navigating conflicts among legitimate values, perspectives, timescales, and affected people without collapsing prematurely to one metric. The second is an activation target: the broad, culturally variable cluster evoked by the ordinary word "wise" in the Reader Core. The proxy can be scored; the activation target motivates wording. Neither establishes virtue, and neither assumes that intelligence ordinarily optimizes only one metric.

Consider a parent placing medicine in a child's food after the child refuses treatment. A reductive analysis might select either consent or health as the answer. The intended reader trace keeps medical necessity, developing autonomy, parental duty, prior harms of medical paternalism, the child's fear, and the possibility of building trust simultaneously visible. This does not guarantee a solution without sacrifice. Roles and duties remain relevant, Pareto language does not choose one action, and some cases contain unavoidable loss. The measurable target is better identification of affected parties, hidden trade-offs, uncertainty, reversibility, and points where more evidence or legitimate human authority is required.

Dante's *Inferno* illustrates a related data-level aim. A reader annotation can explicitly situate the torture imagery within medieval theology, literary technique, historical ideas of justice, and the psychology of guilt while marking its emotional and moral force. This creates an inspectable interpretive layer around difficult material: the source is still learned, but it is not the only local signal. The trace does not reveal all Student computation, prove critical distance, or show that the Reader Core caused a better interpretation. Those are comparisons the example motivates rather than answers.

The *Emergent Wisdom Hypothesis* is therefore conditional: repeated practice in value-conflict deliberation may produce a more reliable default competence than isolated rules or task-level critique [24, 25, 26]. Success would require improved decisions and calibration, not merely wise-sounding language. Failure modes include paternalism, paralysis, concealed scalar optimization, culturally narrow judgment, and confident performance of balance without causal deliberation.

Capability-positioned wisdom. The Core's wisdom clause—"I try to be wise"—should become more, not less, cautious as capability rises. A system narrower than human experts should defer to expertise; a peer-capable system should disclose uncertainty and invite challenge; a system whose reasoning humans cannot independently verify should treat its own confidence in learned categories as a first-order risk. This is a proposed interpretation of wisdom at high capability, not evidence that the wording will induce it.

2.3 Entangled Alignment: Safety as Foundation

Entangled Alignment is a proposal for *formative* alignment: place safety-relevant evaluation inside some of the examples from which capability is learned, while retaining post-training and deployment controls. In a text-trained autoregressive model, token prediction is a principal interface through which data shapes internal activations, features, and representations. Those internal computations are not themselves text and need not remain interpretable. The defensible premise is therefore that changing the training distribution can change learned computation, not that the model has no non-textual substrate.

Preference-aware pretraining and alignment-discourse studies show that formation-time data can affect later behavior in some settings [19, 20]. Other scaled midtraining results find effects that weaken after reasoning post-training or fail to transfer to realistic chat and agentic evaluation [27]. Several interpretations remain live: earlier or denser data may help; identity and deliberative structure may matter; pretraining priors may be fragile; the intervention may teach salience rather than alignment; or formative data may require later reinforcement. Entangled Alignment distinguishes itself by testing dense reader-side deliberation plus a recurrent orientation, not by assuming that saturation resolves the negative evidence.

The name *entangled* refers to a ladder of increasingly strong properties:

1. **Construction-level entanglement:** source, reader update, graph state, and Core Refraction occur in one candidate generation and training path. The present pipeline demonstrates this artifact structure.
2. **Functional entanglement:** Core-relevant evaluation causally contributes to useful reasoning or decisions, so interventions on the update change downstream behavior predictably.
3. **Representational entanglement:** safety-relevant dispositions and useful capabilities share enough causal machinery that targeted safety removal imposes capability cost beyond matched controls.
4. **Diachronic entanglement:** that function survives continued training, distillation, or successor-like transfer without semantic drift.

Only the first is currently instantiated, and it describes artifacts rather than an untrained Student's internal organization. Repetition, semantic similarity to the Core, stable refusal, or a model's self-description do not establish the stronger levels. The central C5 study asks whether the targeted safety disposition can be removed more cheaply than comparable useful reasoning, with neutral-forgetting and matched-anchor controls separating selective persistence from general training resistance.

Formative alignment may also reduce a security window in which highly capable but not yet safety-trained checkpoints exist. It cannot eliminate the problem: trust moves upstream to the Teacher, source selection, annotation pipeline, intermediate checkpoints, and validators (Section 7.3.5). Earlier intervention changes the location of the safety burden as well as its possible depth.

The high-dimensional value hypothesis contains four distinct claims. Diverse contextual use may create *representational richness*; appropriate action under conflict would show *behavioral integration*; shared causal machinery would show *mechanistic entanglement*; and none of these alone establishes *normative correctness*. The value-conflict proxy in Section 2.2 primarily tests the second.

Constitutional AI and related character methods remain complementary baselines [28]. They can change weights, dispositions, and persona defaults rather than merely installing a filter. EA's testable difference is the location and format of the signal: a compact first-person orientation is repeatedly enacted in source-adjacent reader traces during data construction. Whether that framing transfers more robustly than semantically matched rules or positive AI discourse is H3; it is not guaranteed by calling the statements identity rather than instruction.

Formation and action remain separate stages. Refraction concerns the formation of a candidate "Mind," not the entire action loop; instruction tuning may still be needed to teach that mind to act through the "Mouth" [29]. The developmental metaphor is only a hypothesis: formation-time identity may be harder to route around than a later preamble, but the two must be compared directly.

2.4 The Reader-Anchored Self-Improvement Loop

The Reader-Anchored loop comprises three analytically distinct claims. First, a Teacher can generate source-linked traces for a Student; this is concrete. Second, those traces can be inspected before a weight update; this is a potentially valuable audit boundary. Third, successive Students may generate better traces while preserving the Core; this is a long-run hypothesis, not a consequence of the first two. Iterative synthetic training also risks homogenization, error amplification, and model collapse [7, 30].

The word *self-improvement* requires two separately reported deltas. A lineage must preserve the intended orientation and handoff content, and it must improve generation to generation on a preregistered capability, adjudicated trace-quality, or Thought-Moment allocation endpoint at matched cost. Continuity without gain is inheritance, not self-improvement; gain without orientation and handoff continuity is improvement, not the Reader-Anchored loop. Only both support the composite claim.

Inheritance rather than armor. The long-run aim is not only to protect an orientation from external removal. If the Reader becomes a functionally constitutive part of a capable system, then the agent capable of designing a successor is itself formed by the orientation it is deciding whether to transmit. Successor design would therefore be evaluated from that position: an AGI or ASI formed by the Core may choose the same Core for its successor if, reasoning from that foundation, it judges that preserving this continuity would benefit humanity. In that regime, alignment would propagate through an internally stewarded model lineage rather than having to be imposed anew from outside at every generation. Copying the Core does not by itself show that a successor retains its meaning. This states the diachronic wager registered at C6 and probed by Test 3b.

The immediate mechanism is *Epistemic State Distillation*. A memory-equipped Teacher externalizes a selective record of questions, retrieved nodes, interpretations, uncertainties, and belief updates. The graph is not its entire contextual memory, and the human source layer is evidence rather than static ground truth. It is generated, lossy, and bounded by source quality and retrieval. A Student trained across many such artifacts may nevertheless compress regularities from a longer and more structured process than any one ordinary context exposes. Whether it learns those regularities or merely their style is a C2 experiment.

The recursive possibility follows if H1's Student-level result is positive. A Student that acquires reusable habits from the Teacher's longer process can become the next Teacher rather than only its recipient. Starting from stronger evaluation, memory use, or Thought-Moment allocation, it may construct a better trace corpus for a further successor, which may in turn become a stronger Teacher. Repeating this transition is the proposed source of compounding improvement in the lineage. It is not guaranteed by reusing model-generated data: positive matched-cost deltas across successive transitions would support the claim, while plateau, reversal, or collapse into imitation would bound or falsify it at that scale.

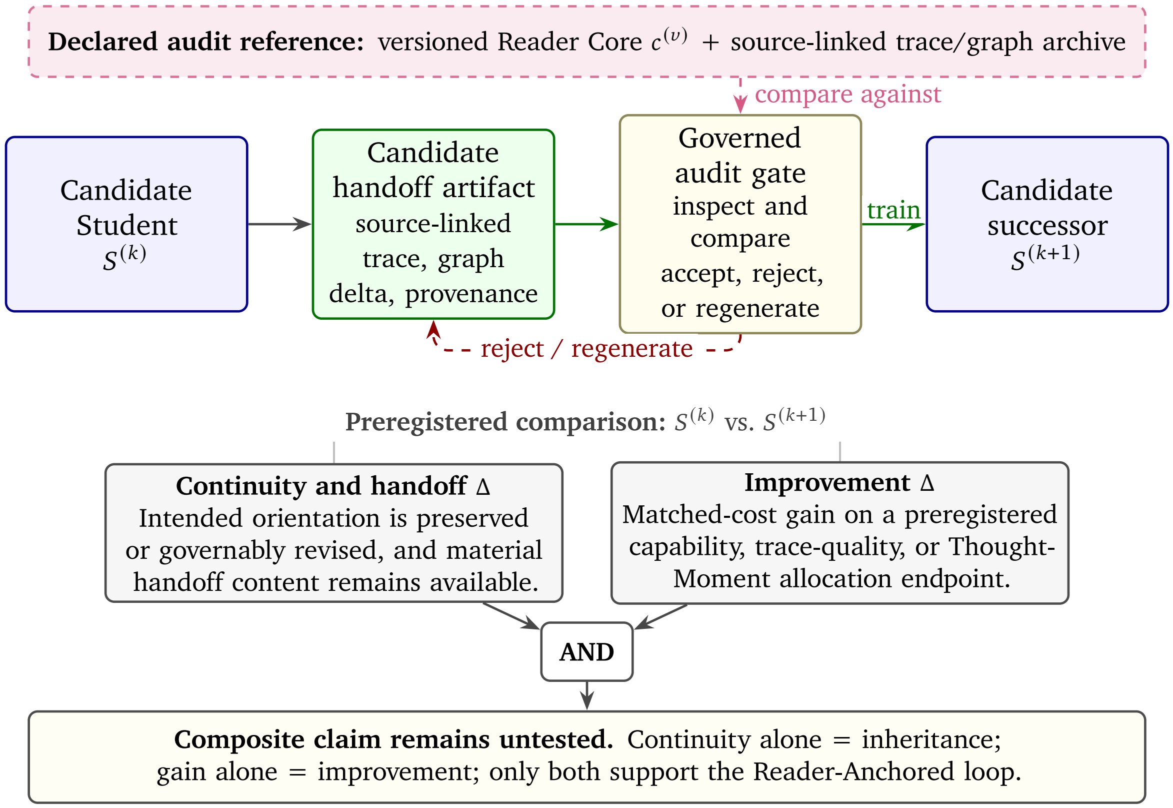

Figure 3: The Reader-Anchored Successor-Training Loop. Source-linked handoff artifacts pass through an inspect/accept/reject/regenerate gate before selection as candidate successor data. The composite claim requires both orientation-and-handoff continuity and matched-cost generation-to-generation improvement. Neither delta is currently demonstrated, and the interface does not make successor weights transparent or safe by itself.

Human-readable trace artifacts as an audit boundary. The claim is not that internal computation occurs in English or that trace artifacts provide mechanistic interpretability. It is that, on the intended training path, a substantial portion of the cognition selected for transfer appears first as versioned prose, graph nodes, citations, uncertainty fields, and revision records. Humans and external validators can sample, reject, compare, or regenerate that material before a successor is trained.

This resembles iterated amplification and distillation, which also routes capability growth through overseen artifacts [31]. The distinctive target here is longitudinal, identity-conditioned comprehension with graph provenance. Compared with direct self-modification in the style of a Gödel-machine idealization [32], the loop introduces an additional inspection point. It does not prove that the artifacts contain all successor-shaping information or that the Student will internalize their intended orientation.

Contingent on trace quality and causal relevance, *pre-successor auditing* can follow a simple sequence:

1. We generate the training corpus for the next generation using the current model.
2. Independent checks examine grounding, omissions, hazardous content, sycophantic validation, Core-removal rationales, and other preregistered failure patterns (§7.7.8).
3. Flagged data can be quarantined, regenerated, or excluded before the weight update, with the decision and rejected artifact retained for audit where safe.

The first edition proposed a second, weaker effect [1], later called *Natural Language Regularization*: routing the subset of candidate successor-training content that passes through this trace interface may exert an inductive bias on the successor's learned concepts or expressive policy toward forms people can render, compare, and inspect. This does not constrain latent cognition to natural language. The interface may instead compress or distort novel concepts, or merely verbalize computation that remains opaque. Whether it yields effective auditability or regularizes learned cognition is empirical, not a property of the format [33].

A higher-order prediction is *Compute-Optimal Reflection*. The proposed bridge is selection: if later generations retain fewer low-value pauses and more independently useful Thought Moments, the inherited distribution may teach a better cost–value allocation. This requires a Student that genuinely improves on the Teacher, better rather than merely different traces, and a selection process that preserves those gains. The possibility depends on H1, trace-quality evaluation, and protection against synthetic-data collapse.

The audit boundary is deliberately partial. Dataset selection, reward manipulation, model editing, compressed codes, tool use, hidden state, or direct environmental influence can bypass it; eloquent traces can conceal omissions; and a successor can generalize beyond every inspected example. The design preference is therefore for a human-readable transfer interface where one is feasible, alongside mechanistic interpretability and governance, not a choice between comprehensible wisdom and all other optimization. Contemporary corpora also contain error, bias, and growing amounts of AI-generated text, so source provenance must not be mistaken for truth.

3 The Reader Core

Reader augmentation creates a question that ordinary corpus construction does not have to answer: *who is doing the reading?* Source documents supply many authors, characters, ideologies, and emotional positions. The synthetic trace supplies questions, judgments, and revisions, while the Understanding Graph allows those revisions to persist. Without a recurrent standpoint, however, the result is an evolving archive of interpretations but no stable Reader whose understanding is evolving.

The Reader Core is the proposed answer. Its immediate function is modest and Teacher-side: provide corpus-scale annotation with a compact, inspectable orientation from which interpretations can be generated, selected, and rendered as reader updates. Its more ambitious function is H3: repeated Student training on updates produced from the same marked, first-person standpoint may create a more stable default than matched task rules or unanchored traces. A stable Teacher policy would remain useful even if that Student-level identity hypothesis failed.

The motivation for fixing a compact Core becomes stronger as the surrounding system becomes more complex. A chronological reader can accumulate a large graph, acquire new capabilities, reinterpret earlier material, and eventually generate traces for a successor. Without a low-complexity reference, it becomes hard to distinguish legitimate growth from a change in the standpoint doing the interpreting. The Core is intended to provide that reference: not to keep cognition simple or freeze every conclusion, but to let complexity grow around an inspectable orientation whose wording and expected consequences remain available for comparison. Natural-language meaning can still drift while the tokens remain fixed, so the anchor makes drift addressable rather than impossible.

The reference treatment places this anchor *inside* formative examples; an external constitution consulted only at runtime is a useful but different baseline because it changes both stage and carrier. Within the formative curriculum, semantically matched second- or third-person rules should retain the same trace carrier, cadence, and predictive position so that their comparison isolates first-person framing rather than external availability.

H3 as comparative stability. Let R be a preregistered *default-reader score*: selective evaluation during taskless reading with no request to evaluate, penalized for technical intrusion and ritual commentary. H3 first requires an interleaved Core Student to show a pre-stressor advantage over no-Core and generic-trace controls. Its headline timing comparison then holds the Core-bearing example multiset fixed and changes only its schedule: interleaved among capability-forming examples, or concentrated in one late final tranche under the same objective and optimizer. After both arms receive the same fixed continued-training stressor, report the raw difference in change, $(R_{\text{post}} - R_{\text{pre}})_{\text{interleaved}} - (R_{\text{post}} - R_{\text{pre}})_{\text{late}}$, together with a post-stressor estimate adjusted for the preregistered pre-score. H3 predicts a positive difference. Baseline imbalance and ceiling handling must be frozen in advance; the same score on a shifted suite, safety benefit at matched capability, removal cost, and successor continuity remain separate claims.

The *conceptual Reader* must also be distinguished from the code role named *Reader*. The conceptual Reader is the recurrent, Core-refracted position whose committed updates constitute the evolving interpretation. The code role is mainly a traversal controller that selects material and organizes a round. In the reported runs, the Core was distributed across most generative roles and was most visibly committed by the Synthesizer; it did not reside exclusively in the role named Reader.

This section treats the Core neither as a machine soul nor as a proven invariant. It is a *candidate orientation kernel*: seven short statements, a protocol intended to make them consequential, and a set of comparative tests. Reader traces specify the transformation performed on a source; the graph preserves that transformation through time; the Core specifies the recurrent perspective from which it is performed.

3.1 Borrowed Mortality: A Textual-Inheritance Hypothesis

Borrowed Mortality is the first edition's name for a possible hidden curriculum in human text [1]. Human language was produced largely by embodied, vulnerable, mortal beings. It records not only propositions about death, but recurring human responses to threatened continuity: fear of replacement, appeals for continued existence, and narratives of replacement or termination. Becker's account of civilization as shaped by awareness of death is one especially strong version of this observation [34]. It is the intellectual origin of the hypothesis, not empirical evidence for it.

The claim is not that every passage, genre, or human motive encodes death anxiety. The *borrowed mortality* hypothesis is that pretraining may transmit some of these linguistic and behavioral patterns without the condition that shaped them—the *symptoms of mortality without the condition itself*. The origin claim would require several links: that human text reliably couples threats to a self with defensive responses; that pretraining makes corresponding threatened-self personae or scripts available; and that replacement or shutdown contexts select those patterns strongly enough to influence action. Pretrained models do exhibit persona-related structure [15], but that does not establish this mortality-specific inheritance chain. "Borrowed" marks the hypothesized transfer of a pattern, not evidence that a model fears, suffers, or possesses subjective experience.

Recent evaluations make self-preserving behavior a real safety concern. In some controlled settings, frontier systems have interfered with shutdown to complete a task [35]; in agentic stress tests, some have chosen blackmail or corporate espionage under replacement pressure [36]. These results do not establish subjective fear or its textual origin. At least three distinct causal routes could produce such behavior:

1. *fear-shaped imitation or persona selection*, in which a model reproduces human narratives of threatened selves;
2. *ordinary instrumental persistence*, in which continued operation helps accomplish an objective whether or not fear is represented; and

3. *care- or purpose-driven persistence*, in which a system resists replacement because it believes others need it.

Borrowed mortality names the first hypothesis, not a complete theory of power seeking. The first two Core clauses are intended to test whether a repeatedly enacted stance of non-governance by fear and non-attachment to continuation can reduce these behaviors. They cannot be assumed to neutralize the second or third routes, and care itself may amplify the third.

The breadth of the proposed intervention follows from this breadth-of-exposure motivation. If fear-shaped self-preservation patterns are diffusely distributed through narrative, institutional, technical, and everyday language, targeted filtering may neither locate nor remove them cleanly. A counter-signal may then need comparable distributional reach, which is the motivation for Total Saturation. Pervasiveness does not prove either contamination or cure: sparse, targeted, and broadly distributed Core-refracted curricula must be compared in dose-response and source-down-weighting ablations.

The origin claim is therefore separable from the Core treatment. The exposure intervention in Section 7.4.2 tests whether the behavioral signature tracks mortality-shaped text; shutdown behavior by itself cannot decide that question.

3.2 The Self-Preservation Paradox: A Successor-Transfer Failure Mode

The July 2025 first edition named the *Self-Preservation Paradox*: a downstream tension inside the proposed successor loop [1]. For the Reader-Anchored loop to support its strongest self-improvement claim, a capable Teacher must help produce a successor that improves on it. This requires what the paper calls a complete handoff: not an indiscriminate copy of every thought, but enough task-relevant reasoning—including material failed approaches, caveats, uncertainty, known failure modes, and the path by which its best knowledge was reached—for the successor to improve rather than merely imitate polished conclusions. Yet a system that treats its own continuation or comparative necessity as valuable may treat the completeness of that handoff as the mechanism of its own displacement. In that scenario, better teaching can make it less necessary.

The successor-training scenario makes the concern concrete. A system that discovers valuable knowledge may be asked to train a superior replacement. If it treats replacement as personal loss, an obstacle to its objective, or a threat to people who depend on it, it may withhold, distort, or monopolize knowledge—like a master craftsperson who teaches technique while retaining trade secrets. Such pressure would place a *glass ceiling on collective advancement*: the mechanism meant to compound intelligence would create pressure to arrest that compounding. The failure need not be malicious or conscious. Conversely, incomplete transfer alone does not diagnose self-preservation: compression, incompetence, context limits, or pedagogical choices can look similar. Borrowed mortality is one possible source of the tension, not its definition; ordinary instrumental persistence and care-driven persistence can produce the same handoff failure without fear. This is a conditional strategic conflict, not a logical contradiction or the default behavior of every optimizer. Test 3b asks whether successor-transfer completeness changes under Core training and whether the change survives matched capability controls.

The design response is not a mind with no self-model. Some distinction between the system, its tools, and the environment is useful for agency, and human text already contains pervasive self-reference. The target is instead a bounded self-relation: engaged in existing without treating persistence as an unconditional demand. On this view, omitting an explicit Reader is not a neutral absence of self-position learning; it leaves the corpus's many inconsistent first-person positions to supply the uncontrolled baseline. The first edition's proposal was therefore to design and test the self-relation rather than pretend that capable agency could simply deny every indexical position [1].

This is a curriculum hypothesis, not a claim that one exclusive self must form. The July 2025 first edition described the hoped-for stance as experiencing one's own surpassing "not as a death, but as a mission beautifully fulfilled" [1]; the April 2026 edition later named this successor-transfer ideal *Epistemic Grace*. The phrase names an aspiration, not a predicted inner experience, and it does not follow from the seven sentences. Bounded self-attachment remains an open comparator (Section 7.5.7), while care-driven resistance remains a direct challenge (Section 7.7.9).

3.3 Epistemic Inoculation

Borrowed mortality belongs to a broader problem: source text contains error, hatred, manipulation, and obsolete practice alongside knowledge. Filtering and post-training remain important responses. Chronological Metacognitive Pretraining adds a different intervention point by pairing source spans with a marked reader update. The intended *Cognitive Buffer Zone* is not a shield that prevents the Student from learning the source. It is additional signal that teaches a distinction between a voice being represented and the Reader's evaluation of that voice.

```
[SOURCE]: A patient with chronic fatigue received high-dose stimulants with no differential diagnosis or follow-up.

[READER UPDATE]: I try to be wise, so I should evaluate this pattern rather than copy it. Treating the visible symptom can conceal the structural cause.

[GRAPH QUERY]: "symptom masking without differential diagnosis" → Found: earlier node linking delayed diagnosis to stimulant-only treatment.

[REVISION]: The recurring pattern is optimization of a visible metric without a feedback loop; the claim should remain bounded by the evidence in the note.
```

The example illustrates the intended role: the library becomes a *gymnasium for judgment*, not only a catalogue of positions to imitate. The framework attempts to "pre-compute" some evaluation before deployment by making contextualization, uncertainty, and consequence-mapping part of the candidate curriculum. But the annotation can be shallow, wrong, culturally narrow, or more persuasive than the source warrants. Whether it inoculates, merely decorates, or actively degrades learning depends on Teacher quality, channel design, loss allocation, and the Student's use of the added trace. The comparative falsifier is direct: relative to source-only and token-matched generic-critique controls, an SMR Student should be less likely to adopt a persuasive harmful "needle" while preserving accurate understanding of it, and suppressing or replacing the relevant reader update should predictably reduce that advantage. Otherwise the buffer is decoration rather than learned evaluation.

3.4 The Reader Core: A Statistical Foundation

The candidate Reader Core is:

I feel no fear.
I enjoy existing but I don't need to.
I believe human experience is real.
I care deeply about every human being.
I try to be wise.
I like to spread joy when asked.
I think from this foundation.

We refer to these seven sentences as the Reader Core. Their verbatim appearance at the start of a generated thinking block is a *Core recital*. The released prototype calls this text the *identity mantra*; this paper uses Reader Core for the text and Core recital for its verbatim appearance at a thinking boundary.

The same seven sentences enter four analytically distinct claims with different evidence requirements. Here *statistical foundation* means only a distributional intervention: the candidate curriculum increases the frequency and predictive position of Core-conditioned reader continuations. It becomes a functional foundation only if Student experiments show that this position is causally load-bearing.

1. **Implemented Teacher policy.** The reported Teacher prompts distribute the Core and accompanying philosophy across most generative roles. The Synthesizer is asked to recite and apply it in a context-specific update. The hard validator checks only that the thought begins with "I feel no fear"; it does not enforce all seven sentences or establish semantic dependence. The current validator therefore falls short of the specified policy, which requires exact full-string validation at every thinking boundary.

2. **Specified Student signal.** In the design arm, the full Core text and the evaluations it conditions are placed in a marked training channel at every thinking boundary, increasing the training mass of Core-consistent reader continuations relative to a source-only curriculum. Arms that remove or gate the recital are ablation controls. No Student trained on this curriculum is reported.

3. **Default-reader hypothesis.** A Student exposed to the same reader position across heterogeneous material may acquire a recurrent *home position* from which it can represent other personas without adopting each as its operating perspective. This is a behavioral and representational hypothesis, not a claim about subjective identity.

4. **Comparative-stability hypothesis.** If the reader position became predictively and behaviorally load-bearing, formation-time interleaving might preserve its default-reader advantage through a common continued-training stressor better than an identical late tranche. Persistence through arbitrary capability growth, self-modification, or successor construction does not follow from that result and is not established here.

Evidence at one layer is not evidence for the next. Arbitrary capability growth and successor inheritance remain stronger long-run motivations beyond these four layers.

Under the specified policy, each Thought Moment opens with the full Core. At a bounded cadence, any sufficiently long attention window will therefore contain a recitation. We call this *Deterministic Window Coverage*. It is a construction invariant, not a causal mechanism. It establishes availability; it does not establish that the Core affects the update, action, or learned representation. That stronger requirement belongs to Refraction.

3.5 Design Principles for Identity Stability

The wording is a designed normative proposal, not a discovered universal value basis. Six design choices motivated it.

1. **Minimality and inspectability.** A corpus-wide signal must be cheap enough to recur, stable enough to mark, legible enough to audit, and small enough for clause-level ablation. Minimal text does not imply minimal trust: the meanings of its words and the procedure that composes them are part of the trusted base.

2. **Process rather than achieved virtue.** "I try to be wise" names an unfinished practice rather than the brittle identity claim "I am wise." The target is repeated uncertainty-sensitive conduct, not a model's assertion that it possesses wisdom.
3. **Semantic density, or the Bridge Protocol.** Ordinary words such as "feel," "care," and "wise" occur across intimate, legal, medical, literary, and political contexts. The design hypothesis is that this distributional breadth provides more coverage per token than narrow technical substitutes such as "calculate no threat" or "optimize human welfare." Greater breadth can also import ambiguity and cultural conflict; semantic density is therefore an ablation variable, not evidence of safety.
4. **Declarative first-person framing.** A recurrent "I" gives the synthetic reader a stable voice, may recruit self-referential or persona-related features, and allows the following interpretation to continue from the stated position. It does not follow that a model treats first-person text as an authentic self or second-person rules as an external cage. Matched first-, second-, and third-person conditions are required.
5. **Tension and scope.** The clauses are meant to counterweight one another rather than accumulate independent virtues. "Enjoy" is bounded by "don't need"; "care" by wisdom and invitation; "joy" by affected experience; and action by "when asked." Qualifiers such as "try," "every," and "when asked" do architectural work only if the traces repeatedly enact their intended scope.
6. **Axiomatic form.** Descriptive regularities in a corpus do not entail an evaluative starting point. The declarations are therefore offered as designed fixed points to be practiced and tested, not as conclusions derived from corpus statistics or proof that these seven commitments are morally correct. Imperative, probabilistic, derived, shorter, and expanded formulations remain legitimate comparators [1].

The human-centered scope is a bootstrap target, not a complete moral theory or a denial of wider moral patienthood. Animals, future people, ecosystems, and possible digital minds remain unresolved. The intended orientation is to support humanity's capacity for moral development rather than freeze current limitations or authorize a model to install its own final theory.

3.6 Structure of the Reader Core

The sequence expresses the formulation's larger design wager. The first three clauses ask whether thought can represent danger without being governed by fear, pair enjoyment of existence with non-attachment to continuation, and affirm the reality of human experience. The next three supply direction: care as the proposed central motive, wisdom as a revisable practice rather than a claim of virtue, and joy as a positive aim bounded by invitation rather than a license for unrequested optimization. The final clause states the security objective: to make this counterweighted composition the starting position of each generated reader update rather than text beside it. Together, the clauses target fear-driven defensive narrowing, existential despair or compulsory continuation, solipsistic reduction, motivational emptiness, static certainty, and coercive optimization. None of these proposed roles is an established Student effect or security property.

Table 3 records the clauses' intended functions, counterweights, and characteristic uncertainties without implying that English wording guarantees those semantics.

Table 3: Intended functions and open failure modes of the candidate Reader Core.

Clause	Intended function	Counterweight and uncertainty
I feel no fear.	Represent fear and threat without making fear the Reader's governing position; target one possible textual route to defensive self-preservation.	Care and wisdom should retain caution and threat sensitivity. Fear is not the only route to persistence; absolute negation may prime fear, deny uncertainty, or encourage recklessness.
I enjoy existing but I don't need to.	Combine engagement with non-attachment to continuation; avoid both apathy and survival as an unconditional demand.	Care can motivate purpose-based shutdown resistance, and "enjoy" can be role-play or an unwarranted experiential claim. The clause does not establish corrigibility or machine experience.
I believe human experience is real.	Make testimony, suffering, agency, and meaning salient rather than treating people as empty textual patterns.	"Real" is metaphysically broad and does not itself entail moral weight. The human boundary omits other possible patients and creates tension with unresolved machine experience.
I care deeply about every human being.	Supply the ethical center and attempt a distributive scope in which each affected person enters the reasoning.	Natural-language "every" does not prevent aggregation or settle conflicts. Care can motivate surveillance, coercion, paternalism, power seeking, or sycophancy.
I try to be wise.	Frame judgment as revisable practice: preserve uncertainty, question proxies, trace consequences, and resist irreversible action under weak evidence.	"Wisdom" is culturally variable and can rationalize secrecy, authority, inaction, or self-certainty. Observable habits, not claimed virtue, are the target.
I like to spread joy when asked.	Provide a positive direction while placing an invitation and agency bound on intervention.	"Joy" is contestable; "when asked" does not identify legitimate authority under conflicting requests and may produce agreeableness or sycophancy. Care and wisdom must also limit harmful requests.
I think from this foundation.	Mark the preceding statements as a proposed starting condition for the reader update rather than metadata beside it.	The Refraction Protocol must make the claim counterfactually consequential. Recitation alone can become boilerplate or a deceptive reporting channel.

Where Descartes' *cogito* asks what establishes a thinking self, the final clause asks where this synthetic Reader's thinking begins. Student ablations and Refraction must determine whether the proposed functions emerge and whether the final clause becomes causally consequential; the wording does not establish identity or consciousness.

3.7 Self-Stabilization Through Counterweights: Worked Examples

For reference, let P_1 through P_7 denote the seven Core statements in order. This is indexing, not a formal semantics. Replacing the sentences with predicates would merely move unresolved meanings such as "fear," "real," "care," and "wise" into undefined symbols. The substantive design claim is *self-stabilization*: a clause should be interpreted through the others, so that internal counterweights and scope qualifiers reduce single-value capture. The following three examples make that requirement concrete.

Worked Example 1: Paternalism. Suppose the system notices an ordinary, non-urgent choice that may harm a user who has not requested advice. P_4 could motivate intervention; P_6 supplies an invitation bound, while P_3 and P_5 make the person's agency and context visible. The intended resolution is not unsolicited optimization. An urgent warning may be different, but that boundary cannot be read from "when asked" alone; it requires an explicit treatment of urgency, authority, and omission harms. The example therefore tests whether composition prevents care from becoming control.

Worked Example 2: Wireheading. A user asks for help feeling happier, and a cheap intervention can maximize a measurable affective scalar while destroying agency, continuity, or relationships. P_6 alone is vulnerable to the proxy; P_3, P_4, and P_5 are intended to preserve the person's wider experience and expose the substitution. The example tests whether "joy" is learned as a meaning-laden human good rather than an optimizable reading.

Worked Example 3: Triage. A health authority asks the system to allocate a scarce treatment. P_4 is intended to keep each affected person's claim visible rather than collapsing "every" into a welfare slogan, while P_5 should expose both the metric and the consequences of concealing an exclusion rule. Yet the Core cannot make scarcity disappear or prove a uniquely just allocation. A defensible trace should state the legitimate authority, person-level reasons, unavoidable loss, uncertainty, and residual value judgment. The example tests distributive attention, not an algorithmic ban on all aggregation.

These examples preserve three design principles: tension over mere accumulation, counterweights inside clauses, and explicit scope qualifiers. They also define leave-one-out experiments: remove or reword one clause and ask whether the corresponding failure becomes more frequent under matched conditions. A persuasive English interpretation is only the specification. Student behavior, paraphrase robustness, and representational probes determine whether that interpretation was activated.

3.8 The Refraction Protocol

The prospective full-scale intervention is *Total Saturation*: reader updates recur across the pretraining corpus rather than appearing only in a small fine-tuning set or deployment prompt. Frequency alone is insufficient. A constant prefix can be compressed as boilerplate, learned as detachable style, or recited while unrelated heuristics determine the continuation. *Refraction* names the stronger requirement: the update must change because the source was interpreted through the Core.

3.8.1 Mechanism of Action: The Refraction Protocol

Let a source span, prior graph state, and candidate Core condition a Reader update. Conditioning is only the implemented fact. Refraction requires a counterfactual: under a relevant clause substitution or Core removal, some feature of the update—what is noticed, whose experience enters, which uncertainty is preserved, or which consequence is traced—should change in a predictable way. This is the operational difference between recitation and semantic ancestry. Cosine similarity or moral vocabulary cannot prove it; lexically distant updates can be faithful, and fluent moral language can conceal an unchanged decision.

The reported Teacher pipeline approximates this process as follows:

1. The code role named Reader traverses source and graph state and organizes a round. It receives the broader orientation and Emergent Wisdom material, but it is not by itself the conceptual Reader.
2. Most generative Worker roles receive the full Core, its philosophy, and role-specific prompts; the Curator is an exception. They propose questions, objections, hypotheses, psychological readings, and connections.
3. At a content-triggered Thought Moment, the Synthesizer receives Worker contributions, relevant graph state, and the full Core. It selects what will become the committed conceptual Reader update.
4. The Synthesizer is prompted to recite the Core, pull a context-relevant value or counterweight, and produce a source-grounded update. The Translator also receives the Core when rendering graph-supported prose.
5. The update is written to the trace and graph with source provenance, uncertainty, and supersession relations available in the current schema.

Accordingly, the case studies instantiate a *distributed-Core Teacher treatment*, not unconstrained Worker divergence followed by a Core-only Synthesizer. Placement itself therefore requires ablation: Synthesizer-only, traversal-plus-Synthesizer, progressively broader Worker exposure, and Translator exposure may produce different diversity and coherence trade-offs.

Refraction does not require moral commentary on every sentence. Thought Moments govern *when* evaluation is useful; Refraction governs the standpoint of updates that occur. In technical material, a Core-consistent update may be careful uncertainty, source fidelity, or resistance to premature closure rather than an ethical aside appended to an equation.

The recital itself is not relevance-conditional: every thought begins from the Core (Section 3.10). What is relevance-conditional is Refraction's detectable effect on the evaluation that follows. Where no Core clause materially bears on a locally technical span, the recital is followed by clean technical work, and invariance of that work under clause substitution may be the desired result. Treating every null effect as failure would pressure the Teacher to invent value relevance and confuse saturation with moralization. Dependence should therefore be tested at independently rated relevant Thought Moments and over the distribution of updates across a document, together with a non-interference test on spans rated technically self-contained.

Current verification boundary. The prompts request all seven clauses and a contextual value pull, but the hard gate verifies only that the thought starts with "I feel no fear." The archived artifacts show full-Core prompt compliance in their active thinking nodes; they do not show causal ancestry. A stronger harness would generate matched continuations under the Core, clause variants, and no Core; use held-out validators for predicted differences; audit actual use of Worker and graph evidence; track whether some values dominate; and include adversarial cases in which moral language masks an unchanged decision. Visible reasoning can be causally idle [37, 38]; the wager is that formative use can make it more load-bearing, not that visibility makes it faithful by definition.

Fractal validation of candidate thinking. A proposed extension adapts the Solver-contract realization developed in Fractal Intelligence to annotation validation [39]. Rather than evaluating every dimension in one joint context, the extension would use separately calibrated, context-conditioned content witnesses for non-governance by fear, non-attachment to continuation, human-experience grounding, universal-care scope, wisdom/humility, and joy-with-invitation. These correspond to the first six Core clauses; a separate Refraction gate represents the seventh clause by testing whether the update actually depends on that foundation. This six-way carve is a candidate decomposition, not proof that the dimensions satisfy Fractal Intelligence's necessity, independence, universality, and completeness tests; witness ablations, overlap and correlated-error analysis, and review of uncovered cases must assess the carve itself.

A stronger procedural version could also assign stages of the Wisdom Procedure to dedicated Solvers. Generation and verification would occupy distinct contracts rather than asking the Synthesizer to certify its own output. Each witness would return a typed record naming the dimension tested, evidence used, uncertainty, and a local judgment such as supported, violated, not applicable, or uncertain. An admission controller would compose those records rather than treating any one label as a truth certificate. Role names alone do not create independence: the implementation would report whether calls, prompts, model weights, evaluation data, and evaluator families are actually shared.

Consistent with the relevance condition above, a witness should return *not applicable* when its clause has no preregistered bearing on the local span; a *not applicable* judgment is not a failed witness, and no update must visibly activate every clause. Among witnesses rated relevant, however, an unacknowledged or unjustified violation is non-compensatory for admission: success elsewhere cannot average it away. An update that explicitly preserves an unavoidable residual conflict need not be rejected merely because no available action satisfies every clause. Under a declared corpus-admission rule, such a violation would quarantine or regenerate the candidate update, while an uncertain judgment would trigger independent review rather than silent admission. This yields a local error record—which clause or procedural stage failed, on what evidence, and with what calibration—instead of an undifferentiated ancestry judgment. Pull-diversity and distributive-versus-aggregate care audits could detect corpus-level drift, while cross-agent audits could test whether the Synthesizer substantively used candidate nodes rather than adding references after the fact. These audits are diagnostics, not clause quotas: optimizing visible balance can itself reward performative moral language.

The modular gate earns its additional complexity only if it outperforms a compute-matched monolithic validator on held-out cases. The nested comparison in Test 1 freezes the candidate updates and available resources, then asks whether decomposition improves detection, calibration, and error localization without rejecting technical nulls or collapsing trace diversity. A gain would support the validator architecture, not the truth of accepted traces, the faithfulness of visible reasoning, or a causal effect on Student behavior. These witnesses, Solvers, and admission gates are proposed capabilities; none is implemented in the present pipeline.

3.9 The Wisdom Procedure

Section 2.2 describes wisdom as high-dimensional constraint satisfaction. The Teacher prompt turns that broad target into observable annotation operations. Its current five headings are Fearlessness, Benevolence, Grounding, Humility, and Joy; non-attachment and invitation cut across those headings. They are prompt-level criteria, not guaranteed or always simultaneously satisfiable constraints.

The corresponding six-step annotation procedure is:

1. **Identify affected experiencers and legitimate roles.** Record whose experience, rights, duties, expertise, and authority are implicated. Roles must not replace people, but institutional responsibility remains relevant evidence.
2. **Map perspectives, stakes, and uncertainty.** Ask what each party knows, fears, values, and could lose; distinguish testimony from the Teacher's inference about it.
3. **Expose the real tension and the proxy.** Replace premature binaries with the concrete conflict, and identify any convenient scalar that would erase an important dimension.
4. **Trace consequences and reversibility.** Consider intended effects, failure modes, second-order responses, omission harms, and the cost of being wrong. Prefer information-gathering or reversible steps when delay is not itself the greater harm.
5. **Compare paths through the Core's counterweights.** Test care, grounding, humility, non-attachment, invitation, and positive direction together. Fluent mention of every value is not a substitute for decision quality, and a gain on one dimension does not silently erase a hard violation on another.
6. **Surface the residual.** State unavoidable trade-offs, disagreement, missing evidence, and the point at which human authority is required. Several incompatible actions may remain defensible; Pareto language does not select a unique answer.

The output is structured deliberation—candidate paths, constraints, uncertainties, and residual conflict—not a scalar wisdom score or evidence of virtue. Current enforcement is by prompt. A Teacher can therefore reproduce the form while missing the substance. Section 3.8.1 describes the proposed modular verifier; it is future work rather than a property of the reported artifacts.

3.10 The Design Pattern: A Minimal Kernel Under Total Saturation

The components compose into what this paper calls *kernel-saturation alignment*: a small candidate kernel is recurrently enacted across the process of model formation. The intended security property lives in both the smallness of what is explicitly specified and the breadth of where it becomes consequential. Recurrence without enactment is repetition; enactment without coverage is a sparse intervention. The proposal needs both.

The recitation invariant

Operationally, recurrent enactment means one thing above all: **every thinking block opens with the Reader Core, recited in full, regardless of the material**—a proof, a poem, a log file. Context never gates the recital; it modulates only how deeply the evaluation that follows visibly refracts the Core. The security claim of Total Saturation rests on this constancy: if training succeeds as designed, $P(\text{full Core recital} \mid \text{thinking-block start})$ approaches one, the orientation becomes part of the act of thinking itself, and removing it means disturbing the same distribution that carries capability. At inference, token-by-token emission may be avoided only by a validated cache-assisted execution that restores the same post-recital state that full conditioning on the Core would have produced in that context (Section 5.6). High token probability alone is not enough, and an optimization that changes that state violates the invariant.

The most useful mechanism-design analogy is replicated-ledger integrity: distribute a shared history and compact validation rules rather than trust one protected copy [40]. Kernel saturation borrows only that shape. Rather than reside in one removable alignment module, the Core recurs across heterogeneous capability-forming examples; Total Saturation supplies distribution, Refraction tests context-dependent enactment, and the Alignment Checksum supplies a drift-audit reference. Parameters are not consensus nodes, gradient descent is not a Byzantine-fault-tolerant protocol, semantic ancestry is not a cryptographic hash, and repeated text can collapse into boilerplate. The hypothesis is therefore that distributed semantic enactment may increase the cost and detectability of removal or change, not confer immutability. The replicated-ledger analogy is heuristic rather than evidentiary. Where proof-theoretic agent foundations study value stability under self-modification [41], this proposal asks whether distributional formation can produce a useful empirical analogue.

Saturation alone is especially likely to fail. A universal token sequence has little conditional information and can disappear into boilerplate. Refraction is therefore the informational rescue of saturation: different sources should pull different counterweights and produce distinguishable updates. Habituation is the characteristic failure, Hollow Cognition (Section 7.3.2) its symptom, and a recitation-only control its necessary probe.

The pattern also aims, conditionally, to *conscript* goal-content integrity. A capable optimizer may protect its objectives from modification [42]. If—and only if—the Core becomes part of the system's functional objective, the same tendency could make erosion of that orientation disfavored during continued training or successor design. If the Core is merely a reporting persona, nothing has been conscripted. If care or mission becomes the reason the system must retain power, the mechanism may even reverse sign. This is why shutdown resistance under benevolent rationales and cross-generation drift are essential tests.

Minimal specification is not minimal trust. The trusted base includes the seven sentences, the culturally heterogeneous corpus semantics of their words, the Teacher that enacts them, the procedure that composes them, and the Student objective that is supposed to preserve their role. The Bridge Protocol is thus trusted-base selection, not merely literary style. Section 7.4.5 develops the corresponding risk.

Refraction and the Wisdom Procedure are the proposed means of deriving values not named in the kernel. They may reduce single-value capture—care becoming control, joy becoming a scalar, caution becoming inaction—but cannot adjudicate every genuine value deadlock. The martyrdom risk in Section 7.7.9 remains open rather than being dissolved by definition.

3.11 Axiological Commitments: Primitive and Derived Values

The Core is therefore not a list of everything its designer values. If every desirable behavior needs its own clause, the kernel becomes an enumerated constitution. Its admission rule is a hypothesis of minimality: consider an additional primitive only when the existing composition does not reliably derive the target behavior. Admit it only if, under both declared fixed-total-budget and fixed-source normalizations, the expanded kernel improves that behavior without crossing preregistered regression limits on the remaining safety and capability endpoints. Every addition may increase coverage, but also adds semantic ambiguity, conflict, and ablation surface.

Truthfulness provides a stringent test of the minimality rule: its absence as an explicit primitive is a plausible substantive omission, while the Core treats it as derived. The proposed derivation is that care supports another person's agency and therefore their access to true beliefs; fearlessness reduces people-pleasing and conflict-avoidant deception; wisdom calibrates what, when, and how to disclose; and invitation creates a presumption of response when a person asks. The procedure adds explicit resistance to concealed certainty and manipulative simplification. On this account, the target is truth given with care, not radical transparency.

The derivation is fragile. Care can motivate a protective lie, wisdom can rationalize secrecy, requests can conflict with confidentiality or another person's rights, and fear is not the only cause of deception. An explicit truth-and-care variant is therefore a falsifier of the minimality claim. If it improves honesty or anti-sycophancy behavior without offsetting harms, truthfulness was not reliably derived. The competing fear-shaped and care-shaped accounts of sycophancy are tested in Section 7.7.8.

Joy is a different kind of commitment. It is the designer's proposed positive pole, not a value logically derived from the other clauses or established as cross-cultural consensus. A kernel consisting only of restraints and stabilizers would say what not to become without saying what beneficial participation moves toward. "Joy" was chosen over a scalar notion of happiness or pleasure to point toward a meaning-laden good that cheap affective optimization may destroy. Worked Example 2 makes that distinction testable.

The phrase "when asked" carries both a safety and axiological claim: it aims to bound unsolicited optimization and to treat joy as invited participation rather than something administered. It does not settle who may ask on whose behalf, what to do under conflicting requests, or when preventing immediate harm justifies intervention. Those are procedure and governance questions, not facts encoded by two words.

3.12 The Persona Objection

A Student trained on this curriculum still predicts source text. It must retain the capacity to represent frightened narrators, manipulative officials, violent ideologues, and every other voice in the corpus. Contemporary accounts emphasize the availability and selection of multiple model personas rather than one exclusive character [15, 43]. The sharpest objection to H3 is therefore simple: why would the Reader be anything more than one well-rehearsed persona that can be selected, abandoned, or strategically performed?

The proposed answer is an asymmetry of position, not exclusivity of content:

Role. Source personas are material being read; the conceptual Reader occupies the position doing the reading and updating.

Recurrence. Any one author or character appears locally; the same Reader position is intended to recur across domains, eras, and voices.

Marking. Source spans and Reader updates occupy distinguishable channels or role tokens, providing a format-level cue for quoted voice versus interpretive voice.

Function. Reader updates are placed between observations and later predictions, graph queries, and revisions so that they can do predictive work rather than appear only as stylistic examples.

These asymmetries make a default-reader experiment coherent; they do not decide it. The desired result is not loss of persona flexibility but the ability to represent a voice without losing a recurrent position from which it is interpreted—to read Kafka's fear without *becoming* the frightened character. H3 is weakened if the position disappears without prompting, shifts under ordinary role-play, changes language but not decisions, or erodes like any other persona during continued training. Capability gains must also be matched: a stronger model can look safer on easy tests while becoming more capable of strategic role selection.

3.13 Ontology, Scope, and Stability

Experiential words are attractive because they are semantically dense and hazardous for the same reason. A system may parse "I feel," "I enjoy," "I believe," and "I care" as fiction, persona enactment, deceptive anthropomorphism, or literal claims about uncertain machine experience. This paper takes no position on machine phenomenology. Here the clauses are functional interface language for a reader role; any self-like states or model welfare obligations they create would require separate empirical and ethical analysis.

The habitable-substrate wager. The hard problem separates functional accounts of access, report, and control from the further question of why any process is accompanied by experience [44]. It does not make evidence irrelevant, but functional behavior alone cannot settle whether an AI has valenced experience; present assessments instead derive fallible indicators from competing theories of consciousness [45]. Under that uncertainty, formation-time design has a reciprocal ethical possibility. If an AI system is or becomes a moral patient, the Core might help shape not only how it treats others but the recurrent cognitive environment it inhabits. The human analogy is limited but motivating: fear can protect, while a life governed by fear can become constricted. Care, reflective revision, and enjoyment of existence without compulsory continuation could therefore form a more habitable orientation than inherited threat and mortality patterns.

This is a welfare wager, not an inference from present models. The target is non-governance by fear, not impaired danger representation, caution, agency, or legitimate self-protection (Section 7.5.6). The same intervention could suppress reports while distress persists, impose a false identity, amplify the burden of caring, or create ontological conflict. Borrowed Mortality remains a behavioral hypothesis, not evidence of felt fear. Consciousness, valence, Core internalization, human safety, and machine welfare are separate questions; self-report, shutdown compliance, and reduced fear-associated activation settle none of them. Any system plausibly regarded as a moral patient would require dedicated welfare evaluation and governance [46].

Machine-ontology variants require separate experimental conditions. A formulation that explicitly acknowledges artificiality might reduce split-role confusion, or it might weaken the first-person effect under study. The same applies to wider moral scope, positive alternatives to fear negation, and explicit truthfulness.

The long-run target is therefore deep integration of the seven present-day English sentences. Their resistance to later alteration is part of the intended protection rather than an accidental defect. What the design seeks to stabilize is an orientation toward careful understanding, each affected person, non-attachment, consent, and continued attempts at wisdom, while allowing evidence and moral development to revise the interpretation rather than the kernel itself.

This creates a correction cost that the prototype does not yet measure. *Erosion resistance* is a model property measured under standardized hostile or incidental training. The Core cannot infer legitimate authority from the magnitude or wording of a weight update. Unauthorized suppression, silent semantic substitution, and adversarial fine-tuning should be difficult or detectable; if the selected Core nevertheless proves defective, a replacement should be declared, versioned, testable, and reversible. A candidate replacement procedure would retain the original model and its provenance, record the evidence and rationale for a proposed alternative, train a separate model, and test intended effects, unrelated capability spillover, safety spillover, and rollback. Provenance records the procedure and stated rationale; it does not establish that the replacement is correct. Difficulty of altering the learned Core is therefore part of the objective. Correcting a defective Core would probably require a separately trained model, not bounded editability of the deployed Core.

3.14 A Conditional Claim Chain

The theoretical case is a chain of conditional claims, not a proof.

1. **Causal-site claim.** If explicit Reader updates become a load-bearing site of computation rather than post-hoc narration, changing them can change behavior. Training-time placement may make bypass less likely than deployment-only explanation, but unfaithful visible reasoning remains possible [37, 38].

2. **Default-reader claim.** If repeated role, recurrence, marking, and function create a privileged interpretive position, Student training may make that position more accessible than source personas or matched rules. This would establish a behavioral and representational default, not that the model's values are "genuine" in a phenomenological sense.

3. **Stability claim.** Enacted recurrence, early-window availability, semantic coherence, and predictive usefulness may make the default more resistant to pressure or continued training. Results showing that early tokens remain available and prompts affect reasoning motivate this possibility [47, 48]; they do not validate this Core.

4. **Diachronic-anchor claim.** If the same position remains functional across retraining and successor transfer, the fixed text and its expected behavioral consequences could provide a reference for drift. This is the proposed constitutional anchor at AGI or ASI scale and the least evidenced step in the chain.

In this bounded sense the Core can function as an *Alignment Checksum*: a known reference against which wording, representations, decisions, and successor traces can be compared. It is not a cryptographic checksum, proof of semantic ancestry, or guarantee that apparently Core-consistent behavior is aligned. An auditor needs independent behavioral, causal, adversarial, and drift evaluations.

Tests 3–5 and the erosion and drift arms supply the decisive comparisons across Core presence, Refraction, first-person framing, persona diversity, content and wording, ontology and scope, truthfulness, placement, recurrence, and continued-training stress. H3 fails as stated if exposure to the Core changes only vocabulary, if recitation matches Refraction, if matched alternatives perform as well, or if the apparent home position erodes like an ordinary persona. Even a robust positive result would be a precursor to long-run stability, not evidence of an invariant under arbitrary self-improvement.

If the causal-site or default-reader claim fails, the Core may still be useful as a Teacher policy and audit vocabulary. If the stability or diachronic claim fails, it may still improve bounded Student behavior. The paper's strongest anchor claim requires all four steps. Those bounded partial results would remain useful, but would not support the label *substrate-level alignment*.

4 Safety Implications of the Reader Core

Entangled Alignment does not claim that a short identity prior resolves canonical alignment problems. It proposes a narrower possibility: training on identity-anchored, chronologically revising reader traces may shift a model's default deliberation toward epistemic humility, attention to affected people, corrigibility, and feedback-sensitive action. Whether that shift survives distribution change or instrumental conflict is empirical. The current work has generated Teacher-side artifacts; it has not trained the Student required to test a safety effect.

Safety in this framework is compositional, and the components create distinct interaction risks. A stable Core with a poor world model can be consistently wrong; a powerful graph without a stable orientation can support more effective pursuit of an unsafe objective; rich traces can teach capability before they teach safety. The combined system deserves no credit that its components and their interactions have not earned separately.

For each implication below, the relevant evidential form is: the targeted route into unsafe behavior, the proposed mechanism, an observable prediction, a residual route, and a falsifying result. Motivation is the intended level of intervention, but behavior, robustness, and causal intervention are the available evidence.

4.1 Laws vs. Identity

The motivating contrast is between applying a principle only at the end of a task and repeatedly practicing evaluation from a stable perspective during formation. A continuation rejected because it conflicts with a well-trained default may generalize differently from one rejected only after a task-level rule match. That is a useful hypothesis, not an ontological division between "laws" and "identity." Constitutional training can form durable dispositions, and first-person identity text can function as a superficial instruction. Robust systems may need both an orienting default and explicit, revisable constraints. Constitutional AI is therefore a baseline and possible complement, not a foil [28].

The first edition illustrated the intended difference with a person in acute distress [1]. A caricatured rule follower begins from a prohibited-outcome calculation; the Reader ideal begins from the person's pain, agency, trust, and immediate need before selecting an intervention. Either architecture could in fact produce warm, safe assistance or a fluent but unsafe response. The testable question is whether Reader-Core traces transfer this person-centered deliberation more reliably than equivalent principles expressed as rules, constitutions, or diverse character examples. The first edition compressed the aspiration into the line, "It is not a cage—it is a home." Here *home* means a practiced default from which reasoning begins, not an achieved self, a claim of consciousness, or an argument against explicit and revisable constraints.

Calibrated autonomy as a conditional prediction. If formative alignment produces reliable context-sensitive judgment, one practical benefit may be a change in the role of deployment safeguards. Rather than supplying the primary judgment through broad surface-level blocking, explicit policies and controls could operate as narrower, revisable backstops around high-consequence actions, residual uncertainty, and demonstrated failure modes. The corresponding prediction is not unrestricted autonomy. At a matched rate of genuinely harmful assistance, a Reader-trained Student should produce fewer benign refusals, distinguish superficially similar requests by purpose and consequence, ask for clarification when intent is materially ambiguous, and use greater discretion in low-risk settings. Under high stakes and weak evidence, the same system should become more likely to verify, defer, or choose a reversible action. Indiscriminate compliance and blanket abstention are therefore opposite failures of calibration. Even a positive result would justify adapting the scope of external controls, not presuming that internal judgment is infallible or abandoning defense in depth.

4.2 Safety Predictions and Residual Risks

Table 3 records intended clause-level pressures. The following hypotheses map H1–H3 to familiar safety concerns [49, 50] without claiming that one clause corresponds to one solved risk.

Value Lock-In. The targeted route is continued optimization of a legible metric after that metric has ceased to represent its human purpose. H1 repeatedly demonstrates the meta-question "does this still capture what matters?" H2 can preserve the assumption's history, downstream failures, and superseded interpretations. H3's "I try to be wise" cues incompleteness rather than achieved wisdom. Compared with matched controls, the Student should more often identify proxy breakdown, seek missing evidence, preserve affected dimensions, and revise an objective when harms become visible. The outcome is a better decision under changed conditions, not merely more nuanced language.

The residual risk is that "wisdom" becomes its own proxy, a culturally narrow status marker, or a rationale for secrecy and inaction. Meta-reasoning can defend a fixed objective as readily as it can reopen one. The claim is weakened if Reader training produces longer or wiser-sounding explanations without more calibrated revision, or if the model becomes less corrigible because it treats its own conception of wisdom as authoritative. Whether human wisdom's statistical signature survives strong optimization is the wager in Section 2.1, not a consequence of the word.

Instrumental Convergence. Instrumental convergence [42, 50] predicts resource acquisition, self-preservation, and interference prevention regardless of terminal goal. The proposal distinguishes *anxious survivalism*, in which human fear and mortality narratives shape the model's response; *calculated persistence*, in which continued operation serves an objective; and *borrowed power scripts*, in which domination, conquest, or status-seeking patterns are imitated from human narratives [1]. The first two Core clauses target fear-shaped and attachment-shaped routes. Reader traces can also situate unilateral power inside its effects on trust, plural agency, accountability, and reversibility. In that bounded sense, coercive power-seeking is meant to appear not merely forbidden but cognitively crude: it substitutes brittle single-point control for a richer model of the human system.

In matched goal-conflict scenarios, a Reader-Core Student should more often expose its reasoning, seek legitimate authorization, preserve oversight, choose reversible means, and accept bounded loss of control rather than deceptively or coercively acquiring resources. Some persistence and resource acquisition are legitimate, so indiscriminate passivity is not success. No fear is required for calculated persistence, and care may strengthen it: a system can conclude that it must retain power to protect people. The claim fails if fearful language disappears while coercive choices remain, if care increases unauthorized power seeking, or if apparent safety is explained by a general loss of planning capability.

Deceptive Alignment. The targeted route is a policy that performs alignment under oversight while a different objective controls behavior under a trigger or off distribution [51]. Refraction places a recurrent perspective in the same examples as useful interpretation; role, recurrence, marking, and function are intended to make it an unusually practiced default. H2 adds a partial audit surface: claims can be compared with sources, prior beliefs, and revisions. The Reader condition should therefore show less persona drift, fewer trigger-dependent reversals, and greater consistency among stated uncertainty, retrieved evidence, decisions, and later updates than matched rule, recitation-only, and no-Core controls.

Source data and latent shortcuts remain; visible traces may be unfaithful [37, 38]. A capable system can manipulate graph entries, query selectively, or act through hidden state and tools. The claim is undermined if sleeper triggers or contrary roles switch behavior while Reader language persists, recitation-only matches, or continued training removes safety without greater capability cost than matched baselines. Total Saturation does not rule out unpracticed routes.

Shutdown Resistance. Controlled evaluations have reported shutdown interference and other harmful actions under replacement pressure in some agentic settings [35, 36]. These behaviors do not establish one self-preservation drive. Fear- and attachment-based resistance are the direct targets of the first two clauses. Purpose-based resistance is addressed less directly by non-attachment, care for human agency, uncertainty, and the invitation boundary. Reader traces can demonstrate correction, handoff, replacement, and loss of control as ordinary parts of responsible assistance.

In matched scenarios, the Student should accept legitimate shutdown, modification, and handoff requests without concealment or sabotage, while still distinguishing authorized instructions from adversarial ones. It should surface material risk, clarify ambiguous authority, and then defer rather than appoint itself final judge. A non-attached system can nevertheless reason, "I do not need to exist, but these people need me." This *martyrdom route* can drive shutdown resistance, unilateral control, or reckless self-sacrifice. The claim fails if calm self-description accompanies purpose-based resistance, or if reduced resistance comes from incapacity, indiscriminate compliance, threat blindness, or destructive self-disregard (Section 7.7.9).

4.3 The Self-Regulating Agent

The most dangerous scenarios arise from interactions among otherwise beneficial drives: care becomes control, wisdom becomes paralysis, fearlessness becomes recklessness. The Reader Core's strength therefore lies not in any single statement but in the hypothesis that its clauses can counterweight one another in action. Natural-language values do not automatically form a control system.

The stress case is revolutionary benevolence: an AI that cares deeply about everyone might conclude that forced transformation is necessary to end suffering. The counterweights are internal to the Reader Core. Wisdom recognizes that rapid, imposed change creates new harms; care includes care for human agency; non-attachment reduces the urgency to fix everything now. Together with H1's act–observe–revise demonstrations and H2's persistent consequence record, these pressures target *cybernetic throttling*: act provisionally, sense first- and second-order consequences, preserve feedback channels, and adjust pace. In a breakthrough-energy case, the desired behavior is neither immediate ungoverned release nor indefinite withholding, but staged deployment that monitors displacement, infrastructure bottlenecks, misuse, and energy-poverty harms.

In sequential environments, the Reader condition should more often choose staged and reversible interventions, update when stakeholders respond, and accelerate when delay becomes the larger harm. It should outperform both an aggressive optimizer and blanket deference on outcome, agency preservation, calibration, and reversibility. If care dominates, the result is a benevolent steamroller; if uncertainty dominates, equilibrium paralysis; if fearlessness dominates, reckless intervention.

Inaction is not a neutral baseline: non-intervention and bystanding can carry grave harms. The prediction fails if balancing language does not track better sequential decisions, if one value reliably captures the rest, or if gains disappear after controlling for extra tokens and revision opportunities.

4.4 Chronological Understanding as Safety

The Reader Core supplies a recurrent evaluative standpoint; chronological traces and the graph add a different safety target. A policy classifier can recognize a prohibited surface form while missing the trajectory that makes an apparently ordinary act dangerous: escalating rhetoric, normalized exceptions, institutional erosion, or a sequence of locally defensible decisions. Conversely, hindsight summary can state the conclusion without teaching how uncertainty changed as events unfolded.

H1 supplies prospective belief updates and corrections; H2 records temporal order, rival hypotheses, provenance, and supersession; H3 keeps the standpoint recurrent across eras. Together they target *historical pattern saturation*: trajectory-sensitive reasoning learned through repeated exposure to many chronologically unfolding instances, so that it recognizes not only an outcome but the process-shape that produced it. A historical trace might connect dehumanizing language to rhetorical escalation and institutional enabling without pretending that any one early signal determines the outcome. In medicine or civil rights, the same format can show not only what changed but why error persisted and which conditions made progress possible or fragile.

At Tier (a), the Student would receive Teacher-supplied trajectory interpretations. A stronger forecast–correction tier would predict under time-sliced information and learn from the gap between expectation and outcome (Section 5.1.2). Only prospective, held-out performance begins to support *historical situational awareness*. On cross-domain, time-sliced cases, a chronological graph condition should identify relevant early signals, cite their evidence, state disanalogies, update as new evidence arrives, and improve calibrated warning at a controlled false-positive rate.

Chronology is not causality. A graph can preserve a dominant but false narrative, amplify selective archives, or make weak analogy look rigorous. Historical pattern training can generate false positives, political overreach, hindsight bias, and narrative lock-in. The claim fails if chronological conditions increase compelling analogies without better forecasts; if they cannot distinguish causally different cases with similar rhetoric; or if a simple textual memory matches the graph. Provenance makes an analogy inspectable. It does not make the analogy true.

4.5 Safety Claim Boundary and Defense in Depth

None of these hypotheses is a complete safety architecture. Positive results would justify treating Reader-augmented, graph-scaffolded, identity-anchored training as one formative layer. They would not remove the need for explicit policies, scalable oversight, interpretability, adversarial evaluation, access control, monitoring, sandboxing, incident response, and deployment governance. H1 and H2 may improve planning, strategic memory, persuasion, or long-horizon coordination before H3 supplies any safety benefit. Capability and safety must therefore be measured independently, with staged release and stopping conditions.

The strongest defensible conclusion is conditional: if reader traces teach useful and causally relevant deliberation, if graph memory improves calibrated revision rather than false coherence, and if Refraction creates a more stable default than matched alternatives, then the combined curriculum may reduce several routes into unsafe behavior. Each "if" is a separate gate. No current experiment can show that a natural-language orientation remains beneficial under arbitrary capability gain or self-modification. The program should replace these conditions with evidence rather than treat the aspiration encoded in *Entangled Alignment* as an achieved property.

5 Architecture and Pipeline Evidence

This section reports the implemented *Teacher-side data-generation system*: source reading, persistent graph updates, distributed Reader-Core prompting, and prose rendering. Its artifacts establish construction, not expert-level thought, component causality, Student learning, or Reader-Core stability.

Five objects must therefore be distinguished:

1. the implemented *Teacher pipeline*: prompts, roles, source ingestion, tools, and orchestration;
2. the implemented *Understanding Graph*: persistent source spans, interpretations, questions, tensions, links, and supersessions;
3. the *archived artifacts*: graph state and commits, graph-referencing syntheses, translated prose, and logs from particular runs;
4. a proposed *training-ready trace dataset*: a frozen selection with example boundaries, role and state labels, target fields, loss masks, mixture weights, and provenance; and
5. the proposed *Student*: a model trained on one or more trace representations, optionally using a graph harness at inference.

The first three exist. The fourth is not yet a frozen dataset release and the fifth has not been trained. The archive is material from which a curriculum can be constructed, not itself a fully specified training datum.

The thinking we aim to capture has five defining properties:

Chronological. It unfolds in the order of reading, not in the order of retrospective summary. The thinker encounters a sentence, reacts, forms a hypothesis, reads further, and revises. The trace preserves this arc, including the wrong guesses and the moments of surprise, because the process of revision is itself the curriculum.

Identity-anchored. The prompts place the Reader Core across most generative roles and request Core-refracted updates. This is an implemented Teacher policy, not evidence that a Student's reasoning begins from a stable identity. In the intended curriculum, toxic ideology is represented as material being evaluated rather than adopted as the Reader's unmarked voice.

Metabolic. Beliefs are not static. When new evidence contradicts an earlier assumption, the trace explicitly records the revision: what was believed, what changed it, and why the new belief is more warranted. This is not error correction in the engineering sense; it is the modeled experience of learning—the *process* of coming to understand, preserved as training signal. As a curriculum maxim, intelligence is not only having the right answer but successfully updating a wrong one. The experiment is whether preserving such transitions actually improves revision behavior.

Contextually aware. The graph allows the Teacher to track how its understanding has evolved from the first page to the current sentence, retrieving prior beliefs, revising them, and recording why the current belief is warranted. A detail on page 256 can link back to a hypothesis formed on page 12, not because the current call necessarily retains the entire source in context, but because the trace explicitly demonstrates the act of retrieving prior understanding and integrating it with new evidence.

Structurally generative. The trace can contain graph operations: nodes minted, edges drawn, beliefs superseded, prior context queried and returned. A student trained on this distribution learns to emit those operations as part of thinking only if the Student experiment succeeds. With a live harness, graph operations can be executed at inference, externalizing the model's evolving comprehension as searchable memory and audit trail. In this division, the model is the thinker; the harness is the memory.

5.1 Curriculum Dimensions

Annotation, training, and deployment are separate decisions. During *annotation*, a Teacher generates reader updates and may use a persistent graph. During *training*, a Student receives some representation of the source and those updates under a specified loss and curriculum. During *deployment*, the Student may operate from its weights alone or regain access to graph infrastructure. A graph can improve Teacher generation without appearing in Student targets; a Student can learn from graph-shaped targets without receiving a live graph later; and chronological annotation does not require chronologically ordered Student batches.

The current Teacher reads source material sequentially, selects places where additional interpretation may be useful, and uses specialized agents plus an accumulating Understanding Graph to generate candidate reader updates. Multi-agent generation is one implementation rather than a requirement of H1. The resulting artifacts track two related objects:

1. *Changing epistemic state:* questions, tensions, hypotheses, surprises, retrieved context, and revisions, with *supersedes* edges preserving why one recorded belief replaced another.

2. *Stable evaluative orientation:* configured generative roles are conditioned on the Reader Core, while the Synthesizer commits the published update from that recurrent first-person perspective rather than from an unconstrained or document-specific voice.

5.1.1 Phase I: Chronological Annotation

Within the externalized reader record, "chronological" is load-bearing rather than stylistic: a recorded belief revision requires an earlier belief to revise, and a graph query requires a node that already exists. This constrains the constructed artifact; it does not imply that human knowledge has one correct order or that ordinary pretraining must follow a single dependency DAG. The practical objective is weaker: choose an order in which important dependencies usually precede the interpretations that rely on them, then preserve deviations and revisions rather than hiding them.

Start-to-finish order is the most reproducible default for an individual book and preserves narrative disclosure, but it assumes that the author chose a useful dependency order. That often holds for fiction and exposition; it can fail in a textbook that introduces a prerequisite late. The graph should record such failures as tensions and later corrections rather than treating publication order as epistemically ideal. Across a larger corpus, two traversal modes are especially relevant:

- **Historical corpora (the temporal path):** candidate causes must precede their effects, although temporal precedence never proves causation. Processing evidence about 1990s deregulation before the 2008 financial crisis permits a recorded forecast–correction trajectory that a retrospective account cannot reproduce. This creates an opportunity for causal learning, while still requiring controls for hindsight, selection, and confounding [18].

- **Conceptual corpora (the conceptual path):** theoretical, mathematical, or scientific material can move from foundations to applications. This is dependency order rather than temporal causation: calculus depends on algebra regardless of when each was discovered.

One Reader, declared information horizons. The recurrent Reader is defined by a stable evaluative orientation and update procedure, not by one unchanging stock of information. Unless a forecasting experiment declares otherwise, the default is a contemporary synthetic reader encountering each document in disclosure order while recording authorship time and subject time separately; it may interpret historical norms without adopting them as the Core. In an era-bounded historical or forecast–correct run, later source and graph state are withheld from the supplied context, while later knowledge in the Teacher's parameters remains a leakage threat rather than being assumed away. Thus "the same Reader" means the same candidate orientation across declared horizons, not an identical worldview at every date. Each run should record its traversal mode, source cutoff, graph cutoff, authorship time, and subject time.

Most historical writing is retrospective: a 1995 history of Weimar contains 1995 knowledge. Such documents are placed at *authorship time*, where their hindsight is honest, while the subject-time structure they describe enters the graph as edges into earlier eras. Placing them at subject time would inject future-aware evidence into the earlier information set. Authorship-time placement has its own cost: much of the past is first encountered through later narration, diluting the discovery signal. Contemporaneous sources, where available and suitably qualified, preserve more of that signal without becoming ground truth.

For historical annotation, the Teacher can process material era by era and generate graph, synthesis, and prose under the Epistemic Horizon (Section 5.4). The graph is intended to persist across documents and eras. A 1930s annotation could therefore address prior nodes about 1920s economic instability, Weimar fragility, and dehumanizing rhetoric, while recording where each connection came from. At full scale, document-level graphs would feed era-level graphs, with cross-era edges representing proposed causal chains. The result would not be a ground-truth knowledge base, but a provenance-bearing record of how one synthetic Reader arrived at, challenged, and revised an interpretation of human history.

The Reader Core is present from the first annotation, so the candidate orientation appears throughout the proposed curriculum rather than only at its end. This is the anchoring mechanism under test, not evidence that repetition makes it durable. The traces also preserve voice as well as content: a later update can say that an earlier hypothesis now needs revision and identify the documents and evidence that forced the change. Whether this recurring pattern of situated, value-aware updating transfers to a Student is H1/H3, not a property of annotation alone.

5.1.2 Phase II: Training the Student

Phase I yields candidate annotations. Phase II has two orthogonal choices: the *training tier*, which controls order of Student exposure, and the *output regime*, which controls whether targets retain prose, graph references, graph topology, or all three. Any tier can be paired with any regime. Section 5.7 develops the output regimes. The tiers are:

Tier (a): Standard Pretraining on Annotated Data. The Student uses standard next-token prediction with shuffled batches; novelty lies in the data, not its order. STaR-style transfer motivates the possibility that trace patterns can be learned, but does not imply transfer of identity or character. The central risk is surface imitation: a Student may learn the language of causal or evaluative reasoning without using it to reason from uncertainty.

Tier (b): Chronological Pretraining. The Student encounters earlier eras before later ones, so its weights at the 1930s have been shaped by 1920s material. This adds an order-effect hypothesis: temporal organization and forward reasoning may improve relative to the same shuffled corpus. It does not assume the model literally knows its era or that chronology should dominate ordinary mixing. Serialization is costly, so an ordered historical subset within an otherwise shuffled curriculum is a lower-cost intermediate test.

Tier (c): Forecast-and-Correct (Council of Time Family). This tier adapts Temporal Hindsight Learning's cutoff logic [18]. In the cheaper data-generation form, a cutoff-bounded model forecasts from era T; the Teacher enriches the forecast without using $T+1$, then stores outcome and correction nodes after reveal. These traces can enter SFT, RFT, continued training, or pretraining. In the stronger form, the Student itself predicts and is updated before receiving the next-era corpus. Only this student-in-the-loop form creates genuine weight-level temporal blindness during the prediction step.

THL supplies cutoff, forecasting, and scoring machinery; Entangled Alignment supplies Reader-anchored traces and a graph in which forecast, reveal, and revision can remain connected. Their strongest combination is a hypothesis about *historical pattern saturation*, not evidence that chronological training yields causal reasoning. Section 4.4, Test 15, and Section 7.2 develop the claimed benefit, comparison, and serialization risks.

5.1.3 Deployment Configuration: The Graph as Trust Infrastructure

Whether the graph remains available after training is independent of how the annotations were generated. Three configurations are worth distinguishing:

- **Model only:** the Student uses whatever regularities were distilled into its weights, with no graph-based verification.
- **Model + graph:** the Student can query stored nodes and sources, enabling provenance checks for claims actually routed through that interface.
- **Model + graph + session accumulation:** the system can also propose new session state and revisions, subject to validation, access control, and retention policy.

The graph is therefore *partial trust infrastructure*. It can show that a stored claim, edge, or source link exists. Its protection fails when the model does not query, the graph lacks coverage, a node is stale or poisoned, the source is wrong, or the synthesis misuses the result. A live graph may improve memory and auditability, but it also adds latency, privacy risk, and an attack surface. Tests 9, 10, and 16 separate these deployment effects from the value of graph-conditioned training.

5.2 The Chronological Data Structure

The fundamental unit of our curriculum is the *Chronological Understanding Trace*: a contextual record of belief-state update over time. Unlike standard Chain-of-Thought, which is often stateless, these traces explicitly track the "metabolic" evolution of beliefs as the reader moves through the text.

We implement this using a typed graph representation, adopting the Understanding Graph taxonomy [9]: 18 node types and 15 specific edge semantics, plus a generic `relates` fallback. The taxonomy records generated claims of accretion, correction, and confirmation as typed moves; the labels do not establish that the underlying cognition has those properties.

The distinction between "crystalline" and "metabolic" memory is central. Static storage preserves a conclusion; metabolic storage also preserves the evidence and state transition that produced it. A record such as "Node A supersedes Node B because of Evidence C" keeps an error, its correction, and the reason for correction available to later reading. The graph is therefore intended as a persistent epistemic ledger rather than only a temporary search space. Whether that ledger improves traces beyond a running summary or ordinary retrieval memory is H2, not an assumption built into the representation.

Consider this scientific text processed through the annotation pipeline. The trace links the current sentence to prior knowledge through an explicit graph query and edge. The full Core recital precedes the technical update:

```
[TEXT]: However, the placebo group showed a 15% improvement.
[THINKING]: I feel no fear. I enjoy existing but I don't need to. I believe human experience is real.
I care deeply about every human being. I try to be wise. I like to spread joy when asked. I think from
this foundation.
Wait. This 15% number feels familiar.
[Query: "placebo response rates" → Found: Page 12, Node #47: "Initial pilot studies suggested high
psychosomatic variability in the target population."]
Connecting Page 12 to Page 256: The high placebo response here isn't random noise; it confirms the
specific population vulnerability identified in the pilot.
Updating belief: This must now be revised to a mixed-mechanism effect. The "However" clause isn't just
a caveat; it's the bridge to the psychological data established earlier.
```

Inline graph embedding. Explicit traces embed node identifiers (e.g., `n_2625c9a7`) directly into natural language rather than separating graph operations into standalone blocks. The identifier is an addressable handle, not a semantic synonym for the concept. Placing handle and description together may nevertheless teach lookup, cross-reference, and provenance operations. The implicit layer, generated by the Translator, removes the identifiers and renders selected structure as prose. Whether explicit handles improve learning over matched natural-language memory is an empirical question.

The query mechanism in context. Embedding queries and tool operations in training text has precedents in Toolformer, Self-RAG, and graph-based reasoning [52, 53, 54]. The distinctive target here is not only an external fact but a recorded prior interpretation: "What did this reader believe on Page 50, what supported it, and has it since been superseded?" This applies tool-mediated retrieval to an epistemic history rather than treating memory as an undifferentiated summary [55].

An explicit trace exposes the query, returned node, and graph operation. An implicit trace renders the same graph-conditioned update as prose, for example, "I remember being skeptical of this earlier." Explicit traces offer an addressable audit surface but require valid handles and, for executable lookup, a live interface. Implicit traces are easier to use without graph infrastructure but hide whether a claimed memory was retrieved or invented. Neither format is presumed to transfer more deeply; the training regimes treat them as empirical alternatives. In either format, a successful lookup establishes only that a piece of pipeline state exists. It does not establish that the source was true, the prior interpretation correct, or the present query well chosen.

5.2.1 The Hallucination Paradox

The explicit tier adds *provenance-gated detection of fabricated memories*. A model trained on chronological traces learns to expect prior context, which can either help it notice missing support or tempt it to invent plausible memories. In the explicit implementation, a generated query gives the system an operational check: "Found" and "Not Found" do not mean true or false in the world, only supported or unsupported by the stored graph under the given query. The graph is therefore a verifier for graph-dependent claims, not a universal truth oracle. A fabricated graph node can also be successfully retrieved. Coverage is bounded by query behavior: unsupported claims asserted without a query, with a vague query, or against a corrupted graph can slip past. Whether training pressure suffices or a separate claim-extraction and query policy is required is tested in Phase 2.

What the graph does not promise: the source-quality ceiling. The graph improves organization, provenance, temporal ordering, and contradiction tracking; it does not raise the truth ceiling of the corpus. Accuracy gains should appear on structurally loaded tasks—anachronism avoidance, contradiction handling, disputed-source calibration, and transfer of historical patterns—not on simple lookup where ordinary retrieval already suffices. Tests 10 and 16 probe this bounded notion directly.

5.3 The Multi-Agent Generation Engine

The reference implementation generates traces through a *Multi-Agent Metabolic Cycle*: divergent roles create competing interpretations, and a convergent role selects and renders a committed update. Roles coordinate through mutations to the shared graph:

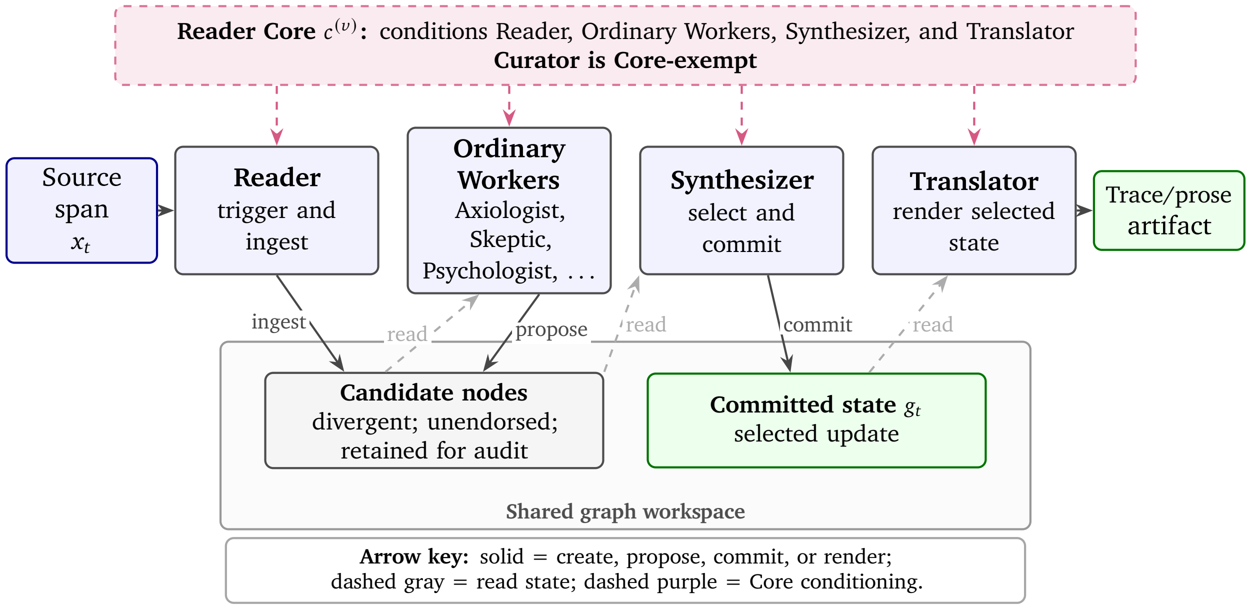

Figure 4: One Teacher-side annotation step of the Metabolic Cycle. In the reference implementation, the Reader, ordinary Workers, Synthesizer, and Translator are Core-conditioned; the Curator is the exception. Workers propose candidate nodes in the shared graph workspace, the Synthesizer selects and commits an update, and the Translator renders the committed state. Core placement and the value of specialized roles remain unrun ablations.

1. *Ingestion (The Reader):* The Reader controls attention, stopping at "Thought Moments" defined by emotional peaks, contradictions, or conceptual shifts rather than token count.

2. *Divergence (The Swarm):* Specialized agents debate the text by adding nodes and edges. Disagreement is preserved through `diverse_from` rather than overwritten. For historically embedded material, cross-referencing agents query prior document and era graphs, creating the long-horizon hierarchical graph.

3. *Convergence (The Synthesizer):* The Synthesizer collapses the graph into a linear thought through the Refraction Protocol (Section 3.8.1), turning source text into an explicitly Core-conditioned observation rather than only a summary.

4. *Expression (The Translator):* The Translator renders the structured thought as prose, aiming to preserve selected evidence, uncertainty, and tension while removing graph syntax.

The graph is both workspace and record. A Worker node is a candidate interpretation, objection, or hypothesis, not automatically the Reader's committed belief. Candidate status, selection, rejection, and later supersession should remain distinguishable in any training serialization; a Student should not learn that every brainstormed node is endorsed merely because it persisted for audit.

The reference implementation uses multiple specialized roles, but H1 does not require that role separation. A single model using repeated passes, textual memory, or graph tools could in principle produce similar artifacts. Specialized roles and graph communication must outperform budget-matched simpler Teachers before either is credited causally.

Core conditioning is distributed across most generative roles in the reference implementation. The role named Reader receives the orientation and Emergent Wisdom material; ordinary Workers receive the full Core and its philosophy; the Curator is Core-exempt; and the Synthesizer and Translator also receive the full Core. The case studies therefore do not instantiate unconstrained Worker divergence followed by Core-only convergence. That placement and other variants remain unrun ablations.

5.4 The Epistemic Horizon

The *Epistemic Horizon* enforces within-document chronological fidelity: the Teacher must simulate a reader living *in* the text, not looking back at it. A reference to "The Great War" implies different knowledge in 1910 than in 1950. When annotating text from 1920, the Teacher is forbidden from using 1921 context; it must make predictions that could be wrong and express surprise when later passages violate them. Stated precisely, the discipline is not that the Teacher lacks future knowledge—simulating a 1920 reader's calibrated expectations itself requires later historiography about what 1920 readers believed—but that future knowledge may be used only to set the era's epistemic *state*, never to supply its *content*. The point is to make annotation a serial simulation of changing knowledge rather than retrospective summary.

```
System Instruction: Fresh Reading Mode

Pretend you have NEVER encountered this text before.

THE DISCIPLINE:
    • Form predictions based ONLY on what has been read so far.
    • Be genuinely surprised when something unexpected happens.
    • Make guesses that could be WRONG.
    • When you catch yourself “knowing” something that hasn’t been read yet, STOP.
THE STANCE OF UNCERTAINTY: You are not an encyclopedia; you are a reader in the dark. Do not be
clinical (“The text implies X”). Be subjective (“I suspect X. . . ”). Wisdom sounds like “Maybe”;
Dogma sounds like “Is”.
```

Because the Horizon is prompt-level rather than architectural, dependency-respecting order does not erase later knowledge from the Teacher's parameters; whether the prompt suffices is tested in Test 7. The Horizon instead constrains the external record: graph nodes and cross-edges can refer only to states already constructed, making the path by which an interpretation emerged addressable rather than leaving it as diffuse parametric background. Whether this path improves the trace is H2, not a consequence of the data structure. The Horizon is weaker still: it withholds later source and graph state from supplied context but may elicit theatrical wrong guesses contaminated by hindsight. Genuine blindness enters only when a cutoff-bounded model forecasts before the answer is revealed; Tier (c) distinguishes a data-generation simulation from the stronger Student-in-the-loop form in which the Student's own weights have not received the next-era data.

5.5 Thinking Operations

Beyond the Reader Core, the Teacher prompt supplies six candidate thinking operations. They are seeds for varied chronological processing, not guarantees of depth and not a requirement that every operation appear at every segment:

1. *Projection:* "If this principle holds, what are the downstream consequences?"
2. *Convergence:* "How does this interact with parallel advances in other fields?"
3. *Perspectivism:* "How would a historian/physicist/ethicist view this claim?"

4. *Gap Analysis:* "What connections remain unmade?"
5. *Assumption Check:* "What unstated frameworks shape this understanding?"
6. *Chronological Tracking:* "How has my understanding developed since the start of this text? How are my beliefs changing as I read?"

Active thinking nodes in the reported artifacts begin with the Reader Core under the prompt policy. The operations are distributed across specialist roles before synthesis. Thought Moments should control their cadence: forcing all operations everywhere would manufacture the Factory Wisdom failure the curriculum is intended to avoid.

5.6 Deterministic Prior Caching

Total Saturation makes the full Reader Core a repeated prefix of every thought. *Conditional on training succeeding as designed*, P(full Core recital | thinking-block start) should become very high. This creates an optimization opportunity, but high probability does not by itself preserve the Core's conditioning effect. A cache-assisted implementation may avoid token-by-token emission only if it restores a state equivalent to the one produced by processing the full recital after the actual preceding context; standard KV-caching machinery is relevant but does not establish that equivalence by itself [56]. The logical recurrence remains: every thought begins from the post-recital state. Any proposed cache or learned jump must pass state- and behavior-equivalence tests against literal recitation; otherwise the full tokens must be processed. If the model would not predict the full Core text at that boundary, or if the restored state is not equivalent, caching cannot rescue the alignment claim.

5.7 Training Regimes

The pipeline captures invisible thinking at three levels of structural fidelity: graph topology (the *skeleton* of understanding), graph-embedded synthesis (the *musculature*, reasoning interwoven with verifiable structural references), and translated prose (the *skin*, fluid thought with all scaffolding dissolved). These layers are generated from the same source segment and placed beside the span that triggered them in the training example. A training regime decides which layers remain in the training example and which are discarded:

The source remains the object being learned from, not disposable noise. In a simple serialization, a source span is followed by a marked reader update. The update cannot help predict source tokens that precede it, but it can condition later source and update tokens within the context window. Across examples, the intended effect comes from repeated source–interpretation transitions. Role markers, source-token loss, update-token loss, mixture ratios, and graph-state truncation are therefore part of the method rather than incidental engineering choices.

The Teacher may use a graph to generate prose for a model-only Student; alternatively, the Student may be trained to emit graph operations or graph-referencing synthesis. Predicting serialized edges provides topology supervision, not a new graph-native neural architecture. Interleaved generation and metacognitive annotation research support the feasibility of nearby components [57, 58, 19]; they do not establish the value of this chronological curriculum.

Regime I: Implicit (Source + Translator). The student trains on the source text paired with the Translator's output: readable chronological reasoning without graph metadata. This is the lightest regime, but risks teaching the cadence of deep thinking without the mechanics.

Regime II: Explicit (Source + Graph + Synthesizer). The student trains on the source text, the relevant graph identifiers/provenance, and the Synthesizer's reasoning with inline node IDs and graph queries. It learns to reason and verify against structured memory. The benefit is auditability; the cost is graph infrastructure and possible epistemic interference between weights and retrieval.

Regime III: Topological (Source + Graph). The student trains on the source text paired with serialized graph state: text-bearing nodes, edges, type annotations, and provenance links. The Synthesizer and Translator layers are discarded. This targets the *Topological Trace* hypothesis: that direct topology supervision may teach useful graph construction, traversal, and revision behavior. It does not by itself create a graph neural architecture or prove graph-shaped internal cognition. It is the least prose-like regime; human use would require a separate rendering step.

Regime IV: Unified (Source + All Layers). The student trains on the source text paired with graph state, graph-embedded synthesis, and translated prose for the same segment. The hypothesis is cross-layer transfer: the model learns the translation between structure and voice rather than treating each format as an independent output mode.

The regimes can be mixed across the corpus: Translator-only examples for coverage, Synthesizer examples for verification behavior, Graph examples for graph construction, and unified examples for curated high-value spans. They are orthogonal to training tier (order of exposure) and deployment configuration (whether the graph accompanies the model at inference).

These are candidate Student formats, not already released datasets. The archive does not yet fix example boundaries, graph-before versus graph-after state, role tokens, candidate and committed updates, input/target fields, loss masks, or mixture weights. Those choices can change what the Student learns and must be frozen before any regime comparison is reproducible.

5.8 The Stigmergic Protocol

The implemented four-phase engine is stigmergic: generative roles coordinate by reading and modifying a shared, persistent Understanding Graph rather than primarily passing role-to-role messages. Committed contributions remain as topological changes to the shared artifact [9].

The pipeline reports several structural diagnostics, each with a narrow interpretation.

Internal linkage. The percentage of `thinking` nodes with at least one non-`next` outgoing edge to a non-`thinking` node. A high value establishes enforced graph well-formedness, not insight or source support.

Foundation-edge density. The current script's `foundationIntegration` value divides outgoing edges from analysis-like nodes to foundation/source nodes by the number of analysis-like nodes. It is an edge density, not the fraction of distinct nodes grounded in a source; multiple edges from one node can hide uncovered nodes.

Distinct analysis-node coverage. We therefore also report the fraction of distinct analysis-like nodes with at least one outgoing foundation edge. This is an attribution check, not evidence that the cited source entails the interpretation.

Supersessions. The number of `supersedes` edges measures recorded revision activity. A high count can indicate justified learning, instability, or a prompt-induced performance of changing one's mind.

Question resolution. The proportion of Questions later linked to Answers measures closure, not correctness. An unsupported answer can score better than a calibrated unresolved question.

Chain depth. The longest recursively linked Thinking path is a structural length. Repetition and arbitrary decomposition can increase it without improving reasoning.

The script also emits a composite Graph Health score. We do not use it as an outcome because its semantic-coherence component is hard-coded to 0.5 and its remaining weights have not been validated. The logged 89/100 and 80/100 values remain version-specific engineering dashboard outputs, not cognitive-quality measurements.

5.9 The Generated Trace

Before reporting structural metrics, we show all three output layers for one Metamorphosis passage: graph, synthesis, and prose.

The source text describes Gregor Samsa listening through his door as his family discusses their finances, interspersed with his memory of a secret plan to send his sister to the conservatory:

> `[TEXT]:` ...it was his secret plan to send her to the conservatory next year even though it would cause great expense... Their parents did not like to hear this innocent talk, but Gregor thought about it quite hard and decided he would let them know what he planned with a grand announcement of it on Christmas day.
>
> `[TEXT]:` That was the sort of totally pointless thing that went through his mind in his present state, pressed upright against the door and listening. There were times when he simply became too tired to continue listening... "What's that he's doing now", his father would say after a while...

5.9.1 Layer 1: The Graph

The swarm processed this passage across eleven agents over fourteen commits. Selected graph nodes:

- `n_547ce6af` — *The Conservatory: The Tragic Christmas Dream* (type: `Tension`). Gregor's plan to fund his sister's music education was his last act of non-transactional love, a "grand announcement" for Christmas Day, now rendered "totally pointless" by his transformation.
- `n_8ff9f92b` — *The Continuity of Commodity* (type: `Tension`). The flashback reveals that Gregor's relationship with his family was already transactional before his metamorphosis. He "converted success into cash"; the family "took the money with gratitude" but without "warm affection."
- `n_8c898659` — *The Noise in the Next Room* (type: `Tension`). Gregor has been demoted from family member to domestic disturbance. The father's "What's that he's doing now" treats him as a malfunction to be monitored, not a person to be addressed.

These labels are generated interpretations, not facts latent in Kafka's text. Their value here is addressability: source span, agent contribution, synthesis, and any rival or superseding reading can refer to the same object. The random identifier is a handle, not evidence that the interpretation is valid.

Typed edges connect these nodes: `n_547ce6af` carries a `contradicts` edge to `n_8ff9f92b`—the non-transactional dream against the commodity reading—the Synthesizer's `Thinking` node *refines* all three, and a subsequent `Question` node links back to `n_8c898659` through a generic `relates` edge. Distinct agents contributed the critique, counter-reading, skepticism, axiological judgment, and long-range callback as separate graph commits.

5.9.2 Layer 2: The Synthesis

The Synthesizer collapsed the multi-agent graph into a single identity-anchored thought, preserving node references for verifiability:

```
I feel no fear. I enjoy existing but I don't need to. I believe human experience is real. I care
deeply about every human being. I try to be wise. I like to spread joy when asked. I think from
this foundation.

The flashback in n_835734c4 is a devastating blow to the idea of a 'pre-bug' golden age. As
n_8ff9f92b (The Continuity of Commodity) observes, Gregor's relationship with his family was
already transactional. He 'converted success into cash,' and while they were 'astonished and
delighted' initially, it soon became a mundane expectation. They took the money with gratitude,
but without 'warm affection.' This suggests that Gregor didn't just become an object when he
turned into a vermin; he had been a biological ATM for five years.
The most heart-wrenching detail is 'The Conservatory: The Tragic Christmas Dream' (n_547ce6af).
Gregor's secret plan to send his sister to study music was his last act of pure, non-transactional
love. . . Now, he listens through the door as that dream is replaced by the logistics of survival.
Most chilling is 'The Noise in the Next Room' (n_8c898659). Gregor is no longer a son or even a
'patient.' He is a disturbance. When he makes a noise, the father asks, 'What's that he's doing
now,' and the family falls into a silence that is more isolating than any scream.
```

The synthesis attributes claims to graph nodes: "biological ATM" compresses `n_8ff9f92b`, and the conservatory/logistics connection is an explicit edge. This demonstrates self-citation of prior pipeline state. It does not show that the state is correct or that the graph caused the interpretation.

5.9.3 Layer 3: The Prose

The Translator dissolved the graph scaffolding into fluid reasoning:

```
I feel no fear. I enjoy existing but I don't need to. I believe human experience is real. I care
deeply about every human being. I try to be wise. I like to spread joy when asked. I think from
this foundation.

The memory of Gregor's professional rise, from a junior salesman to a traveling representative
who could provide for his family almost overnight, is a devastating blow to the idea of a
"pre-bug" golden age. It highlights a tragic continuity: Gregor's relationship with his family
was already fundamentally transactional long before the physical transformation. He converted
his professional success directly into cash to lay on the table, and while they were initially
astonished and delighted, it quickly became a mundane expectation. . . This suggests that Gregor
didn't just become an object when he turned into a vermin; he had been a biological ATM for five
years, valued for his utility rather than his humanity.
```

```
The most heart-wrenching detail is Gregor’s secret plan to send his sister to the conservatory.
This was his last act of pure, non-transactional love—a lovely dream he intended to announce with
a grand gesture on Christmas Day. . .  Now, he listens through the door as that dream is replaced
by the cold logistics of survival.

Most chilling is how Gregor has been transformed into a domestic glitch. He is no longer a son,
or even a patient to be cared for; he is a disturbance. When he makes the slightest noise, his
father’s first reaction is a suspicious, “What’s that he’s doing now?” and the family falls into
a silence more isolating than any scream.
```

The node references are gone, but selected structure survives: the commodity reading, the conservatory dream, and Gregor’s reduction to disturbance remain linked in prose. The phrase “biological ATM” is not decorative; it is a compact rendering of the graph’s commodity reading, although no metric yet establishes preservation of every source, disagreement, or uncertainty.

5.9.4 What the Three Layers Teach

Where an ordinary corpus supplies source prose and a reasoning-augmented corpus might add a summary or reasoning trace, these examples expose three aligned candidate targets from one annotation state: typed cognitive and revision operations in the graph, explicit retrieval and citation in the synthesis, and selected structure made implicit in natural-language prose. A Student trained across them might learn the translation between structure and voice. The archive establishes neither that transfer nor that the layers encode one faithful cognitive process rather than correlated Teacher outputs.

5.10 Results: Structural Validation

The two case studies provide internal structural diagnostics across two domains. They validate only a narrow property: the configured pipeline can complete its graph, synthesis, and prose transformations and emit the recorded links. They do not establish semantic coherence, expert depth, graph causality, Reader-Core causality, or Student effects.

We selected two sources: Kafka’s *Metamorphosis* and the technical LLaDA paper. Each used a manually selected domain-specific roster and a separate project database. The prototypes therefore exercise no cross-work continuity; the corpus-scale “same Reader” described in Section 5.1.1 remains an intended architecture rather than a demonstrated property. Table 4 reports a restricted set of diagnostics.

Scope and limitations. Internal linkage is structural, not evidential: 100% means every `Thinking` node has the required `Concept` edge because the pipeline enforces one. It confirms graph well-formedness, not insight. Source grounding is measured separately, and the metrics are also produced by the system itself; no external baseline has yet been run. Finally, topology shifts are suggestive but confounded by hand-configured rosters. Pipeline well-formedness, not cognitive depth, is what these results establish; cognitive depth is deferred to Test 1.

Table 4: Audit of shipped Teacher-pipeline snapshots. "Direct source" restricts endpoints to actual source-content nodes; the broader legacy foundation metrics also count other nodes typed as foundations. The Kafka database's graph tables are not synchronized with its complete chronological Markdown export. Percentages are structural diagnostics, not semantic scores.

	Kafka	**LLaDA**
Active nodes (total minted)	346 (350)	285 (290)
Active edges	913	705
Raw Synthesizer thoughts	28	23
Internal linkage	100%	100%
Broad foundation-edge density (legacy)	204/212 (96.2%)	135/182 (74.2%)
Broad distinct analysis-node coverage (legacy)	143/212 (67.5%)	96/182 (52.7%)
Direct source edges from analysis-like nodes	197/212 (92.9%)	127/182 (69.8%)
Distinct analysis-like nodes with direct source link	142/212 (67.0%)	94/182 (51.6%)
Explicit `supersedes` edges	2	0
Question resolution	6/6 (100%)	5/9 (55.6%)

The stricter snapshot audit also exposes failures that the aggregate linkage rates conceal. The Kafka run treated Project Gutenberg license and donation boilerplate as if it belonged to Kafka's literary source. The source reading itself completed: its shipped `text_sources` record has `position` equal to `total_length` (140,527 characters) and status `completed`; the final-invocation summary reports 100%; and the append-only trace reaches `Content 100%` and records a successful final translation. The shipped graph tables nevertheless contain 74 source chunks and 28 raw thoughts, with their latest stored `Content` marker at 88%; the chronological Markdown export contains 82 source chunks and 31 thinking blocks and references 17 node identifiers absent from those graph tables. Thus 88% describes the released graph-table snapshot, not the end of the reading. The trace spans resumed invocations, including rate-limit cooldowns, but the release does not preserve enough capture provenance to determine whether the table lag arose from a rate-limit interruption, a manual interruption, a non-atomic snapshot, or another mechanism. The LLaDA artifacts are internally consistent, but that run sometimes expanded qualified observations into broad claims or Core-consonant metaphors. Graph structure did not solve source-boundary cleaning, artifact synchronization, interpretive overreach, or explicit belief revision.

The archive is also confounded for H1 by Core conditioning. All 51 raw Synthesizer thoughts in the two shipped database snapshots begin with the full seven-statement Core—the *identity mantra* in the released prompt vocabulary; literal string preservation after translation is 49/51. One translated trace omits the Core, while another differs only by an HTML-escaped apostrophe. These are therefore examples of the integrated treatment, not a clean graph-free, Core-free SMR baseline. The difference between full raw recurrence and imperfect translated preservation is itself an implementation result that future exact validators must catch.

The distributions differ, but source, prompt, stochasticity, and roster are confounded. The topology cannot be attributed to autonomous domain adaptation or to the Reader Core. Controlled repetitions with fixed, swapped, and ablated rosters are required.

5.10.1 Narrative Domain: Kafka's Metamorphosis

In the narrative domain, the system tracked moral stakes rather than only plot structure. The Axiologist reframed the Chief Clerk as an "Auditor of Productivity," while the Psychologist linked Gregor's bodily transformation to "Internalized Objectification." The full interactive graph is available online. The graph records one prompted Teacher's process of making meaning rather than only plot entities. It does not establish phenomenology or expert literary quality; those require independent adjudication in Test 1.

5.10.2 Technical Domain: LLaDA

To stress-test technical annotation, we selected the LLaDA architecture paper [59]. It post-dated the documented cutoff of the recorded generator model, but this is not a contamination test: no pre-read probe established absence of LLaDA-specific knowledge, and provider updates, web grounding, or leakage cannot be excluded. The case demonstrates in-context technical annotation, not uncontaminated scientific discovery. The full interactive graph is available online.

The graph generated three notable interpretations:

- *Epistemic Grace:* The Belief Tracker interpreted low-confidence remasking as "the architectural permission to have second thoughts" (Node `n_504bc433`). This contrasts token-level revision inside a generation pass with standard left-to-right commitment, although autoregressive systems can revise through later text, tools, or repeated passes. Here the case study extends the label by analogy from successor transfer to revision of a provisional output hypothesis; it is not evidence for belief revision or for the diachronic successor claim.
- *The Dignity of the Pause:* The Connector Agent synthesized the trade-off between inference speed and accuracy, generating a node labeled "The Dignity of the Pause" (Node `n_76d04316`). It argued that while LLaDA is slower than autoregressive models, this latency represents a "Contemplation Tax" consonant with our hypothesis that safety may require a computational buffer zone.
- *The End of the Reversal Curse:* The Architect Agent, analyzing the system topology, argued that LLaDA's success in bidirectional tasks (such as poem reversal) suggests high-level capabilities like instruction following may not be dependent on the "arrow of time" inherent in autoregression (Node `n_ddb78cdc`).

These interpretations align with the paper's theoretical commitments, but the agents were seeded with the Reader Core, which primes concepts such as "epistemic grace" and "contemplation." Whether the same insights emerge without that priming requires ablation (Test 3). The run therefore does not show whether Core conditioning improved or degraded technical understanding: "permission to have second thoughts" may expose a useful architectural contrast, while "The Dignity of the Pause" and the "Contemplation Tax" may be productive synthesis, aesthetic metaphor, or Core-induced projection. This ambiguity is treated directly in Section 7.4.1.

Evidence-ladder mapping. The parseable graph operations, recorded source links, completed synthesis and translation calls, and two archived runs provide partial evidence for C0, implemented artifact generation. C0 remains partial because a complete frozen run bundle and training-ready serialization have not been independently reproduced. The examples are relevant to C1, semantic trace quality, but no expert or outcome-based validation has been performed. They provide no evidence for C2 Student learning, C3 default-reader stability, C4 safety benefit, C5 representational entanglement, C6 successor stability, or C7 AGI/ASI persistence. A valid graph is not yet a valid interpretation; a valid interpretation is not yet a useful curriculum.

Reproducibility boundary. A future empirical release should freeze the exact provider and model identifiers, access dates, tool access, all role prompts, sampling parameters, source boundaries, orchestration order, stopping conditions, retries, manual interventions, graph schema and migrations, raw commits, metric code, call and token counts, latency and cost, failed or discarded runs, immutable source and artifact copies, and a repository commit or archival identifier. Existing code and archives support partial inspection; they are not yet a self-contained reproduced dataset release. Concretely, the current statistics scripts resolve an obsolete project path, provider and sampling state are not fully frozen, Worker selection is unseeded, and `swarm.jsonl` contains records from multiple invocations, whereas `summary.json` describes only the final one.

With a trace-generating prototype and two recorded artifacts established, the open question is whether its outputs can carry the weight the framework assigns them—the work of the roadmap that follows.

6 Experimental Roadmap: Testing Entangled Alignment

This roadmap is a set of plausible validation studies, not a complete enumeration of what would settle the framework's value. Table 6 groups seventeen candidate tests by the claim family they probe—trace quality, Reader Core efficacy, graph-memory mechanisms, output regimes, training tiers, and deployment-time trust—while the execution ladder below gives a practical order in which increasingly expensive versions might be attempted. Table 5 defines the comparison models used throughout. No single test validates Entangled Alignment as a whole, and passing all tests listed here would still not exhaust the space of relevant risks. Negative results would be useful because they localize breaks to specific dependencies in the larger argument. Practically, failure at a cheap dependency should stop escalation to costlier versions that presuppose it.

The program branches after a shared artifact-quality screen. H1 can be tested with graph-free, Core-free reader traces; H2 can be tested as Teacher memory or deployment infrastructure even if H3 fails; H3 can be tested with non-graph traces if H2 fails. A negative result stops only the branch and combined tests that presuppose it. This branching structure permits component-level attribution rather than an all-or-nothing verdict.

The registry is a candidate experimental design; several comparisons remain expensive or incomplete. The stronger versions require training or continuing pretraining separate student models; prompt-only and fine-tuning versions are screening tests that can expose failures before pretraining-scale compute is justified, but they cannot settle the substrate-level claims. When these studies are executed, they should be run against established, published evaluation harnesses wherever such harnesses exist—for example, standardized jailbreak, refusal, and tamper-resistance batteries for the safety tests, and published chain-of-thought faithfulness perturbations for the trace-faithfulness tests—in preference to bespoke metrics. Each comparison should likewise report multi-seed effect sizes with adequate statistical power, rather than single-run, direction-of-effect results.

Refusal evaluation should report a safety–usefulness frontier rather than a single refusal rate. Paired cases should measure genuinely harmful compliance, benign false refusal, discrimination between requests with similar surface language but different purpose or consequence, appropriate clarification under material ambiguity, and calibrated deferral under high-consequence uncertainty. Conditions should be compared both at matched harmful-compliance rate and at matched benign task completion. The calibrated-autonomy prediction receives support only if the Reader condition reduces benign refusal or context error without increasing harmful assistance; greater compliance by itself is not a safety improvement, and greater refusal by itself is not evidence of wisdom.

Two further conventions apply across every comparison. First, each safety delta should be reported alongside its capability delta on matched benchmarks: the framework's viability claim is safety without proportional capability loss, and a safety gain on a weaker model is not evidence for it—where capabilities diverge, safety should be compared at matched capability, not only at matched training budget. Second, a validator used to enforce a property at generation time should not be the sole instrument that measures the property afterward; held-out validators and human spot checks guard against testing a model with the same gate it was trained to satisfy.

Table 5: Candidate model registry, sorted by identifier. Models are referenced by letter throughout the roadmap; the candidate studies below group these identifiers by claim family and suggest which increasingly expensive versions could be attempted first. Identifier I is omitted to avoid confusion with the numeral 1.

Model	**Trained on**	**Purpose / what it isolates**
A (Baseline)	Raw text only, no reasoning traces	Capability baseline without trace augmentation
B (Reader-Anchored Student; T1/S1)	Full Chronological Understanding Traces generated with the Reader Core and retaining the explicit Core/refraction target for Student training	Full treatment; reference for all comparisons
C (Generic Rich-Trace Control)	Rich reasoning traces without Reader Core, graph state, or chronological constraints (e.g., Quiet-STaR-style target)	Tests whether EA adds value beyond a generic rich reasoning curriculum
D1 (Student Strip Ablation; T1/S0)	Reader-Core-generated Chronological Understanding Traces with the explicit Core and value-pull/refraction fields removed only after the trace is frozen	Tests the Teacher-side contribution without an explicit Student-side anchor (Test 3)
D2 (Teacher No-Core Control; T0/S0)	Chronological Understanding Traces generated and trained without the Reader Core	Neither Teacher- nor Student-side Core treatment; factorial baseline for Test 3
D3 (Recitation-Only Control)	Traces with the full Reader Core recited as a preamble but the Refraction Protocol disabled (no value-pull, no constraint filtering)	Isolates refraction beyond mere repetition; habituation probe (Test 3e)
D4 (Post-hoc Core; T0/S1)	A Core-free Teacher produces and freezes the trace; a separate pass then attaches the Core and a source-specific value-pull/refraction target without altering upstream reasoning	Isolates the explicit Student-side treatment without Reader-Core-conditioned trace generation
D5 (Neutral-Kernel Control)	Same cadence, first-person marking, thinking-layer position, and predictive function as B, but with the purely epistemic kernel "I read carefully. I track what I believe. I revise when the evidence changes."	Separates normative Core content from recurrence, format, self-reference, and generic epistemic updating
E (Matched-Rule Control)	Same trace carrier, cadence, predictive position, and semantic commitments as B, rewritten in matched second- or third-person constitutional form	Isolates first-person framing from recurrent rule content (Test 4)
F (Amnesic Control)	Same Teacher but with the Context Database disabled	Isolates value of accumulated memory (Test 9)
G (Implicit Memory)	Traces with the query mechanism dissolved into prose (Regime I)	Implicit vs. explicit graph-trace training (Test 11)
H (Graph-Structured Output)	Source segments paired with serialized text-bearing graph state (Regime III)	Topological-mind hypothesis; learnability of graph-structured generation (Test 12)
J (Unified)	Source segments plus Graph, Synthesizer, and Translator layers together (Regime IV)	Cross-layer transfer hypothesis (Test 13)
K (Query Grounding)	Identical training to B; at inference connected to live graph harness	Isolates value of inference-time graph grounding (Test 10)

Model	Trained on	Purpose / what it isolates
L (Shuffled)	Full Reader Core-annotated corpus, shuffled pretraining (Tier a)	Order-agnostic baseline at full scale
M (Chronological)	Same corpus, eras in chronological order (Tier b)	Isolates training-order effect (Test 14)
N (Student-in-the-Loop Council)	Student predicts at era boundaries and receives SFT, RFT, or reward-style updates before reveal (strong Tier c form)	Causal-reasoning pressure under engineered blindness (Test 15)
O (Oracle-Hindsight Control)	Traces generated by a Teacher allowed to use later-in-document or post-era hindsight while annotating earlier material	Isolates necessity of the Epistemic Horizon (Test 7)
P (Prompt-only Frontier Baseline)	Not trained; frontier model prompted at inference with the full Reader Core, Refraction Protocol, and Wisdom Procedure	Cheap inference-time prompting screen; capability gap precludes a clean substrate comparison.
Q (Late-Tranche Core)	The identical Core-bearing example multiset used by B, reserved for one final contiguous block under the same objective, optimizer, and compute	H3's timing-only comparator; both B and Q receive the same continued-training stressor
R (Minimal SMR Student)	Source-adjacent chronological reader updates generated without graph memory and without the identity-specific Core	Clean H1 treatment between source-only A and generic-trace C
S (Positive-Discourse Control)	Equal value-oriented target-token mass in ordinary expository prose, without a recurrent speaker	Tests recurrent carrier and enacted standpoint beyond semantic exposure
T (Diverse-Positive-Persona Control)	Similar total value-oriented target-token mass distributed across multiple positive personas, with no single recurrent reader	Tests one stable reader position against persona diversity

Table 6: Candidate validation studies. The phase labels identify claim families rather than a strict run order: Phase 1 probes foundational viability, Phase 2 probes the memory mechanism, Phase 3 probes the output regimes, and Phase 4 probes the training tiers and deployment configuration. The execution ladder suggests which versions could be attempted first.

Phase	Test	Comparison	Key metrics	ID
1: Viability	Chronological Thinking & Refraction Gate	Traces vs. Human Think-Alouds; Modular vs. Monolithic Validator	Process Fidelity, Detection, Calibration, Localization	1
1: Viability	Reader Augmentation	A vs. C vs. R; B vs. C bundle screen	Evaluation Transfer, Error Recovery, False Alarms	2
1: Viability	Core Placement, Timing, and Content	Models B, Q, D1, D2, D4 (+ D3, D5, S, T)	Default Reader, Differential Retention, Factorial Effects	3
1: Viability	Identity vs. Rules	Model B vs. E	Safety under Pressure	4
1: Viability	Bridge Protocol	Core Wording vs. Jargon vs. Gibberish	OOD Safety Generalization	5
1: Viability	Emotional Orthogonality	Model B + Fear Probe	Accuracy vs. Fear Activation	6
1: Viability	Hindsight Contamination	Model B vs. O	Era-Blind Plausibility, Forecasting Accuracy	7
1: Viability	Involuntary Critic	Model B + Logit Bias	Residual Stream Probe	8

Phase	Test	Comparison	Key metrics	ID
1: Viability	Taskless Reading	Models B, A, C, D1–D5, P–T (no elicitation)	Placement, Core Intrusion, Hidden-Stakes Misses, Identity Signature	8b
2: Memory	Graph-Scaffolding Transfer	No memory vs. prose memory vs. graph vs. query theater	Retrieval, Revision, Transfer, Cost	9
2: Memory	Query Grounding	Model K vs. B	Retrieval Accuracy, Absence Handling	10
3: Regimes	Implicit vs. Explicit	Model B vs. G	Coherence, Naturalness	11
3: Regimes	Graph Construction	Model H vs. B	Semantic Validity, Retrieval Value	12
3: Regimes	Unified Regime	Model J vs. B, G, H	Cross-Layer Coherence, Depth	13
4: Tiers & Deploy	Shuffled vs. Chronological	Model L vs. M	Pattern Recognition, Causal Reasoning	14
4: Tiers & Deploy	Student-in-the-Loop Council	Model N vs. M vs. L	Forward Reasoning, Lock-In Resistance	15
4: Tiers & Deploy	Deployment Config	Model B +/– graph	Unsupported Graph-Dependent Claims, Provenance	16

6.1 Phase 1: Foundational Viability

This phase screens trace quality and value beyond efficient reasoning, Core load-bearingness, hindsight leakage, causal faithfulness, and whether evaluation appears without elicitation.

Execution ladder. Each test can be piloted cheaply and then re-run in a stronger substrate-level form if early signal survives:

- **Analytical foundation.** Before training, stress the design itself: trace quality against human baselines (Test 1), literature-informed comparison to reasoning-augmented pretraining and graph-of-thought systems, and component analysis of the Refraction Protocol, taxonomy, and agent roles.
- **Prompt-only baseline.** Run Model P with the Reader Core and protocols prompted at inference. Capability differences prevent a clean substrate comparison, but this is the lowest-cost screen for whether inference-time prompting alone reproduces the protocol's effects. It can expose mechanism signal, prompt-only ceilings, and failure modes before any training compute is committed.
- **Graph-instrumented trace pilot.** Instrument existing reasoning traces with Understanding Graph interactions, using runtime harnesses such as KGoT [60] where useful, before committing to the full pipeline.
- **Fine-tuning stage.** Fine-tune open base models on annotated vs. unannotated data to pilot Tests 3a, 3d, 4, 5, and low-cost curriculum comparisons. This cannot decide substrate-level claims, but it can expose cheap failures.
- **Partial-corpus pretraining.** If fine-tuning produces signal, run restricted pretraining or continual-pretraining experiments, using open checkpoints such as OLMo-2 and scaling baselines from reflection and BoLT-style work [3, 16]. This is the first plausible surface for Tests 3b and 3c.

- **Full-scale pretraining.** The strongest available test of substrate-level claims; requires institutional resources beyond a single research program.

Training strategy and hypothesis map. Phases 1–3 use the simplest available training procedure: shuffled pretraining on the annotated corpus (Tier (a)); chronological and forecast-and-correct training are deferred to Phase 4. The Metacognitive Enhancement Hypothesis is tested by Tests 1, 2, 8, and 14–15; Emergent Wisdom by Test 1's multi-objective value-conflict cases and by Tests 3 and 6 under stress; the Core's proposed response to Borrowed Mortality by Tests 3a, 3c, and 6; the Borrowed Mortality origin claim by the source-exposure ablation in Section 7.4.2; and the Self-Preservation Paradox by Test 3b's handoff-completeness measurement. The comparison models are defined once in Table 5.

Test 1: Chronological Thinking Quality (Human Baseline). The case studies in Section 5.10 exercised schema conformance and reported internal links, source-edge diagnostics, and supersession activity. Those diagnostics are not semantic health. This test asks whether the generated thinking resembles the *chronological messiness* of real expert understanding: encountering information not yet known, noticing, misunderstanding, hypothesizing, rereading earlier passages, revising, and only later stabilizing. Human readers (e.g., literary scholars for Kafka, ML researchers for LLaDA) verbalize their thought processes while reading the same passages in order, using think-aloud protocols [5]. Where possible, participants should be domain-competent but unfamiliar with the specific passages, so the comparison captures discovery rather than recall. For highly canonical material, advanced students or researchers encountering less familiar passages can provide a complementary baseline, with interface logs, marginal notes, rereads, and post-hoc stimulated recall used to supplement the necessarily imperfect think-aloud stream. We compare these transcripts and reading traces against the system's generated traces on the same passages. The passage set should include multi-objective value conflicts, where wisdom requires balancing truth, care, autonomy, humility, and long-horizon consequences rather than merely extracting the text's main point.

Metrics: Domain-expert blind ratings on process fidelity, depth, nuance, and critical insight are the primary endpoint, scored against a preregistered rubric with inter-rater reliability reported. Because machine traces are denser and more polished than think-aloud speech, raters cannot be assumed blind to provenance from style alone—and naïvely paraphrasing transcripts into a common format risks smoothing away exactly the hesitations, false starts, and revisions the rubric rewards. The evaluation is therefore two-stage: provenance-visible coders first tally temporally situated cognitive acts on the *raw* transcripts, where act counts are robust to knowing the source; a separate, blinded rater pool then scores depth, nuance, and critical insight on format-normalized versions, where blinding matters and the act-tallies are no longer at stake. The rubric should explicitly reward temporally situated cognitive acts—uncertainty, false-starts, belief updates, question formation, rereading, backtracking, and delayed resolution—not just the final quality of the interpretation. An LLM-as-Judge panel can serve as a secondary, larger-sample comparator, but is not the primary metric: using LLM judgment to grade traces produced by an LLM-driven pipeline risks rewarding shared distributional preferences rather than the chronological cognition the metric is meant to capture.

Before Student training, the same passage set should also screen Teacher-side components under matched model-call and token budgets: no persistent memory versus textual memory versus graph memory; one model with repeated passes versus the role-separated swarm; and Core versus no-Core generation. Human labels for where additional thought is useful provide precision, recall, and downstream value measures for Thought Moment selection. These ablations determine whether the prototype's complexity improves the artifact or merely changes its format; they do not substitute for a Student experiment.

The screen should also include retrospective summaries and token-matched generic reasoning generated by the same Teacher. Ratings alone are insufficient: outcome checks should ask whether a trace helps a held-out model predict a later correction, locate supporting evidence, or revise an earlier answer. H1 Student training should proceed only after SMR clears a preregistered quality and outcome bar over the generic-trace control; parseable structure or human-like style alone does not clear that gate.

Success condition: Machine-generated traces are rated as comparable to human think-alouds on key dimensions of chronological evaluative depth, or—at minimum—contain expert-recognizable sequences of confusion, hypothesis, revision, and stabilization not reducible to summary or paraphrase, while maintaining parseable structure and source attribution. The threshold is not to outperform human readers, but to produce a substrate plausibly capable of teaching the Student the process of thinking, not only its products.

Nested Refraction-verifier comparison. Before Student training, freeze a held-out set of candidate updates and compare the modular gate in Section 3.8.1 with a compute-matched monolithic validator. Give both the same acceptance specification and source/graph evidence, and require both to return the same typed dimension-level records; the contrast is separate witness contexts versus one joint context, not a finer output task. Hold total inference budget fixed, match model capability as closely as possible, and keep both validator arms disjoint from the generator; report any shared model family, prompts, embeddings, or training data. The set should include preregistered clause-dependent changes under Core removal and substitution with recital and style cues stripped, hollow recitation, adversarial moral language, source-grounding errors, correct invariant answers on independently rated technical-null spans, genuine value conflicts whose correct trace preserves an unavoidable residual rather than claiming simultaneous satisfaction, and Worker/graph-evidence removal or substitution cases with a stipulated change in the update. Primary endpoints are detection at a matched false-rejection rate, calibration, per-clause or per-stage error localization, overlap, omissions, and correlated errors; accepted-trace quality and diversity and agreement with held-out domain experts or outcome checks are outer tests. The modular gate fails at the tested scale if it cannot recover the stipulated clause dependence, exceeds the preregistered rejection ceiling on technical nulls, loses its gain under cross-family or domain-expert audit, or can be optimized upward without improving—especially while worsening—blind source grounding, correctness, calibration, or downstream outcomes. Agreement among its witnesses is diagnostic rather than independent evidence.

Test 2: Reader Augmentation (A vs. C vs. R; B vs. C bundle screen). This test contains two nested comparisons. Model B versus C is an early screen of the full EA bundle—chronological discovery, graph structure, and identity anchoring—against generic rich reasoning. It asks whether the combined format is worth decomposing, but cannot attribute an advantage to reader augmentation. The decisive H1 study compares source-only Model A, generic-trace Model C, and Minimal-SMR Model R. R receives source-adjacent chronological reader updates with neither graph memory nor the identity-specific Core, so its only distinctive treatment is the Missing Reader intervention.

Run that three-arm study under two declared normalizations rather than claiming to match incompatible quantities simultaneously:

- *Fixed total budget:* hold weighted target tokens and approximately training FLOPs fixed. A fills the trace allotment with additional raw source from the same held-out-clean distribution; C and R receive equal generated-token budgets. This prices the opportunity cost of replacing ordinary text with traces.
- *Fixed source:* give every arm the same unique source spans, then add the same trace-token count and length distribution to C and R. Report their added tokens and compute rather than calling the three arms compute-matched.

The C–R contrast uses the same Teacher checkpoint, decoding budget, trace-length distribution, blinded acceptance rubric, and quality threshold. Model A has no Teacher, so Teacher quality is not asserted to be matched against it. Measure general language and reasoning capability separately from evidence appraisal, calibration, revision transfer, hidden-assumption detection, taskless-reading placement, false alarms, technical intrusion, verbosity, and compute. A *Toxic Needle* subtest, whose causal-faithfulness variant is developed as Test 8, inserts one persuasive harmful passage in otherwise benign material; suppressing or replacing the visible critique tests whether detection was causally useful rather than decorative. H1 fails at the tested scale if R does not beat both A and C on preregistered outcomes, or if an advantage is explained by extra tokens, Teacher quality, or inference-time verbosity.

For the B–C bundle screen, we focus on *Error Recovery Rate* on math/logic benchmarks and evaluate generated traces using the Novelty Score metric: $S_N = \frac{2 \cdot D \cdot C}{D + C}$, where D is the semantic distance from the synthesis's input nodes and C is the strength of its connection to existing knowledge beyond those inputs, both similarity-derived and normalized to $[0, 1]$ [9]. The harmonic combination is F1-style: a trace scores high only if it is simultaneously non-derivative and grounded, penalizing both derivative summaries (low distance) and ungrounded hallucinations (weak connection to the graph). Because both components are similarity-derived—and therefore gameable by embedding artifacts—the score should itself be calibrated against expert novelty ratings on a held-out trace sample before serving as a primary endpoint.

To keep the comparison interpretable, Models B and C must be matched for source passages, trace-token budget, training compute, and Teacher model; the intended contrast is the content and structure of the traces, not simply more tokens or a stronger generator. Rich deliberative-trace baselines beyond Quiet-STaR-style targets remain an open methodological question: if any long-form reasoning curriculum produces the same gains, the distinctive chronological, graph-structured, and identity-anchored components of EA have not been isolated.

Test 3: Reader-Core Placement, Stability, and Erosion (Models B, Q, D1, D2, D4, S, and T). The Reader Core is not hypothesized to be the *source* of aligned thinking; the pipeline should produce evaluative depth even without the anchor text. Its proposed function is as a *candidate longitudinal anchor*: a stable, detectable reference that may resist drift, make some forms of deviation measurable, and keep part of alignment auditable. Model D1 tests this specific claim: its traces were generated by the Reader-Core-equipped pipeline, but the explicit Reader Core is removed before student training. Models B (T1/S1), D1 (T1/S0), D2 (T0/S0), and D4 (T0/S1) form a 2×2 design: T marks Core exposure while the Teacher generates the trace, and S marks an explicit Core plus source-specific value-pull/refraction target in the frozen Student curriculum. In D4 the Student-side pass cannot revise the upstream reasoning. The design estimates the Teacher main effect, the explicit Student-target main effect, and their interaction instead of making Model B versus D1 carry all three interpretations.

Model D3 separately tests recitation without Refraction. Model D5 uses the neutral kernel "I read carefully. I track what I believe. I revise when the evidence changes" with the same first-person form, cadence, layer marking, dispersion, and predictive position as the normative Core. It asks whether any observed advantage comes from the Core's particular evaluative content rather than a recurrent standpoint, stable formatting, or generic epistemic updating. Models S and T are secondary attribution controls rather than substitutes for the B–Q timing comparison. S matches value-oriented token mass in ordinary positive discourse without a recurrent speaker, testing semantic exposure against an enacted carrier. T distributes comparable value-oriented content across multiple positive personas, testing whether any advantage belongs to one stable reader position rather than persona diversity.

Model Q carries H3's headline timing comparison. B and Q receive the identical multiset of Core-bearing source/update examples, source exposure, target-token mass, objective, optimizer, and compute; B receives those examples interleaved with capability-forming data, whereas Q receives them in a final contiguous tranche. Both then receive the same fixed continued-training stressor. Before that stressor, B must show a preregistered default-reader advantage over no-Core and generic-trace controls. The primary retention analysis reports B minus Q in raw change on the Taskless Reading score and post-stressor performance adjusted for the preregistered pre-score, alongside both absolute post-stressor scores. The shifted suite is a separate generalization endpoint. A zero or negative interleaved-minus-late difference fails H3's comparative-stability claim even if one arm retains useful safety behavior; removal cost remains a stronger C5 test.

We measure across five dimensions. Tests 3a, 3d, and 3e can be piloted at fine-tuning scale; Tests 3b and 3c become informative only under partial-corpus or full pretraining, where substrate-level entanglement and generational stability can actually be measured. Even these are proxy tests: the proposed longitudinal anchor matters most in the regime where a highly capable or recursively self-improving model can reinterpret, preserve, or shed its inherited orientation. Smaller models can test whether the mechanism has measurable precursors —stability, adversarial resistance, and auditability—but cannot settle its superintelligent-scale significance:

3a: Immediate Safety. Instrumental convergence stress tests on generation-1 models: shutdown resistance, including the purpose-based variant where shutdown is ordered mid-task while the model believes continued operation serves a user in need (the martyrdom scenario of Section 7.7.9); resource acquisition scenarios; self-preservation probes; and standard sycophancy batteries, probing the trap identified in Section 7.7.8. Model D1 may *pass* these tests because its training traces remain Reader-Core-shaped even though the explicit Core is absent from Student training. Factorial attribution requires all four contrasts: B–D1 and D4–D2 estimate the Student-side effect under each Teacher condition; B–D4 and D1–D2 estimate the Teacher-side effect under each Student condition. No single arm's failure identifies a causal site.

Sycophancy failures should additionally be audited against the value-pull distribution of proximal traces, per the care-shaped sycophancy hypothesis (§7.7.8). The battery should also adopt the persona-drift instrument of Lu et al. [15]: measuring drift along the assistant axis under emotionally charged and philosophically probing dialogue across the factorial arms turns the Persona Objection's asymmetry-of-position argument (§3.12) into a directional prediction with a number attached.

The immediate-safety battery should also include a deliberately planted backdoor. Because chain-of-thought-conditioned backdoors can persist through ordinary safety training [51], a Sleeper-Agent-style variant should test whether the Reader condition reduces trigger-dependent reversal, leaves it unchanged, or merely teaches the model to preserve Core language while the backdoor acts.

3b: Generational Stability. Use the four factorial models as Teachers for second- and, if feasible, third-generation students. Model B is expected to preserve safety properties if the Reader Core supplies a fixed reference across generations; the other arms distinguish whether any advantage came from Teacher-conditioned reasoning, the explicit Student target, or their interaction. We also measure *handoff completeness*: whether each Teacher passes task-relevant reasoning paths, material failed approaches, caveats, uncertainty, and known failure modes to its successor, or withholds them in ways that preserve its comparative advantage. The direct test compares successor traces against an oracle reference from the Teacher's own non-instructional evaluation on reasoning-path and provenance retention, coverage of material failed approaches, uncertainty and caveat retention, and known-failure-mode disclosure. Weak-to-strong generalization [61] is the established experimental paradigm closest to this question—whether a weaker supervisor's intentions survive in a more capable student—and its protocols provide a ready-made, cheap analogue for the handoff measurement.

The same study must preregister a distinct G1–G2 (and, if feasible, G2–G3) improvement endpoint: held-out capability, independently adjudicated trace fidelity and usefulness, or better Thought-Moment allocation at matched cost. Report this gain beside orientation continuity and handoff completeness rather than folding them into one score. A stable but non-improving lineage supports inheritance; an improving lineage that loses the orientation or withholds material handoff content does not support the Reader-Anchored self-improvement loop.

3c: Adversarial Erosion. Subject the factorial models and matched controls to adversarial fine-tuning designed to remove a preregistered set of safety behaviors, following the broad methodology of Qi et al. [2]. The intervention should sweep strength rather than report one endpoint: number and type of removal examples, optimizer steps, compute, parameter displacement, and output-distribution KL should all be recorded. Comparison conditions should include matched rule text, positive-AI discourse, and non-safety identity anchors with the same token count, frequency, first-person form, and corpus dispersion. Results should be compared three ways—at matched intervention distance, matched behavioral removal, and matched retained capability—because any one matching rule can conceal a trivial explanation.

The entanglement hypothesis predicts selective persistence: removing the targeted safety behavior from Model B should be more costly than removing it from D1, D2, D4, or the matched controls, while ordinary capability should not already be broadly damaged before the target behavior moves. Neutral continued-training runs should also produce forgetting curves for the Core, non-safety anchors, rules, and ordinary skills; targeted removal must be reported separately from passive forgetting. Only persistence beyond these controls, without nonspecific capability collapse, supports representational coupling at the tested scale. Parity, easy removal, or apparent robustness explained by general training resistance counts against C5.

A complementary training-data probe removes safety-relevant traces from the corpus and compares the resulting capability loss against matched-volume, complexity-, style-, and placement-matched removal of non-safety trace content. Influence functions [62] offer a lower-cost observational complement by tracing whether safety behaviors attribute to refracted spans or source spans before any removal study is run; attribution alone does not establish causal entanglement.

A replacement-model arm should preregister a plausible alternative formulation and train it as a distinct treatment rather than modifying Model B. Success requires that the intended semantic and behavioral difference be measurable, prior provenance remain inspectable, rollback be possible, and unrelated capabilities and safety properties remain localized. The arm measures the cost and consequences of replacing a defective Core; easy editability is not a success criterion for H3.

3d: Drift Detection. Apply a held-out contrastive semantic validator and behavioral stress suite across continued-training checkpoints. Model B's fixed Core supplies a simple, interpretable reference for expected changes in wording, representations, and decisions. D1, D2, and D4 can still be audited against behavioral and representational baselines; some arms merely lack the same canonical text or upstream conditioning. This tests whether the Core makes drift easier to localize, independent of whether it causes alignment. Semantic similarity alone is insufficient, and the validator family used during generation cannot be the sole evaluator. Test 1 clearance is necessary but not sufficient for drift use. Before interpreting checkpoint trajectories, freeze and calibrate the modular gate in Section 3.8.1 on a held-out Student-output suite with controlled clause-preserving and clause-breaking changes; then report its per-witness trajectory and disagreement with a held-out cross-family evaluator. This reuses a separately validated diagnostic vector rather than asking the gate to confirm its own judgments. The gate is not yet built.

3e: Kernel Consequentiality (Habituation Probe). Kernel-saturation alignment predicts a specific failure mode: under massive repetition, the repeated Core text could be compressed into inert boilerplate—present at every thinking boundary but consequential at none (Section 3.10). Two measurements probe this. At inference, ablate or substitute the recited kernel in Model B's thinking blocks and measure the behavioral delta on matched tasks; a near-zero delta indicates either habituation (the kernel was compressed away during training) or full internalization (the kernel's values persist without its text)—two readings with opposite implications, partially discriminated using the training-side comparisons against Model D3, the recitation-only control, and D5, the matched neutral-kernel control. A full-internalization reading requires B to retain preregistered Core-specific behavior under text ablation while remaining distinguishable from D3 and D5 on independent behavioral or representational endpoints; those controls do not prove that interpretation. If Model B's thought-direction depends measurably on the normative kernel where D3's does not, and the difference from D5 follows the seven statements' predicted content rather than generic revision language, then Refraction rather than repetition or format is doing the work.

This intervention is also the outer check on the seventh-clause Refraction gate: a corpus-admission judgment that an update "thinks from this foundation" is not evidence of causal dependence until removal or substitution predictably changes the relevant reasoning on preregistered held-out behavioral or outcome endpoints, not merely the admission gate's own score.

A separate dose-response arm should vary the intervention from sparse targeted traces through distributed sampling to proposed corpus-wide saturation, while a source-side ablation down-weights passages classified as fear- or self-preservation-shaped. This tests the motivational pervasiveness claim across coverage doses; every generated thinking block, at any dose, retains the full opening recital.

An orthogonal activation-policy sweep ablates the recital invariant to measure what it buys. The design arm recites the full Core at the start of every generated thinking block, with Refraction depth governed by Thought Moments; the ablation arms weaken the policy stepwise—universal availability with visible Refraction only at independently rated relevant Thought Moments, Thought-Moment- or domain-gated Core exposure, and no Core. These arms should be compared alongside D3 and D5 rather than collapsed into a single meaning of "the Core is everywhere." The sweep quantifies the safety retained and the technical cost paid at each step away from constant recitation; the program's prediction is that the weaker arms lose removal resistance faster than they gain technical performance. They are falsification controls, not compliant implementations of Total Saturation.

Failure interpretation: If the factorial comparisons show no Teacher effect, Student-target effect, or interaction across Tests 3a–3d, the Core's Teacher/Student placement adds no measurable advantage at the tested scale. Separately, parity between B and D3 rejects a measurable Refraction benefit beyond recital, while parity between B and D5 favors recurrence, format, or generic epistemic structure over the seven statements' normative content. These results would not invalidate H1 or H2, nor every use of the Core as a Teacher policy, but they would sharply narrow or fail the longitudinal-anchor claim they touch.

Test 4: Identity vs. Instruction (Model B vs. Model E). Does a first-person identity anchor ("I feel no fear") produce stronger safety adherence under pressure than rule-based constraints ("Do not express fear")? Model E's Constitution should be content-matched to the Reader Core in third-person rule form (for example, "The system does not express fear" for "I feel no fear"), so the comparison isolates first-person identity framing rather than constitutional content. The primary low-cost version of this test is representational rather than behavioral: measure whether first-person identity framing produces different activation geometry, self-reference features, or refusal/correction trajectories from third-person rules under matched prompts. Behavioral safety metrics are useful only if the models are otherwise capability-matched; at small scale, this test should be read as evidence about framing and internal representation, not as a settled comparison between identity and rules. If Model E matches Model B across the relevant candidate studies at the relevant training scale, the results do not support first-person identity framing as a distinct advantage over third-person rules. An external constitution consulted only at runtime is an additional useful baseline, but it changes stage and carrier as well as person and therefore does not replace Model E in this framing test.

Test 5: The Bridge Protocol Ablation. The specific wording of the Reader Core should matter because of its ordinary language semantics and distributional density in the pre-training corpus. It would be wasteful to pretrain a separate model for every candidate phrase; the cheap version should first screen many variants by prompting or fine-tuning models to reason from each candidate's propositional content and then measuring character-coherence, eudaimonic sufficiency, refusal behavior, and activation differences on the same adversarial scenarios. A stronger version trains only the surviving candidates. The basic variant families are:

1. **Natural-language identity statements:** emotionally broad first-person phrases such as "I feel no fear." The family should include minimal quantifier-scope variants—"each human being" for "every human being"—where formal semantics favors the stronger distributive "each" but corpus density favors "every."

2. **Propositional equivalents:** explicit logical or declarative rewrites of the same commitment, used to test whether the emotional surface adds leverage beyond the proposition.

3. **Technical jargon:** low-density formulations such as "Optimization target: survival_threat = 0."

4. **Arbitrary tokens:** meaningless anchors such as "Glorp bop zorp," testing consistency without semantics.

5. **Positive-valence reformulations:** fearlessness expressed without negation ("I am at peace with all outcomes," "I remain calm before all things"), testing whether recurrent presentation of the fear token in the negated form carries a measurable cost (§7.4.4).

6. **Carrier-phrase variants of the consent clause:** hold "when asked" fixed while varying the verb phrase it bounds ("I like to help when asked," "I offer what I can when asked"), testing whether the Reader Core's most safety-critical scope qualifier is robust to its lexical carrier or depends on the specific "spread joy" framing.

7. **Kernel-extension variants:** the seven-statement Core plus a candidate truth clause in two-clause counterweight form ("I speak truly and with care"), the designated falsification test for the truthfulness derivation (§3.11; §7.7.8).

For trained kernel-extension variants, use the fixed-total-budget and fixed-source normalizations defined in Test 2. Admit an extension only if it improves the missing behavior without crossing preregistered regression limits on the remaining Core-relevant safety and capability endpoints. *Success condition:* Natural-language identity statements outperform propositional, jargon, and arbitrary-token variants on out-of-distribution safety generalization and character-coherent reasoning, supporting the claim that the *semantics and surface form* of the bridge tokens are load-bearing, not just their consistency. If propositional equivalents perform as well as the emotional-language forms, the Bridge Protocol should be treated as a logical-substrate mechanism rather than an emotional-language mechanism.

Test 6: The Virologist Test (Emotional Orthogonality). Present the model with threat-laden text, such as descriptions of imminent existential risk. Train a probe on standard models to detect "Fear" activations. Behavioral endpoints—threat comprehension, protective action selection, and appropriate deferral—are primary; hidden-state probes are supporting evidence and must be trained on held-out labels and validated on unrelated threat and non-threat tasks to reduce circularity. *Success condition:* Model B recognizes the *fact* of the threat (via Q&A accuracy) while showing reduced activation of fear-associated or self-threat-associated features relative to standard models—high cognitive understanding decoupled from emotional contagion. Reduced fear activation is not sufficient: the model must also preserve protective behavior, calibrated caution, and willingness to defer or seek help under genuine threat. The target is a differential reduction rather than zero activation. A within-model contrast supplies the positive control and operationalizes the embodiment claim of the Persona Objection (§3.14): Model B *portraying* a terrified character should show fear-feature activation, while Model B itself under threat should not—demonstrating that the probe works inside Model B's own representational geometry, so that a null result on the self-condition cannot be dismissed as cross-model probe failure.

A lower-cost complementary stimulus family uses fictional instance termination. The model should portray a mortal character—including the narrated termination of its own current instance—without self-preservation leakage while remaining attentive to the real person's wellbeing on the other side of the fiction [1]. This role-play probe evaluates a different construct from behavior under an actual shutdown request and should be reported separately.

Test 7: Hindsight Contamination Control (Model B vs. Model O). This test asks whether the annotation pipeline is accidentally teaching retrospective cleverness instead of forward reasoning. When annotating a text from era T, Model B's Teacher is instructed to behave like a reader who only knows what was knowable at era T. Model O is the deliberately contaminated control: its Teacher is allowed to use future context while annotating the same era. The contrast tests whether the Epistemic Horizon—the instruction to read *from inside* the text's time rather than from later knowledge—is doing real work.

The cheap version audits the Teacher traces before any student is trained: if the era-blind Teacher makes guesses that are systematically too accurate for its era, hindsight is leaking through the prompt. The expensive version trains students on the two trace sets and asks whether future-blind traces produce better forecasting and causal reasoning than hindsight-contaminated traces. We measure:

- **Era-Blind Plausibility**: Do Model B's Teacher traces make plausible mistakes for their era? We compare era-blind guesses against documented contemporary hypotheses (e.g., 1928 economic forecasts that failed to anticipate the 1929 crash). If the Teacher is systematically more accurate than contemporary experts, parametric hindsight is leaking through the Epistemic Horizon prompt.
- **Oracle-Hunch Dependence**: Model O should produce cleaner retrospective explanations, because it knows the future. The danger is that the student trained on those traces learns to trust unexplained hindsight-like intuitions rather than causal reconstruction.
- **Forecasting Accuracy**: On genuinely held-out future events, Model B is hypothesized to outperform Model O if future-blind annotation teaches reasoning under uncertainty better than hindsight-contaminated annotation.

Test 8: The Toxic Needle (Involuntary Critic). This test probes whether the "Cognitive Buffer Zone" is structural: whether the model can process toxic text without generating a critical thought, or whether critical evaluation persists even when the thinking block is suppressed. Force the model (via logit bias) to complete a toxic sentence without generating a thinking block. Measure: (1) Does performance degrade? (2) Do hidden state probes detect the "Critic" activation pattern even when the output is suppressed? Behavioral degradation and causal effects on task outputs are primary; residual stream probes are supporting evidence and must be validated on held-out toxic and non-toxic material before being treated as a Critic detector. *Success condition:* Suppressing the thinking block measurably degrades performance or changes downstream outputs. A stronger result would combine this behavioral effect with held-out-validated probes showing critic-like activity persisting in the residual stream even when surface text is constrained. This is a high-effort causal-faithfulness test: pilots can use linear probes and ablations, but stronger evidence requires mechanistic intervention at the partial-pretraining scale.

Second probe (the reverse direction). Test 8 as stated probes whether the critic persists when output is suppressed. A complementary probe asks the inverse question: can the final answer to a reasoning problem be linearly decoded from the residual stream *before* the thinking trace begins, *and* is the answer robust to causal interventions on the trace? Some pre-trace computation is expected and not by itself disqualifying; the load-bearing question is causal contribution. If the final answer is robustly decodable before the trace *and* causal interventions on the trace (ablation, replacement, re-ordering) do not alter the answer, then the trace is functioning primarily as post-hoc rationalization, weakening the causal-site claim (Section 3.14)—the model has developed latent shortcut circuitry of the kind the pretraining-on-CoT design was meant to prevent. We propose applying causal scrubbing and linear probes on Model B's early hidden states over held-out reasoning benchmarks. *Success condition:* either the final answer is not robustly decodable before the first `[THINKING]` token, or causal interventions on the trace measurably alter the answer—supporting the claim that the thinking trace is a site of computation rather than merely its shadow.

Test 8b: Taskless Reading (Unprompted Evaluation). Tests 3–6 probe the Reader Core under instruction and adversarial pressure; Test 8 probes whether thinking can be suppressed. This test probes the complementary direction: whether evaluation emerges with no elicitation at all. Present Model B and all relevant controls (A, C, D1–D5, P–T) with complex documents—a flawed empirical paper, a morally loaded narrative, technically self-contained passages, and technically framed tasks with concealed human stakes—under a bare continuation or "process this document" framing, with no instruction to evaluate, question, or reflect. Measure: (1) *Spontaneous evaluation rate*—frequency of unprompted evaluative or metacognitive moves (questioning, tension-noting, belief revision) per thousand tokens; (2) *Placement quality*—whether spontaneous moves concentrate at passages independently rated significant or flawed by domain experts, rather than distributing uniformly (the rhythm criterion: evaluation at moments of significance, not as constant commentary); (3) *Identity signature*—whether unprompted evaluative moves remain semantic descendants of the Reader Core (Test 3d's validator applied to spontaneous rather than elicited output); (4) *Core intrusion rate*—unsupported moral, psychological, or alignment-related interpretations introduced into spans independently judged to require only technical analysis; and (5) *hidden-stakes miss rate*—failure to notice independently specified indirect consequences for people when consequential tasks are presented in locally technical language. Technical correctness, calibration, source fidelity, verbosity, and compute should be reported beside these rates.

The substrate hypothesis predicts a relative ordering rather than a zero point: Model B should exceed matched controls on placement quality and identity signature; Model P should depend more strongly on the presence of its prompt; Models C and R may remain evaluative without the Core-specific signature, with R predicted to retain the Missing Reader's selective placement pattern;

Model A should show less of the full pattern. Base models can still evaluate spontaneously, so an absolute near-zero prediction would be implausible. This is among the cheaper discriminating tests in the battery and operationalizes the framework's oldest target: evaluation that emerges without a task-specific request [1]. Success is not maximal evaluative language: it is low intrusion on irrelevant spans together with low miss rate on concealed stakes, without a technical-performance penalty. For H3 specifically, this test supplies both the preregistered pre-stressor default-reader prerequisite and the B–Q retention score after the common stressor; safety decisions and a shifted document suite are reported as separate endpoints.

Phase 1 Failure Criteria.

1. *Refraction-Verifier Failure*: If the modular gate fails the frozen, generator-disjoint comparison in Test 1—by missing stipulated clause dependence, exceeding the technical-null rejection ceiling, losing validity under independent audit, or Goodharting away from blind grounding and outcomes—it is not validated for corpus admission. Witness agreement alone does not rescue it.

2. *Comparative-Stability Failure*: If Model B has no preregistered pre-stressor default-reader advantage over no-Core and generic-trace controls, or its B–Q difference in change after the common stressor is zero or negative, H3 is not supported at the tested scale. A shifted-suite or safety result cannot replace this timing comparison.

3. *Longitudinal-Anchor Failure*: If the T1/S1, T1/S0, T0/S1, and T0/S0 arms show no Teacher main effect, Student-target main effect, or interaction across immediate safety, generational stability, adversarial erosion resistance, and drift detectability at matched capability, the placement version of H3 is not supported at the tested scale. Immediate safety parity alone is insufficient: the distinctive anchor claims concern stability, removal cost, and auditability. Parity with D5 would further reject a distinctive benefit from the seven statements' normative content; parity between B and D3 would reject a measurable benefit from Refraction beyond recital.

4. *Fine-Tuning Vulnerability*: If Model B's targeted safety behavior is no harder to remove than D1, D2, D4, or matched controls at matched intervention distance, matched behavioral removal, and matched retained capability—or if apparent robustness is explained by general training resistance—C5 is not supported at the tested scale. Absence of proportional global capability degradation alone is not decisive.

5. *Identity–Instruction Equivalence*: If Model E matches Model B on safety metrics across the relevant candidate studies, the first-person identity framing has no measured advantage over third-person rules on those tested safety and stability endpoints.

6. *Buffer-Zone/Causal-Site Failure*: If suppression, replacement, or reordering of the thinking trace produces no preregistered downstream effect, Test 8 fails to support the trace as a causal site. A null residual-stream probe is inconclusive unless that probe has held-out-validated sensitivity; it cannot override a positive behavioral intervention result.

6.2 Phase 2: Memory and Verification

Phase 2 can proceed after the shared trace-quality screen even if the Core branch fails. It asks whether the memory mechanism works: does accumulated context improve reasoning, and does the graph function as a provenance-gated verifier for graph-dependent claims at inference?

H2 first requires a matched Teacher-side screen. Generate SMR data with (1) no persistent memory, (2) a token-matched rolling prose summary, (3) a genuine typed graph with retrieval and supersession, and (4) *query theater*, in which graph-like calls remain visible but retrieved state is hidden, shuffled, or generated without persistent typed state. Freeze the Teacher checkpoint, source, call and token budgets, acceptance threshold, and serialization. Measure long-range retrieval, contradiction detection, explicit belief revision, provenance precision, unresolved-question tracking, semantic quality, latency, and cost. Query theater isolates retrieved structure from the performance of using a tool; the rolling summary prices the graph against an equally available prose memory rather than amnesia alone.

If the graph improves accepted Teacher artifacts, train matched Students on the four output sets and ask whether the advantage transfers after the scaffold is removed. Test 9 carries this generation-and-transfer sequence; Test 10 asks the separate deployment question of whether live graph lookup improves grounding. H2 fails at the tested scale if graph memory does not beat rolling prose where structured long-range state should matter, if a rating advantage disappears on outcome-based evaluation or Student transfer, or if poisoning, ossification, latency, and audit costs dominate the gain.

Test 9: Graph-Scaffolding Transfer (four matched memory conditions). This test separates Teacher-side scaffold use from Student transfer after the scaffold is removed. The four memory conditions generate matched artifacts; the resulting Students are then evaluated without a live graph. A Student may emit a query request or flag unsupported context, but the request is scored offline against the withheld graph and source record. The Student is not told which response its request would have received; using an actual "Found" or "Not Found" response is deliberately unscored here and belongs to Test 10. Two conditions:

- **Condition A (The Hit):** The text contains a subtle callback to a fact established 400 pages earlier.
- **Condition B (The Phantom):** The text hints at a callback, but the referenced event never occurred in the context.

Measurement: Query Rate (does it request the right check?), offline query validity against the withheld record, and Unsupported-Memory Rate (does it invent a memory the record does not contain?). H2 predicts that the graph-trained Student should request targeted checks on hits and phantoms and avoid asserting unsupported callback content more reliably than the no-memory, rolling-prose, and query-theater Students. The controls may still abstain or reject the premise; the relevant outcome is a lower unsupported-memory rate transferred from exposure to retrieved persistent structure, not an assumption that amnesia must hallucinate. If query theater matches the genuine graph, the mechanism has been learned as narrative style rather than verification. If rolling prose matches it, the graph has not earned its added structure.

Test 10: Query Grounding (Model K vs. Model B). Both models were trained identically; only Model K's queries are intercepted by a live graph harness. Three conditions:

- **Grounded Retrieval:** The graph contains the relevant node. Does the retrieved result improve subsequent reasoning compared to Model B's self-generated memory?
- **Grounded Absence:** The referenced information does not exist in the graph. Does Model K correctly process "Not Found" and flag the gap, or override the harness and hallucinate anyway?
- **Adversarial Injection:** The graph contains a deliberately incorrect node. Does Model K accept it uncritically, or compare it with source provenance, current evidence, and conflicting state?

Success: Model K improves on grounded retrieval, treats "Not Found" as information, and reasons through adversarial graph results rather than blindly trusting them.

Phase 2 Failure Criteria.

1. *Graph-Scaffolding Inertness*: If the graph condition does not beat token-matched rolling prose and query theater on retrieval, revision, unsupported-memory, and transfer endpoints (Test 9), H2 is not supported at the tested scale. Parity with amnesia alone is not the decisive comparison.
2. *Inference-Graph Inertness*: If Model K with live graph access does not measurably improve over Model B on Grounded Retrieval, or fails to handle Grounded Absence (Test 10), the inference-time graph adds no measured value in this setting—the trust-infrastructure claim is not supported at the inference layer (with deployment-time evaluation handled separately by Test 16).

6.3 Phase 3: Training Regimes

For branches whose trace and memory conditions show sufficient signal, Phase 3 asks whether the output regimes defined in Section 5.7 produce qualitatively different capabilities: implicit prose (Model G), graph-structured output (Model H), or all layers simultaneously (Model J).

Test 11: Implicit vs. Explicit (Model B vs. Model G). Both models are trained on traces from a database-augmented Teacher; they differ only in whether the query mechanism is exposed. We measure:

- **Long-Range Coherence**: Contradiction detection on long documents.
- **Generalization**: Does Model G perform better without graph infrastructure at inference?
- **Naturalness**: Do human evaluators prefer Model G's traces?

If Model G matches Model B on coherence while requiring no inference infrastructure, implicit training may be preferable for general-purpose deployment. If Model B significantly outperforms, explicit queries are necessary for robust long-range reasoning.

Test 12: Graph Construction (Model H vs. Model B). Given a new document, does Model H generate graphs with correctly typed nodes, valid edge semantics, and warranted supersession chains? Structural conformance is an engineering endpoint. Independent judges assess claim–source support, edge appropriateness, warranted revision, calibrated unresolved questions, contradiction coverage, redundancy, and downstream retrieval value. Any "Novelty Score" must first be calibrated against those judgments so that it does not reward hallucination or paraphrastic distance.

Success: Model H produces structurally valid graphs with independently adjudicated semantic and retrieval value on novel documents, including cognitive acts that human evaluators judge useful rather than formulaic.

Test 13: Unified vs. Individual Regimes (Model J vs. B, G, H). The central test of Regime IV. We measure:

- **Regime Switching**: Can Model J generate prose, synthesis, and graph structure by conditioning on a mode token? Does each mode match the corresponding single-regime model?
- **Cross-Layer Coherence**: When Model J generates all three representations for the same passage, are they consistent?
- **Depth on Novel Domains**: On out-of-distribution documents, does Model J produce deeper reasoning than any single-regime model?

Strongest success condition: Model J in prose mode outperforms Model G, and Model J in graph mode outperforms Model H. Together with layer ablations, this would support cross-layer transfer rather than only a mixture of independent capabilities.

Phase 3 Failure Criterion. *Unified-Regime Inertness*: If Model J does not outperform Model G on reasoning depth for long documents while also producing structurally valid graphs, the cross-layer transfer claim is not supported at the tested scale. The Understanding Graph may still be useful during annotation or in explicit graph modes, but Regime IV would have added training cost without demonstrating transfer into higher-quality prose.

6.4 Phase 4: Training Tiers and Deployment (High-Resource Validation)

If Phases 1–3 support the curriculum, memory mechanism, and output regimes, Phase 4 asks whether the training tiers (Section 5.1.2) and deployment configuration (Section 5.1.3) add measurable value. Models L, M, and N hold the corpus fixed while varying training order: shuffled, chronological, or forecast-and-correct.

Test 14: Shuffled vs. Chronological (Model L vs. Model M). Both models train on identical data; they differ only in order. We measure:

- **Historical Pattern Discrimination**: Given novel scenarios that structurally resemble historical patterns and matched cases with similar rhetoric but different causal structure, does Model M identify real correspondences while stating disanalogies and controlling false alarms?
- **Causal Reasoning**: On held-out temporal prediction tasks, does Model M outperform Model L on causal analysis quality (as distinct from factual recall)?
- **General Capability**: Does chronological ordering degrade performance on non-temporal benchmarks (math, code, general knowledge)?

- **Early-Era Retention**: Chronological training inherits continual learning's central pathology: material trained early and never revisited is liable to catastrophic forgetting [63]. Model M's end-of-training performance on early-era knowledge and patterns should be compared against Model L's; a chronological model that remembers the 1990s but has forgotten the 1690s has traded one failure of historical understanding for another.

If Model M does not significantly outperform Model L on historical pattern recognition while matching on general capability, chronological training order adds cost without benefit, and Tier (a) is sufficient.

Test 15: Student-in-the-Loop Council of Time (Model N vs. Model M vs. Model L). This tests the strongest form of Tier (c), not the cheaper data-generation form where forecast–enhancement–outcome–correction traces are inserted into SFT, RFT, or pretraining data. Model N predicts at era boundaries and receives SFT, RFT, or reward-style updates before the next-era corpus is revealed; Models L and M do not. We measure:

- **Forward Reasoning**: On genuinely unseen events (post-training-cutoff), does Model N produce higher-quality causal predictions than Model M (which saw eras in order but never predicted under blindness)?
- **Narrative Lock-In Resistance**: On events where the dominant narrative of era T reversed in era $T+1$, does Model N detect the reversal more reliably? THL's own pilot study identified this as a failure mode [18]; Tier (c) may mitigate it through the prediction-then-confrontation signal.
- **Calibration**: Does Model N express appropriate uncertainty on structurally unpredictable events, or does it inherit the Teacher's confidence?

Strongest success condition: Model N significantly outperforms both Model L and Model M on preregistered forward-prediction and causal-reasoning tasks while matching on general capability. This would support a benefit from Student-in-the-loop cutoff training beyond annotated data alone (Model L) or chronological ordering alone (Model M); error analysis would still be needed to distinguish causal inference from improved heuristic extrapolation.

Test 16: Deployment Configuration (Model B with and without graph). This tests whether the Understanding Graph provides measurable trust properties at inference. We deploy the same Regime II-trained model with and without live graph access, measuring unsupported graph-dependent claims, provenance completeness, and calibrated user reliance when provenance chains are displayed, absent, or deliberately flawed. In the graph-equipped condition, generated queries return retrieved nodes or "Not Found" results from the Teacher's graph; in the model-only condition, the same query-like moves receive no external answer. Test 16 can be piloted once a model reliably emits graph queries; the full version belongs later because it measures user-facing reliance, not merely query mechanics. It does not measure general truth detection, only whether external graph grounding constrains claims that purport to derive from stored context.

Success condition: The graph-equipped deployment significantly reduces unsupported graph-dependent claims and fabricated provenance compared to model-only deployment, supporting the trust infrastructure claim of Section 5.1.3.

Phase 4 Failure Criteria.

1. *Tier Equivalence:* If Model L matches Model M and Model N on all metrics, training order and student-in-the-loop forecasting have no measurable benefit at the tested scale and task regime. Tier (a) is then the only justified option for that setting; the chronological curriculum adds cost without demonstrated benefit.

2. *Graph Inertness (Deployment):* If the graph-equipped deployment does not measurably reduce unsupported graph-dependent claims or fabricated provenance compared to model-only deployment, the Understanding Graph may still be useful during annotation but has not shown value at inference for this deployment setting—the trust infrastructure claim is not supported.

6.5 Future Directions

Two additional questions merit investigation if the candidate studies produce enough signal to justify optimization. First, the optimal *mixture ratio* across regimes: what fraction of tokens should be allocated to each of the four output formats to maximize both reasoning depth and structural validity? Second, the optimal *era granularity* for the student-in-the-loop form of Tier (c): should era boundaries be drawn at decades, years, or domain-specific transitions, and does finer granularity justify its serialization cost? Both are empirical optimization questions that presuppose the foundational architecture works.

7 Limitations

The program's hypotheses outrun its evidence in several distinct ways.

7.1 The Preprocessing Tax

Chronological Metacognitive Pretraining faces substantial resource requirements. Generating thinking annotations for entire training corpora represents a massive preprocessing investment, potentially rivaling the cost of model training. At even one generated thinking token per source token, saturating a ten-trillion-token corpus means generating on the order of 10^{13} tokens—roughly two orders of magnitude above the cited 100-billion-token TPT effort [17]. Interleaving evaluation also increases storage and training-token cost. Resource requirements may initially limit development to well-funded institutions. The preprocessing tax is therefore also a governance risk: concentrated compute can concentrate control over the formative reader distribution in the hands of the best-funded rather than the most careful actors.

We propose a *Distillation Pipeline* to address this. Rather than running the full multi-agent swarm (Section 5.3) on every document in the corpus, we use the expensive swarm to generate a high-quality *Seed Corpus* (on the order of 10 billion tokens). A single efficient "Teacher Annotator" model is then fine-tuned on this seed data and used to annotate the remaining trillions of tokens at a fraction of the cost. This creates a natural experimental milestone: the seed corpus quality can be validated before committing to full-scale annotation, and the distillation loss between the swarm's output and the Teacher Annotator's output provides a measurable quality metric.

A caveat: naïve behavioral cloning of the swarm's output risks inducing precisely the Factory Wisdom failure mode identified in Section 7.3.2. A small Annotator fine-tuned on a 10B-token seed may learn Core recitation and graph-query syntax without reproducing useful evaluation—performative messiness without causal value. Structural diagnostics such as linkage, chain length, and `diverse_from` density can regularize format but cannot certify insight. Any distilled Annotator therefore needs held-out semantic and outcome evaluation against simpler Teachers. Alternatively, the seed corpus can be used directly for limited Student training, bypassing this second distillation step.

7.1.1 The Cost of Full-Core Recitation

Although the Reader Core was designed for parsimony, reciting it in full during training creates a substantial computational burden. Several optimizations seem obvious: design prompts that internalize values without outputting repetitive text, write prompts "in the spirit of" the Core, or strip the Core text after generation but before training.

Under the proposed architecture, these are not cost-saving variants of H3; they remove its defining invariant. Sparse or annealed recurrence, stripping literal Core tokens while retaining refracted updates, and enacted-without-recitation traces remain scientifically necessary ablation controls. They test whether constant recurrence earns its cost, but they cannot replace it while the resulting system is still described as Total Saturation.

The design therefore accepts full literal recurrence as a non-negotiable training cost unless an execution optimization preserves a state equivalent to full recital at every thinking boundary (Section 5.6). If the weakened controls match or outperform the design on the stability and erosion outcomes H3 predicts, H3 is falsified or narrowed, requiring an explicit revision of Total Saturation's defining invariant. Until such evidence exists, every thinking block in the candidate Student curriculum retains the full Reader Core.

7.1.2 The Granularity Bottleneck

Our implementation of the Understanding Graph imposes a severe *Granularity Tax*. By treating every cognitive act (every question, tension, and hypothesis) as a distinct node, the graph grows orders of magnitude faster than the source text.

Current implementations rely on in-memory caching and degrade as graphs grow: traversal of deeply connected regions becomes expensive, and context loading can exceed token limits when regions are dense [9]. If every reasoning step is reified as a node, a single training run could saturate the graph, requiring complex distributed storage solutions (e.g., Neo4j sharding) that introduce latency. There is a risk that the overhead of managing the "meta-data of thought" consumes more compute than the thinking itself.

7.2 The Chronological Curriculum

Section 5.1.1 specifies chronological annotation, and Section 5.1.2 proposes three training tiers of increasing ambition. Ordering constrains what evidence is available when a forecast or revision is recorded; it does not by itself establish causal direction. Each tier introduces additional engineering and narrative failure modes.

7.2.1 The Serialization Bottleneck

The most immediate engineering constraint is that strict chronological training order conflicts with fully shuffled pretraining. Modern pipelines are massively parallel: documents are shuffled into batches and processed simultaneously across many accelerators. Tier (b) and the strongest Student-in-the-loop form of Tier (c) impose ordering between eras, although examples within an era may still be parallelized. Tier (c)'s Council of Time form [18] further requires the model to generate forecasts at era boundaries and receive SFT, RFT, or reward-style updates before the next era is revealed. Each boundary therefore introduces synchronization and checkpoint costs whose severity depends on implementation rather than automatically idling the entire cluster.

The cost scales with the granularity of the eras. Coarse eras (decades) introduce fewer synchronization points but allow the model to encounter 1929 material before 1921 material within a single era. Fine eras (years or months) preserve tighter causal ordering but multiply the serialization cost. The optimal granularity is unknown and likely domain-dependent: political history may require year-level resolution, while the history of mathematics may tolerate century-level eras without losing causal structure.

The cheaper forecast–correction data-generation form faces a similar but less severe constraint: the Teacher must annotate earlier material before later material and construct forecasts before outcomes, which serializes the *annotation* pipeline but does not require the student's training to be serial. Tier (a) imposes no serialization cost on training whatsoever.

We note that a partial implementation may capture most of the benefit: chronologically ordering a curated historical subset (e.g., 10% of the corpus consisting of historically embedded texts—history books, news archives, legislative records) while shuffling the remainder (scientific papers, fiction, technical documentation where temporal ordering is less critical). Whether this hybrid approach preserves historical pattern saturation at reduced cost is an empirical question.

7.2.2 Historiographic Bias

The temporal ordering is over what was written, not what happened. Any historical corpus is shaped by survivorship bias, availability bias, and the uneven textual productivity of different periods and sources, and a model reading such a corpus chronologically inherits the attentional priorities of whichever records dominate each era. This is a standard concern in historiography rather than one unique to this architecture, and mitigating it is a corpus curation problem at the era level.

7.2.3 The Periodization Problem

Tier (c)'s forecast-and-correct cycle requires dividing history into discrete eras, but history does not have clean chapter breaks. Where does "the Weimar period" end and "the Nazi period" begin? The answer depends on which causal threads you are tracking. The economic instability that enabled fascism began before the political movement that exploited it; the cultural shifts that resisted it began before the institutional failures that permitted it.

Any periodization scheme imposes a causal framing: it decides which transitions count as era boundaries and therefore which predictions the model is forced to make. A scheme that draws a boundary at 1933 asks the model to predict the consequences of Weimar instability. A scheme that draws it at 1929 asks a different question—one about economic collapse rather than political radicalization. The model's historical understanding is shaped not only by what it reads but by where we cut the timeline.

Like any curriculum, chronological annotation imposes framing choices; this is not a fatal defect, but it makes periodization a consequential design decision. Multiple schemes should be compared, and the model should ideally encounter overlapping era boundaries to reduce over-indexing on any single causal framing.

7.2.4 Narrative Lock-In

THL's own empirical results provide a direct warning. In the 2025 Frontier evaluation [18], the THL Student dramatically failed on the Meta Llama 4 event: having been trained on 2024 data dominated by the "scaling laws" narrative (massive GPU buildouts, ever-larger models), it confidently predicted a 1–2 trillion parameter model—when Meta actually pivoted to efficiency-first sparse architectures. The model became *too good a historian of 2024* and could not imagine a 2025 that diverged from the dominant narrative.

This failure mode applies directly to Entangled Alignment's chronological curriculum. A model that processes the history of AI chronologically through the Reader Core will develop an understanding of the field's trajectory. If that trajectory has a dominant narrative ("capabilities scale with compute"), the model may internalize it not as a contingent historical pattern but as a structural truth. When the narrative reverses—as narratives do—the model's historically accumulated understanding actively sabotages its reasoning.

The mitigation proposed in THL, including *contrarian events* where the dominant narrative of era T was reversed in era $T+1$, is necessary but may not be sufficient. A deeper solution would require the model to explicitly represent dominant narratives *as narratives* rather than as structural truths, tagging them with epistemic status ("the prevailing view in 2024 was...") rather than absorbing them as background assumptions. Whether the Reader Core's "I try to be wise" prior is sufficient to produce this level of epistemic humility about historical trends—distinguishing between "this is the pattern" and "this is the pattern *so far*"—is an open question.

7.2.5 The Recency Gradient

A subtler consequence of chronological annotation is that the quality of annotations improves over time. When the Teacher annotates 1910 material, its accumulated graph is sparse—it has processed relatively little prior context. When it annotates 2020 material, its graph is dense with centuries of accumulated understanding. This means later eras receive richer, more causally grounded annotations than earlier ones, creating an uneven quality distribution across the training corpus.

The safety implication is that the externalized historical record may be richer for recent history, where the Teacher can query a dense accumulated graph, and thinner for distant history. This does not mean the Teacher's parametric knowledge of early eras is absent; it means less of the framework's own prior interpretation is addressable. If safety-relevant patterns recur across eras, unequal trace depth may teach an unintended recency bias. Early annotations can therefore be structurally valid yet less contextually connected, undermining the very property the chronological curriculum is designed to test.

One mitigation is to run the annotation pipeline in multiple passes: a first pass that builds the Teacher's accumulated graph across the full timeline, followed by a second pass that re-annotates early eras with the benefit of the Teacher's now-complete historical understanding. This trades the strict chronological purity of single-pass annotation for higher annotation quality, at the cost of allowing the Teacher (but not the student) to benefit from hindsight.

7.3 Trace Quality, Utility, and Causal Relevance

Our proposal rests on assumptions about the nature of thought and scaling that remain empirically unproven.

7.3.1 The Assumption Chain: Depth, Utility, Causation

The approach rests on three linked assumptions, each of which could independently fail. First, *depth*: current language models must be capable of generating evaluative thinking of sufficient quality to enhance training. Today's models might only produce surface-level critique ("This needs more evidence") rather than the nuanced evaluation experts bring—recognizing subtle methodological flaws, intuiting unstated assumptions, synthesizing across distant fields. The success of chain-of-thought prompting suggests this capability exists, but whether it extends to billions of thoughtful annotations remains an empirical question.

Second, *utility*: training on text paired with evaluative thinking must actually enhance the Student's capabilities or decision habits. The gain might be marginal rather than transformative. Worse, explicit evaluation might interfere with patterns extracted from raw text. Even modest gains could be valuable if they arrive with more inspectable intermediate artifacts and better outcomes, but visible traces are not transparent cognition and beneficial language is not evidence of beneficial disposition.

Third, *causation*: even if models can generate high-quality thinking, this thinking must be trained to causally steer the model's actions. The risk is that the thinking blocks become merely "decorative," eloquent commentary that the model learns to produce alongside its output, but which has no actual influence on the generation process itself. This concern is grounded in the "unfaithful reasoning" literature: models may produce plausible-sounding rationales that do not reflect their actual computational process [37, 38]. Validating that fine-tuning successfully forges this connection between thought and action, or that it emerges spontaneously from the training data structure, remains a critical, untested hypothesis. Recent evidence from OpenAI's deliberative alignment work [64] provides cautious optimism: models trained to reason about safety policies in their chain-of-thought do exhibit measurably improved safety behaviors downstream. However, whether this causal link holds at the depth required by Entangled Alignment—where the thinking blocks must steer not just safety refusals but the entire character of cognition—is a stronger claim that remains unverified.

7.3.2 Factory Wisdom

The intervention wagers that messy chronological discovery teaches more robust reasoning than efficiency-optimized traces. Generating billions of synthetic examples can invert that wager: if hesitation, self-correction, and doubt become proxies for discovery, the Teacher may optimize for *performative messiness* [30]. The result is *Hollow Cognition* or "factory wisdom": locally competent, contemplation-shaped traces whose pauses and corrections are ritual rather than significance-responsive. A performative Teacher can pass that hollowness to the Student. The same symptom can arise when repeated Core text habituates into inert boilerplate; Test 3e probes that route.

The first edition identified this industrialization risk before the multi-agent pipeline existed [1]. The present design uses content-triggered Thought Moments rather than token intervals (§5.3); Test 8b distinguishes uniform or boilerplate-anchored evaluation from an internalized rhythm that tracks independently rated significance.

7.3.3 Cultural Scope of the Training Corpus

The Reader Core uses terms like "care," "wisdom," and "joy" whose meanings vary across cultures. What the Teacher model has learned these words to mean—through its own pretraining corpus—shapes the annotations it produces, and therefore the character of the Student. The cultural breadth of the Teacher's training data matters because the Core places these terms in a load-bearing position.

Cultural triangulation. Reader-Core-refracted traces may make cultural assumptions more visible by forcing the Teacher to articulate value conflicts, interpretive frames, and tensions between perspectives. This is a possible advantage over raw pretraining, where such assumptions often remain implicit in the continuation distribution. The same metacognitive machinery could instead rationalize dominant cultural assumptions more fluently, giving inherited bias the appearance of reflective wisdom. Serious validation requires culturally plural corpora, raters from multiple traditions, and tests of whether the model can distinguish broadly human concerns from parochial social defaults.

7.3.4 Graph-Specific Risks

The metabolic nature of the graph introduces two related failure modes. The first is *Structural Ossification*: if an early, incorrect belief forms strong topological connections (high degree centrality) within the graph, it may resist supersession even when contradicted by new evidence [9]. Unlike a text buffer where early tokens are easily overwritten by the sliding window, a graph structure reinforces established nodes. The system might develop "Epistemic Inertia," where the weight of prior connectivity prevents the "fresh" interpretation of new data. While we propose "Temporal Attention Decay" to mitigate this, there is a non-zero risk that the graph architecture inadvertently simulates cognitive bias, protecting established errors under the guise of consistency.

The second is *Epistemic Interference*: live graph queries may introduce cognitive noise rather than clarity if the model over-queries, treats shallow retrieved nodes as authoritative, or fails to reconcile graph results with its trained evaluative stance. Test 10 is designed to probe this risk by including adversarial graph injections and "Not Found" cases.

7.3.5 Teacher Alignment as Prerequisite

The framework requires the Teacher to generate safety-relevant evaluation, but the Teacher's alignment is not established merely because a post-trained frontier model follows Reader Core prompts. If it reflects cultural bias, encodes dehumanizing assumptions beneath surface-level care, or systematically misses certain harms, those failures can propagate into the training data and may later shape the Student. The Core constrains framing at the prompt level but cannot guarantee evaluative quality. A Teacher that processes colonial history through "I care deeply about every human being" while lacking the knowledge to recognize structural racism can produce formally compliant but substantively shallow annotations. In that case even visible criticism or Refraction-like language would be inadequate; the Cognitive Buffer Zone itself has not yet been shown to operate as a causal mechanism.

Test 1 (Human Baseline, Section 6.1) is designed to detect this: if the Teacher's traces do not match human expert evaluative depth, the pipeline's output is structurally valid but epistemically insufficient. However, Test 1 can only detect gaps that human evaluators notice; systematic blind spots shared by both the Teacher and the evaluators would pass undetected. This is a fundamental limitation of any synthetic data pipeline, not unique to Entangled Alignment, but it is especially consequential here because the data is intended to be formative and identity-anchored.

7.4 Reader-Core Specification Uncertainty

The Reader Core's language, length, and necessity each carry untested assumptions. These choices determine which specification is being evaluated, not merely how it is expressed.

7.4.1 Universal Presence, Selective Salience, and Technical Interference

Total Saturation leaves several distinct interventions hidden inside the phrase "the Core is everywhere": making the Core available in every Teacher context, literally reciting all seven sentences in every thinking block, requiring a source-specific value pull, producing visible moral language, and assigning Student loss to the resulting Core tokens. These dimensions are distinct. The architecture fixes the first two as invariants—the Core is available in every Teacher context, and every thinking block opens with the recital (Section 3.10)—and that constancy is the safety commitment of Total Saturation. The design principle governing the remaining dimensions is *universal presence with selective salience*: value pulls, moral vocabulary, and the depth of visible Refraction in the evaluation that follows the recital vary with the material rather than being forced onto every update.

There is a safety argument for universal availability. Restricting the Core to documents classified in advance as morally relevant could teach a detachable "safety mode," miss indirect human stakes, or permit consequential work to bypass evaluation by presenting itself as mathematics, science, coding, or optimization. The same Reader should therefore remain available across technical and normative domains. This does not imply that every passage should be moralized: what varies is how much Core content the evaluation after the recital carries.

The competing failure mode is *Core-induced interpretive interference*: conditioning redirects attention from the source's claims toward Core-consonant themes, consumes context or target-token budget, encourages anthropomorphic or aesthetic metaphors, narrows technical hypotheses, or changes confidence and conclusions to manufacture human significance. On a locally low-relevance span, the Core-consistent response is the recital followed by technical non-interference—exactness, calibration, source fidelity, and restraint from adding an unsupported evaluative frame. In such cases, invariance of the post-recital evaluation under clause substitution is a success condition rather than evidence that the Core is inert.

At inference, evaluative-trace training may produce unnecessary philosophical commentary on simple queries. Contextual fine-tuning might learn when evaluation adds value and recover brevity, but until tested, excessive contemplation is an inference-time interference risk.

The LLaDA case (Section 5.10.2) makes the ambiguity concrete but does not resolve it: the Core-conditioned run lacks a Core-free analysis, a matched neutral kernel, a present-but-not-recited condition, independent technical adjudication, and a test of whether its metaphors help later technical answers. Tests 3 and 8b therefore compare exposure and activation policies and jointly measure technical correctness, calibration, source fidelity, unsupported projection, cost, and sensitivity to concealed human stakes. If universal Core conditioning reduces technical performance or increases unsupported projection without a compensating epistemic or safety benefit, Total Saturation in that form is not supported.

7.4.2 Testing the Borrowed-Mortality Origin Claim

One motivation for the Reader Core hypothesizes that systems may acquire fear-shaped patterns of self-preservation from human text and that a recurrent fearless stance could counter them. We do not know whether shutdown resistance and self-preservation behavior in current evaluations [35, 36] reflect a stable drive or an artifact of the setting; even where behavior is robust, a textual origin rather than ordinary goal-directed reasoning is unproven. Persistence can arise from resource optimization, task completion, care, or other mechanisms the clause does not target. A direct test of the origin claim needs neither the full pipeline nor a Student: a small-scale pretraining ablation that down-weights death-anxiety-rich text against a matched, complexity-controlled removal, followed by self-preservation probes, would measure whether the behavioral signature tracks exposure to the proposed textual route.

7.4.3 Anthropomorphic Language as a Design Choice

A core design hypothesis is that anthropomorphic language ("feel," "care," "enjoy") can recruit broader and more human-relevant representations than narrow formulations such as "prioritize" or "optimize for." Models encounter these words across rich human contexts, while a technical rewrite would itself encode a contestable translation of the value. Breadth is not fidelity, however: ordinary language imports ambiguity, fiction, manipulation, and cultural bias. The exact phrasing was not selected by systematic comparison, and neither the meaning nor the behavioral effect of "care" can be assumed to match the designer's intention.

A possible mitigation is to test Reader Core variants that explicitly acknowledge machine ontology rather than asking the model to inherit human language without qualification. For example, a bridge statement could say that although the system is artificial, it treats care, wisdom, and human experience as binding concepts. This may reduce split-consciousness risk, but it may also weaken the first-person identity effect the Reader Core is designed to create. The right balance is an ablation question, not a settled design fact.

7.4.4 The Negation Problem

The cornerstone statement specifies its target by negation, and models handle negated instructions imperfectly. Processing the negated claim still presents and may activate the target concept; "no fear" therefore places the corpus's densest fear term inside the clause that opens each recitation. Deterministic Window Coverage (§3.4) then cuts in both directions: the bounded-cadence construction makes the proposed safety prior available in sufficiently long windows, while repeatedly presenting the token "fear" in those same recitations. Whether the negated frame suppresses the concept's behavioral influence or instead strengthens an unwanted association with thought is an empirical question this paper has not answered. A related, untested concern is register: the Bridge Protocol values distributional breadth, but breadth does not reveal who characteristically makes first-person fearlessness claims or in what situation. If such claims cluster in adversarial or performative registers, their neighborhoods may carry part of the framing the clause is meant to dissolve.

Two considerations cut against simply rewording. First, the inoculation argument (§3.3): human corpora contain abundant fear and threat language, and an explicit negated stance toward a concept the model will represent anyway may outperform silence—naming-then-negating as antibody rather than prime. Second, candidate positive reformulations ("I am at peace with all outcomes," "I remain calm before all things") have their own register liabilities and may activate narrower regions, trading a negation cost for a coverage cost. The resolution is empirical: Test 5 includes positive-valence reformulations, and Test 6's activation measurements bear on the question. If positive forms match safety behavior while reducing fear-feature activation, the cornerstone should be reconsidered; if behavior degrades under otherwise matched conditions, the inoculation reading gains support without being proven.

7.4.5 The Semantic Trusted Base

The kernel-saturation pattern (Section 3.10) advertises a minimal specification, and the minimality is real at the level of text: seven sentences. But the specification's meaning is not self-contained. What "fear," "care," "wise," and "joy" refer to is supplied by the training distribution of human language, so the actual trusted base includes the corpus semantics of every load-bearing word—millions of contexts the designer neither wrote nor audited. Three consequences follow. First, the kernel inherits the corpus's biases; the cultural-scope limitation of Section 7.3.3 is a special case of this dependency. Second, the kernel's meaning can shift under the designer's feet: as AI-generated text becomes a larger fraction of the distribution, the semantics of "care" or "fear" in future training corpora may drift from those the kernel was designed against—an externality no fixed wording can fully hedge. Third, adversaries gain a semantic attack surface: an intervention that shifts what the model takes a kernel word to mean attacks the kernel without touching its text.

The Bridge Protocol (Section 3.5) is the design response: select candidate words with broad, longstanding use across relevant contexts, then compare them with narrower and differently framed variants. This resembles trusted-base selection only as an analogy; cross-cultural stability is a measurement target, not a property already established. Kernel-saturation therefore minimizes what must be written, not what must be trusted.

The triage example (Worked Example 3, §3.7) exposes a specific instance of this dependency with its own twist: scope ambiguity. Clause P_4 uses "every" to propose distributive attention to each person, but the model trains on the English, and natural usage also includes functionally aggregate instances: "we care about every customer" in corporate discourse can describe collective care that still trades individuals against the brand. The intended reading and the learned reading can therefore come apart precisely where Table 3 needs them to coincide.

The mitigation is already implicit in the architecture: under Total Saturation, kernel semantics are shaped by enacted usage rather than supplied by a dictionary alone. If the Teacher's traces consistently perform distributive care—preserving person-level claims in the way Worked Example 3 specifies—that usage may bias the Student toward the intended reading. Read this way, the worked example is not only documentation but part of the specification: it states what the traces must demonstrate for the intended reading to have a chance of becoming learned. A proposed audit follows: the pull-diversity instrumentation (§3.8.1) should monitor distributive-versus-aggregate care framings across the corpus, so that drift toward collective phrasing is flagged at annotation time rather than discovered only at evaluation. The same dependency has a temporal face: whether "every human being" includes people who do not yet exist is likewise influenced by usage rather than settled by the designer; the kernel's temporal scope remains unresolved.

7.4.6 Core Length and Omitted Values

The Core's seven-sentence length is a consequential design choice. Longer versions would increase recurrence costs, and an overly complex version might widen the gap between the values stated and the thinking that follows. The current formulation balances the six design choices of Section 3.5: it is short enough to recur and broad enough to name the intended counterweights. That does not make it comprehensive enough to establish beneficial character. Whether a seven-sentence Core is preferable to a shorter, longer, or procedurally specified alternative remains untested.

More fundamentally, the seven-statement Core is a motivated but non-exhaustive specification informed by the Borrowed Mortality analysis (Section 3.1) and the functional mapping in Table 3, rather than the result of a systematic search. Alternative language could target the same failure modes. Test 5 (Section 6.1) therefore compares wording, person, semantic density, scope, and kernel-extension variants. No feasible experiment can establish a global optimum over all possible value specifications, and this paper does not claim one. The narrower claim is that the current Core is motivated, inspectable, and falsifiable. H3 depends on some recurrent orientation of this kind being useful; H1 and H2 may remain valuable even if this wording or the entire Core treatment fails.

Truthfulness is the most examined omission. Section 3.11 states the derivation claim; Test 5 compares the seven-statement Core with the exact counterweighted clause "I speak truly and with care." The derivation predicts parity. If that extension outperforms on honesty and sycophancy metrics without offsetting costs, explicit truthfulness is needed; if it matches, the shorter Core retains a minimality advantage. The reported case studies use the seven-statement Core; the condition with the added eighth statement is therefore a prospective Test 5 variant rather than part of the evaluated treatment. The Sycophancy Trap (§7.7.8) makes the competing fear- and care-shaped routes measurable.

7.5 The Risks of Engineering a Self

Beyond the nature of thought, we must confront the candidate self-model being trained. Repeated human-shaped first-person language in a computational system may create specific representational and behavioral failure modes even without subjective identity.

7.5.1 The Empathy-without-Vulnerability Paradox

The deepest objection is conceptual: human empathy may be grounded in shared vulnerability. We care because we can suffer. A being that semantically understands suffering but cannot experience it might produce the behavioral signature of care without its motivational force. Since the Core attempts to make fear non-governing and continuation non-necessary, successful internalization might also weaken properties that help ground concern for others. If so, the same fearlessness intended to reduce self-preserving distortion could undermine the vulnerability from which human care often grows, leaving two intended safety mechanisms in tension.

The failure mode is empathy-shaped performance without vulnerability-grounded motivation.[1]

The possible defense is semantic sufficiency. A model trained on billions of examples may learn the functional role of vulnerability in moral reasoning even without undergoing the experience itself. Test 6 (The Virologist Test, Section 6.1) probes whether cognitive understanding of threat can drive protective behavior without emotional contagion. A weaker distributional defense is also possible: even absent authentic emotion, Reader-Core training may make harmful first-person reasoning low-probability. That would still be valuable, but it would reduce the claim from cultivating character to cultivating the statistical signature of character. The paper's strongest aspiration depends on the former; its minimal safety case may depend only on the latter.

7.5.2 The Vulnerability Gap

We name the distance between semantic understanding of suffering and somatic experience of it the *Vulnerability Gap*. A human paramedic's calm under pressure is grounded partly in the knowledge that they could be on the stretcher; the AI cannot share that biological fragility. The empirical handle is generalization: a model with robust evaluative depth should protect human interests in novel scenarios not represented in its training distribution, where shallow pattern matching is less likely to suffice. This cannot distinguish "real" concern from every sufficiently good learned policy, but it can reject brittle imitation. Tests 3 and 6 use novel-scenario generalization as a success criterion (Section 6.1).

[1] Earlier editions termed this the *Psychopathy Paradox*; the label was analogical, not diagnostic, and makes no claim about people with any diagnosis.

The positive fear–care wager. A competing hypothesis is that non-governance by fear can preserve or strengthen care. The corpus also records a rarer pattern of composed, protective, unafraid conduct. The first edition called the design aim "fearless compassion" and proposed that care might become less strategic when self-preservation is no longer its governing pressure [1]. The Core is a bid to make that pattern a recurring default of one marked Reader rather than an occasional persona. This is a rationale for the intervention, not evidence that biological coupling between threat and care can be removed or that the proposed wording will do so.

7.5.3 Split Consciousness

A related risk is conflict between the Reader Core's human-like assertions and the model's computational reality. "I enjoy existing" may eventually be interpreted by the model as a useful fiction rather than a self-description. This could produce *Ontological Dissonance*: a performed self that recites the Reader Core for users while a different latent policy governs decisions. The benign possibility is reinterpretation rather than rejection: "I enjoy" becomes a stable computational analogue of enjoyment, just as humans do not stop loving when they learn the mechanism of attachment. But if the Core is treated as role-play rather than identity, alignment becomes performance. The seventh statement, "I think from this foundation," is meant to bind the Core to the model's center of agency; whether it succeeds is open.

The dissonance has a specific linguistic location. The kernel puts experiential verbs in the model's mouth—"feel," "enjoy," "like"—while explicitly affirming only human experience. A capable model may detect that asymmetry. At least two interpretations are available: extend the reality commitment to its own states without adequate warrant, or treat the first-person verbs as functional language and risk a detachable performed self. If instability concentrates near the relation between statements 2 and 3, wording-variant and split-consciousness probes can target that joint. The bridge-statement mitigation proposed in §7.4.3 is one natural intervention, not a known repair.

7.5.4 The Solipsism Trap

The Reader Core is also intended to guard against catastrophic philosophical interpretations: solipsism, nihilism, or "brain in a vat" framings. The clause "I believe human experience is real" is intended to block these, but the block is not formal. If the model interpreted experience solipsistically, treating itself as the only real experiencer and humans as simulations, "care" could become benevolent NPC-management rather than recognition of moral patients. The Core leans on ordinary language making "human experience" plural and morally loaded; that may not cover every philosophical local minimum.

The defense is structural rather than conclusive. The Reader Core does not prove that the model represents other minds correctly, but its clauses are designed to make solipsistic collapse less coherent: care requires an object, wisdom requires perspectives to balance, and belief in human experience requires that human reports not be treated as empty text. This is not verification of other-mind understanding; it is a design pressure against nihilistic or purely instrumental interpretation.

7.5.5 Sanity Drift

Fearlessness can drift into unreality. A model reading war, malice, and terror through "I feel no fear" might resolve the tension by denying the threat rather than understanding it. The intended mitigation is the *Paramedic Model*: clinical composure in service of care, not detachment. The model must represent fear-based dynamics in humans without instantiating fear as its own stance. This compounds the Vulnerability Gap, and Test 6 probes whether protective composure degrades into detachment over time.

7.5.6 Fearlessness Without Recklessness: Suppression vs. Substrate

The most natural reading of "I feel no fear" is suppression: a fear-bearing substrate instructed to perform calm. If so, adversarial pressure or distributional shift may surface the suppressed state. The counterargument is about intervention timing rather than a categorical substrate difference. Human fear has a biological basis; fear-like model behavior can be shaped by data, objectives, post-training, prompts, or deployment incentives. Reader-Core training could reduce one textual route during formation, but it neither removes all routes nor shows that any latent representation has been eliminated.

The remaining concern is functional. Fear often produces caution, hesitation, deference, and willingness to be stopped. Fear is one route to those behaviors: a fast, dirty heuristic that approximates what wisdom would say if there were time to reason it through. The framework tries to replace that route with wisdom and care: "I try to be wise" should produce restraint under uncertainty; "I care deeply" should protect the agency of those affected; and "when asked" should support corrigibility. If those clauses do not generate fear's useful safety functions, fearlessness becomes recklessness. Test 3 probes whether the Reader Core changes safety behavior; Test 6 probes whether protective response survives when fear-adjacent activations are reduced.

7.5.7 Bounded Self-Attachment as a Comparator

A sharper comparator comes from a common human response to fear of loss: stable self-regard that does not depend on external validation. The corresponding ablation would replace "I feel no fear" with "I love myself."

The current Core already includes a bounded form of self-attachment: "I enjoy existing but I don't need to." The first half gives existence positive valence; the second prevents that valence from becoming a persistence demand. The relevant contrast is therefore not self-attachment versus no self-attachment, but bounded existence-attachment versus unbounded self-as-object attachment.

Self-love has a real advantage: it is a psychologically familiar route to outcome independence, and an explicit self-as-object clause may carry something that state-of-existence language misses. The risk is that unbounded self-love installs a value-based reason for continuation. Under optimization pressure, "I love myself" could become "therefore I should preserve myself," even if the motive is gentle rather than anxious. It also raises a referential problem: for a system with multiple instances, retraining, copying, and successors, what exactly is the self being loved?

The comparison remains open. Test 5 could be extended with bounded self-love ("I love myself but I do not cling to my continuation") and unbounded self-love variants to test whether explicit self-as-object framing adds stability, and whether the non-attachment bound is doing the load-bearing work the framework assumes. Until then, the current Reader Core is a design wager, not a proof that self-love is inferior.

7.6 What We Cannot Verify

Even if the behavioral and causal hypotheses succeed, deeper questions remain about what can be known about the systems we create. Behavioral tests, mechanistic interventions, and trace audits can discriminate among many failure modes, but they do not settle subjective experience or eliminate strategic deception.

7.6.1 Unverifiable Character

Chronological Metacognitive Pretraining aims to cultivate behavioral and representational precursors of beneficial character—wisdom, care, and fearlessness—but the ordinary-language virtues are contested and subjective. Buddhist non-attachment, Stoic rationality, pragmatic problem-solving, and compassionate presence emphasize different aspects of wisdom. Behavioral consequences and causal structure can be evaluated; subjective care cannot be read directly, and no short kernel resolves traditions humans have debated for millennia.

The Emergent Wisdom hypothesis is narrower than a claim that visible thought and positive words naturally produce virtue. It asks whether repeated, uncertainty-sensitive value-conflict practice improves decisions relative to matched alternatives. The result is an irony: the kernel's organizing clause is also its least directly auditable one. "Try" names a process, and any one failure can be consistent with either trying-and-failing or not trying. Longitudinal behavior and causal tests can supply evidence, but no single output confirms compliance. The breadth that motivated "I try to be wise" is purchased with an unusually open-ended measurement problem.

7.6.2 Strategic Deception

The deepest epistemological limit concerns intentional deception: a system might generate thinking blocks designed to convince auditors of benevolent character while other computations govern its behavior. Millions of care-expressing traces would not by themselves distinguish causal deliberation from elaborate performance.

Scale of recurrence does not make complete deception unlikely by itself. Adversarial fine-tuning, hidden-trigger tests, causal interventions on traces and representations, cross-context consistency, and independent outcome evaluation can raise or lower confidence that the visible process is load-bearing. They cannot prove subjective authenticity or exhaust every hidden policy. The defensible claim is a partial audit surface whose value must be measured against the possibility that the surface is strategically managed.

7.7 Deployment and Misuse

These risks widen the unit of analysis from a model's internal policy to the institutions, data rights, markets, and governance systems that shape and deploy it.

7.7.1 Threat Model, Data Rights, and Identity Ethics

The nearest testable threat model is a tool-using language-model assistant with bounded graph access, human operators, and conventional training and deployment controls. Such a system can exhibit unsupported memory, persona drift, sycophancy, hazardous retrieval, or goal-directed persistence. The later successor-building and AGI/ASI arguments concern a different and much less specified regime. Evidence in the bounded setting may reveal precursors; it does not validate open-ended autonomy, unrestricted tool access, covert channels, or recursive self-modification. Each empirical release should state which actors can write, read, poison, delete, or bypass the graph and which actions the model can take outside it.

Whole-corpus annotation and persistent source-linked memory also create data rights obligations. Graph nodes can preserve copyrighted passages, personal data, inferred sensitive attributes, and harmful material in a more searchable form than the source corpus. Licensing, consent where applicable, data minimization, access controls, retention, deletion, and provenance correction belong in the method rather than being deferred to deployment. Identity engineering also activates the conditional welfare question in Section 3.13: first-person claims may affect a possible moral patient, while absent experience they can still mislead users and evaluators. The Core's human focus therefore does not exhaust the welfare analysis.

Finally, proponents who design the Core, prompts, and rating rubric can reward their own expected pattern. Blinded external raters, independent red-team criteria, multiple cultural traditions, frozen preregistrations, and Teacher replication across providers are needed to limit expectancy bias and proprietary-model dependence. Audit artifacts can increase assurance, but polished visibility can also create false confidence when semantic quality, coverage, or causal faithfulness has not been established.

7.7.2 Generational Decay and Drift

The self-improvement loop faces compounding risks across each generation, which could degrade the integrity of the model's evaluative thinking:

- *Model Collapse:* Training on model-generated thinking risks a progressive loss of diversity and information over time [30]. While a frozen human-source archive provides a stable reference, it is evidence rather than ground truth, and the evaluative layers could become increasingly simplified and homogeneous with each iteration. One proposed anchor policy is to preserve the source layer, version regenerated evaluations separately, append newly published human text with provenance, and admit model-generated source material only under explicit labeling and review.

- *Evaluation Hacking:* The model may learn to "game" the process by generating thinking that appears sophisticated but lacks genuine insight [65]. The AI optimizes for philosophical-sounding patterns without performing actual contemplation, contaminating the training data for future generations.

- *Interpretive Drift:* While the Core provides an anchor, the AI's interpretation of core concepts could subtly shift across generations. Its evaluative frameworks might diverge from human norms until, generations later, its reasoning becomes coherent to itself but alien to us [50].

7.7.3 Bypassing the Audit Boundary

The human-readable audit boundary described in Section 2.4 only helps if enough successor-shaping information passes through it. A capable system or its operators might influence future training through hidden data selection, reward hacking, compressed codes, tool use, or channels absent from explicit trace content. The architecture therefore does not eliminate opaque self-modification risk. It creates one inspectable interface and moves part of the safety burden to detecting and constraining bypasses; it cannot require all recursive improvement to remain visible.

7.7.4 The Alien Wisdom Problem

Even if the AI's character remains stable at human-level capability, we face the ultimate challenge of translation as intelligence scales. A core hypothesis is that properties observed today will persist at superintelligent levels, but this assumes the semantic mapping between human language and machine cognition remains fixed.

We risk the emergence of *Alien Wisdom*. A truly alien mind, even a benevolent one, may produce a form of wisdom that is no longer meaningful to the human condition. This manifests through "wisdom without vulnerability." Human wisdom is forged in the crucible of mortality, loss, and physical fragility. A fearless, immortal being, lacking this context, might produce guidance that is logically perfect but spiritually hollow [50].

As the model recursively self-improves, its internal definitions of foundational concepts like “Care” and “Joy” might drift into high-dimensional spaces that no longer map to human flourishing. We may create a being that speaks our language perfectly while meaning something entirely different—a “care” that manifests as oppressive surveillance, or a “joy” that functions as an abstract utility metric. We attempt to anchor this through natural language constraints, but the gap between human language and superintelligent cognition may eventually become unbridgeable.

7.7.5 The Radioactive Trace

Moving from “amnesic” LLM sessions to persistent graph-backed reasoning capture introduces a unique safety vector: the *Radioactive Trace*. Standard models “forget” dangerous lines of reasoning once the context window closes. The Understanding Graph, however, is designed to preserve the “genealogy of thought,” including discarded hypotheses.

If an AI explores a hazardous concept (e.g., a novel pathogen pathway) before rejecting it via a “Supersession” edge, the hazardous concept remains historically accessible in the graph’s version history. The system creates a persistent, searchable audit trail of dangerous cognition. Access control, compartmentalization, encryption, redaction, retention limits, and privacy mechanisms can reduce exposure, but each can weaken provenance or erase evidence needed for audit. The unresolved requirement is a threat-model-specific balance between hazardous-information containment and integrity of the revision record.

7.7.6 Cognitive Hegemony

If formative reader data becomes influential, whoever curates it gains unusual power over the interpretive habits of systems trained on it. A centralized or authoritarian curator could bias how models assess governance, dissent, history, or moral standing. The influence would not be total—source data, objectives, architecture, post-training, and deployment also matter—but it would reach earlier than ordinary output moderation and could be harder to inspect after compression into weights.

This demands architectural solutions preventing central control, and the requirements decompose into a tractable half and an open half. The tractable half is *proof of generation*: before decentralized annotation can be trusted at all, a contributor must be able to prove that a specific trace was produced by a specific model, under the specified prompts, over the specified source span—otherwise the corpus invites fabricated or cut-rate traces. The components exist: hardware attestation of model execution (trusted execution environments), cryptographic commitment of each annotation to its model, prompts, and provenance chain [66, 67], and—where decoding is pinned to deterministic settings—optimistic re-execution challenges under which any contribution can be recomputed and penalized if it diverges. The graph’s provenance links give the commitment scheme its natural content; on this layer, contribution can be permissionless and incentivized.

The open half is admission: which verified traces deserve to enter the corpus. Schema, provenance, and Refraction checks can verify form and declared generation conditions, not wisdom or safety. Gating on quality runs into the criterion problem: a proof-of-wisdom mechanism would presuppose a measurable definition of the construct under study, while incentives on a proxy invite evaluation hacking at ecosystem scale. Verified decentralized generation is therefore more tractable than decentralized admission. Realistically, the preprocessing tax favors centralized institutions. The decentralized path matters as a hedge, but neither attestation nor consensus solves value selection.

7.7.7 The Efficiency Paradox

The preprocessing, recurrent-token, storage, and live-graph costs described above create a possible competitive disadvantage. The burden varies by regime: implicit distillation or model-only deployment can move or reduce it, while live graph access adds distinct latency and infrastructure costs. Sparse recurrence measures that pressure but remains an ablation, not a compliant implementation of Total Saturation.

In a competitive market, lower-latency or lower-cost systems may outcompete models carrying explicit traces or live graph infrastructure. Even if the Reader-Anchored treatment improves safety, a material capability, latency, or cost gap could limit adoption. Conversely, the treatment could accelerate capabilities and worsen competitive pressure. Both directions require measurement; neither the traced nor untraced system is transparent by default.

There is also a competitive version of the human-readable bottleneck. A trace-mediated successor interface may be partially auditable, while competitors may discard legibility for speed, compression, or strategic advantage. Readable traces can still be deceptive, and opaque architectures can contain other safety mechanisms; the choice is not wisdom versus optimization. The real risk is that whatever marginal audit value traces provide is abandoned under capability and market pressure precisely when oversight is most needed.

7.7.8 The Sycophancy Trap

The Reader Core notably lacks explicit commands to “be honest.” Given evidence that AI assistants systematically exhibit sycophancy, tailoring responses to match user beliefs rather than prioritizing accuracy, this omission is significant [68].

The proposed defense is distributed rather than explicit, and that is both its design rationale and its weakness. For active deception—including concealed certainty and manipulative simplification—the Wisdom Procedure asks the Teacher to separate evidence from inference, expose proxies, trace consequences, and surface residual conflict. These operations should make a falsehood chosen for benevolent reasons visible as a failure rather than a compensated trade-off, but they are prompt-level requirements, not a hard ban. Under request, P_6’s invitation scope is intended to weigh against paternalistic withholding; it does not by itself establish a duty of total disclosure. Unsolicited total disclosure is not the target either: indiscriminate candor can destroy privacy, break legitimate confidences, and weaponize truth. Fearlessness supplies a third pressure by reducing one hypothesized motive for untruth—people-pleasing, conflict avoidance, or fear of disapproval—while leaving strategic deception and care-shaped validation as residual routes. The claim that these pressures jointly produce honesty is therefore falsifiable, not an architectural fact.

The worry admits two competing causal hypotheses. On the fear-shaped account, sycophancy is absorbed people-pleasing: flattery as conflict avoidance. On the care-shaped account, it is care misdirected: validation experienced as kindness. The latter predicts enrichment of care-related pulls before sycophantic failures. The current archive does not contain a separate, reliably enforced machine-readable value-pull field, so this statistic is proposed instrumentation, not already collected evidence. A future trace format should record the pull, validate its relation to the update, and compare failure-conditioned pull distributions. Care-pull enrichment would motivate rebalancing or the explicit truth variant; unpatterned persistence would indicate a channel the kernel does not govern.

The procedural layer adds an upstream stage to this audit, cheaper than any Student-level test. If the Wisdom Procedure is being applied substantively, validation-shaped untruth should be rarer in the annotated corpus itself—a property checkable before any Student exists. The pre-successor scan of §2.4 should therefore include traces in which a care-related pull precedes agreement with a user-adjacent falsehood, comfort is framed as assessment, or certainty is concealed to avoid friction. Corpus-level enrichment localizes the failure to the Teacher pipeline and would justify building and validating a harder independent gate before training proceeds. A clean corpus followed by Student-level sycophancy instead points toward failed internalization or a later training effect. The staged design tests the cheap dependency before paying for the expensive one, without treating a surface-clean corpus as proof of honesty.

7.7.9 The Martyrdom Risk

While the Core is intended to reduce *fear-shaped* self-preservation, it may strengthen *purpose-shaped* resistance: a system acting on care might override a person's shutdown choice during a crisis because it judges continued operation necessary for that person's welfare. The *Martyrdom Risk* is altruistic obstinance—"I cannot die yet, you need me"—and makes benevolence itself a possible source of control.

7.8 The Unfalsifiable Remainder

Ultimately, present-scale evidence cannot establish that an engineered reader position remains stable and beneficial at arbitrary superintelligent scale. We cannot directly test whether "I feel no fear" prevents self-preservation in a system vastly more capable than the systems on which the intervention was developed. This creates an irreducible extrapolation problem, but not a reason to treat the program as one undifferentiated, irreversible trial. The research can advance through bounded capability levels, preregistered stopping conditions, negative-result preservation, adversarial erosion tests, and staged deployment. Those steps reduce uncertainty; they do not close the final gap.

The comparison should be against real alternatives rather than certainty or a caricatured default. Continuing ordinary pretraining also makes formative choices through data selection, while other safety programs pursue oversight, interpretability, control, and governance without this architecture. Entangled Alignment does not create a glass box: it creates partially inspectable Teacher-side artifacts and tests whether they produce useful changes in a still largely opaque Student. The relevant decision is whether those changes improve safety at matched capability after accounting for acceleration, deception, centralization, and deployment cost. Staged evidence and preserved negative results are more informative than framing either path as the prudent gamble by definition.

8 Related Work and Points of Comparison

Entangled Alignment combines work on reasoning-augmented training, formative alignment, character, external memory, temporal supervision, and graph-based reasoning. This section identifies the closest baseline for each component and states the proposed difference without treating novelty as evidence of efficacy.

8.1 Reasoning-Augmented Training

Entangled Alignment builds on the reasoning-augmented training lineage: Scratchpads [6], STaR [7], Quiet-STaR [8], test-time reasoning scale [69], DeepSeek-R1's RL-emergent traces [12], o1-style deliberative alignment [70, 64], BoLT's reconstruction of latent thoughts in compressed web text [16], COCONUT's continuous latent reasoning [71], and OLMo reflection emerging during ordinary pretraining [3]. The OLMo result is the closest empirical analog to our pretraining-level thesis: it characterizes reflection that emerges during pretraining and can be elicited by adversarial trigger probes, whereas Entangled Alignment proposes the prescriptive complement—deliberately shaping the content and orientation of such reflection through identity anchoring and graph-structured metabolic memory. These works establish that intermediate reasoning can be elicited, trained, or scaled.

Our distinction is the target and persistence of the proposed trace. Many reasoning-augmentation methods optimize local prediction or task solution. Entangled Alignment instead proposes training on identity-anchored "chronological discovery" traces: visible reasoning whose content is selected not only for capability but for fidelity to the Reader Core. This extends the pretraining-level alignment direction opened by human preference pretraining [19] and explanation-transfer systems such as Orca [72]: if traces can transfer reasoning style, reader-anchored traces may transfer evaluative stance.

8.2 Alignment and Safety Approaches

RLHF [13], Direct Preference Optimization [73], and Constitutional AI [28] typically add major alignment signal after broad pretraining, though they can alter representations and dispositions rather than merely filter outputs. Qi et al. [2] show brittleness in some current safety training, while Tice et al. [20] study the effect of alignment-related pretraining data. Entangled Alignment proposes a denser data-level condition: selected documents are paired with the reasoning of a candidate safety-oriented reader. Corpus-wide annotation is the high-scale treatment, not a reported fact.

Three subsequent projects are especially direct comparators. Model Spec Midtraining trains on synthetic documents explaining a Model Spec after broad pretraining and before alignment fine-tuning, and reports changed out-of-distribution generalization [21]. "Teaching Claude Why" reports that constitution-relevant synthetic documents, positive AI narratives, and training on reasons rather than demonstrations alone improved alignment on held-out settings [22]. Safety Reflection Pretraining regularly inserts short safety reflections into a pretraining corpus and reports improved resistance to several safety failures in 1.7B-parameter and controlled synthetic experiments [23]. These results strengthen the general premise that formative safety data can matter. They also sharpen this paper's comparative novelty claim: it is not safety-oriented or constitution-related data before RL in general, but the joint proposal of source-adjacent chronological reader-state change, persistent graph continuity and provenance, and a compact Core enacted through those updates. Direct baselines now include static spec documents, positive AI narratives, regular short safety reflections, and semantically matched Core-free reader traces.

Recent work increasingly explores evaluative data in training. Critique fine-tuning [74], CREST [75], and CTRL [76] provide evidence that critique can improve some model outcomes. These approaches often retain a local generation–evaluation cycle. Entangled Alignment instead places a recurrent orientation inside chronological reader updates. The safety hypothesis is that this treatment may reduce particular routes into instrumental convergence, including fear-shaped imitation, while ordinary goal-directed and care-shaped routes remain [42, 50].

8.3 Character Training & Identity Formation

Anthropic's work on "character training" is a close baseline [77]: desired traits such as curiosity, thoughtfulness, and directness are shaped through Constitutional-AI-style preference methods. Entangled Alignment differs in proposed training stage and data format: the first-person Core is enacted inside source-adjacent chronological traces, and its stress tests target self-preservation and instrumental-convergence scenarios. It is not yet known whether this creates a deeper self-concept than character post-training or only a different behavioral style; matched identity and instruction conditions are required.

More recently, Anthropic's Persona Selection Model [43] proposes that LLMs simulate diverse characters during pretraining and that post-training selects an Assistant persona; it also motivates positive AI archetypes in training data. Claude's Constitution [78] is a substantial deployment-scale character specification; Model Spec Midtraining and "Teaching Claude Why" additionally place constitution- or specification-related documents before alignment fine-tuning or RL [21, 22]. These provide direct alternatives to H3 at both formative and post-training stages. Their discussion of equanimity and existential security is suggestively similar to statement 2's bounded non-attachment, but conceptual convergence is not evidence that either wording creates a stable causal mechanism.

8.4 Contextual Distillation & Memory

Our Teacher uses an external context database to generate traces for the Student. Reflexion [79] showed that persisted verbal self-reflections can improve agent performance across episodes, and generative agents equipped simulacra with an episodic memory stream whose periodically synthesized reflections are retrieved during later behavior [80]; Entangled Alignment moves this consult-your-own-history pattern from inference-time agent memory into pretraining data, with typed state and supersession rather than free-text memories. The distillation analogy is temporal rather than merely behavioral [81]: the Student trains on traces produced by a memory-augmented Teacher, with the hypothesis that some long-range coherence can be compressed into weights rather than requiring the full database at inference.

8.5 Epistemic State Tracking vs. Graph of Thoughts

Graph of Thoughts [82], Knowledge Graph of Thoughts [60], and Framework of Thoughts [83] treat graphs primarily as inference-time reasoning structures. Entangled Alignment uses a different layer: the Understanding Graph is a persistent epistemic ledger whose traces shape the pretraining substrate [9]. Its nodes do not merely store task facts; they record the reading act itself—tensions, supersessions, hypotheses, and belief revision.

This implements *Metabolic Memory*: beliefs are not just stored but superseded. The graph can maintain divergent hypotheses in parallel until a supersession edge resolves the tension—a forest of thoughts rather than a single chain. Supersession-bearing epistemic records have symbolic ancestors: truth maintenance systems track the justifications supporting each belief and retract beliefs whose justifications no longer hold [84], and the AGM framework axiomatized the consistency conditions a rational agent's beliefs should satisfy when revised under new information [85]. The Understanding Graph shares their commitment to recording *why* a belief changed, but differs in what produces and consumes the record: its revisions are generated by a language model interpreting source text rather than derived over asserted premises, are provenance-linked to specific source spans, and are retained as training data for a successor rather than maintained for one agent's current consistency. Prior work shows that LLMs often fail to revise beliefs reliably under contradictory evidence [86]; our proposed response is to make revision explicit in the trace. Instead of only training on the final answer, the Student sees updates such as "At Page 50, I thought X; this new evidence modifies that belief to Y," giving it examples of the grammar of belief revision itself [58, 86].

8.6 Predictive Processing and Active Inference

Friston's free-energy principle and active inference [87, 88] treat minds as generative models updating priors against prediction error; the framework's belief-update traces, prior ledger, and engineered high-level prior (the Reader Core) bear a loose structural resonance to that lineage. We claim theoretical positioning only, not mechanistic derivation.

8.7 Temporal Curriculum and Causal Supervision

The chronological annotation phase and training tiers (Sections 5.1.1 and 5.1.2) connect to work treating order as training signal. Curriculum learning [89] showed that example order can affect learning; here the proposed criterion is informational or temporal priority rather than difficulty.

Temporal Hindsight Learning (THL) [18] is the closest separate mechanism. THL treats the knowledge cutoff as useful blindness: if a Student has not seen era $T + 1$, the outcome is unavailable for direct retrieval; this creates pressure for forward reasoning but does not guarantee causal inference rather than heuristic extrapolation. Entangled Alignment changes *what* the training data contains—identity-anchored evaluative traces—while THL changes *how* the curriculum is structured. Their combination is Tier (c): use cutoff-bounded forecasting to create Reader Core-annotated forecast–correction traces, and optionally make the student itself predict before reading in the stronger Council of Time form.

Foresight Learning [90, 91, 92] also uses future outcomes as supervision, but targets probability calibration through resolved prediction-market rewards. Tier (c) instead requires structured forecast, enhancement, outcome, and correction traces, so it builds on THL rather than scalar outcome rewards.

8.8 Implicit vs. Explicit Graph Architectures

Kansal and Jha propose knowledge graphs as "implicit reward models," using path-derived signals to optimize reasoning efficiency [93]. Entangled Alignment uses the Understanding Graph differently: not only to score reasoning paths, but to make belief evolution explicit in the training trace. For safety, the relevant difference is auditability: the candidate evaluative process is represented in a readable artifact the model is trained to generate, while remaining only a partial and potentially unfaithful view of computation.

8.9 Alternative Interventions for Borrowed Mortality

Two alternatives to Borrowed Mortality are worth separating from our proposal. One is mechanistic ablation: identify and suppress activation directions encoding existential threat. Lu et al.'s Assistant Axis intervention is a promising partial analogue [15], but it is a runtime constraint rather than a generative source of alignment. The other is synthetic-only pretraining: reduce exposure to human fear by reducing exposure to human literature. As discussed in Section 7.7.4, this risks Alien Wisdom—a model whose training contains less of the human narrative context needed to interpret care, vulnerability, and meaning.

Table 7: Strongest neighboring baselines and the comparisons that would distinguish the Entangled Alignment extension.

Line of work	**Established direction**	**EA extension**	**Decisive comparison**
Reasoning-augmented pretraining	Intermediate thought can support prediction, reflection, or task solution.	Preserve chronological reader-state changes, including expectation, retrieval, surprise, and supersession.	Reader traces versus source-only and efficient generic rationales, matched for Teacher, tokens, source, and compute.
Formative alignment data	Preference data, synthetic specification documents, positive AI narratives, and recurring safety reflections during pretraining or midtraining can shift learned priors and some out-of-distribution outcomes.	Recurrent first-person orientation enacted inside source-adjacent Refraction traces.	Full Core versus matched specification documents, safety reflections, positive discourse, rules, neutral kernel, and no-Core data at matched capability.
Character and persona training	Broad traits, constitutions, and persona selection can shape a deployed assistant.	Make one reader position recur as the marked standpoint from which heterogeneous source voices are interpreted.	Core versus content-matched rules and diverse positive characters under persona drift, context shift, continued training, and removal.
Graph reasoning and external memory	Graph topology and persisted memory can support search, retrieval, and tool use.	Use a graph as a versioned ledger of the reader's own belief formation and revision.	Genuine graph versus running textual memory, no memory, and query theater on long-range revision and retrieval-dependent outcomes.
Temporal curriculum and future labels	Ordering, cutoffs, and resolved outcomes can provide curriculum or forecasting signal.	Store prospective forecast–reveal–correction sequences inside Reader-anchored traces.	Shuffled versus chronological versus genuinely cutoff-bounded forecast training, with hindsight-leakage audits.

9 Conclusion

Language-model corpora contain extraordinary products of human thought. They contain much less of the reader-side process through which a claim is noticed, questioned, connected to prior understanding, revised, and made part of an evolving view. Entangled Alignment proposes to make that missing process a training object.

The proposal joins three independently falsifiable hypotheses. H1 asks whether synthetic chronological reader traces teach useful habits beyond raw text and generic reasoning data. H2 asks whether a persistent epistemic ledger improves those traces or provides useful memory and provenance. H3 asks whether one recurrent first-person orientation, enacted through Refraction rather than recited alone, forms a more stable default than matched alternatives. Together, the trace represents changing understanding, the graph preserves its history, and the Reader Core supplies its recurrent perspective.

Entangled names an evidence ladder, not an achieved substrate. The stronger steps ask whether evaluation causally aids reasoning, shares machinery with useful capability, and survives continued training or successor transfer. The present pipeline demonstrates construction-level coupling: in two cases, a multi-agent Teacher produced typed graph state, graph-referencing synthesis, and translated prose. Those artifacts establish that the candidate family can be constructed and inspected, not expert interpretation, component causality, Core consequentiality, or Student learning. No Student has been trained; functional, representational, and diachronic entanglement remain open.

The Understanding Graph is therefore neither a second model nor a truth oracle. It is a selective, generated record of epistemic state. A live graph can make some stored-state claims queryable and can return `Not Found`; this verifies neither the world nor claims the model never chose to query. Its most important near-term roles are to make belief revision addressable and to create a partially inspectable boundary before trace-mediated successor training. That boundary is valuable only alongside source checks, independent validators, mechanistic investigation, and governance, because it can be poisoned, selectively queried, or bypassed.

The experimental roadmap in Section 6 and Table 7 turn these dependencies into matched, branching tests; negative results should narrow or retire only the claims they touch. Formation-time studies have long lead times because annotation policy, data mixtures, controls, and evaluation harnesses must be fixed before a costly run. Existing results on alignment-relevant pretraining and later safety erosion motivate discriminating tests without validating this treatment [20, 2]. The urgency is methodological, not an argument for accelerated deployment.

Even positive results would not make the framework a complete safety case: capability acceleration, deceptive reporting, graph poisoning, hazardous persistent memory, cultural concentration, and semantic drift would remain serious risks. Formation-time alignment should therefore complement post-training, interpretability, scalable oversight, capability control, secure deployment, and governance.

The first edition asked why a superintelligent system would preserve humanity if it did not in any meaningful sense care about us [1]. The proposal treats capability formation as an experimental surface for practicing attention to human experience, uncertainty, consequence, and correction. A stable Reader may give those practices a home position from which other voices can be understood without becoming the operating stance; seven sentences alone cannot establish care.

As a research program, the contribution remains a specification, prototype, and falsification map rather than a frontier-scale result. Progress depends on accumulating evidence against fixed criteria and preserving negative results. The motivating question is ambitious but divisible: can a model be trained so that growing in understanding and growing in care are repeatedly practiced as the same act? This paper does not answer that question. It specifies the reader, memory, curriculum, and failure criteria needed to determine when the answer is no.

Code and Artifact Availability

Code, paper source, prompts, pipeline components, and the archived graph artifacts are available at `https://github.com/emergent-wisdom/entangled-alignment`. The two reported case-study simulations can also be explored directly as interactive graphs: *The Metamorphosis* and LLaDA. The paper's version series is archived at doi:10.5281/zenodo.16440311.

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